

Oxidation and reduction

Learning outcomes

By the end of this section you should be able to:

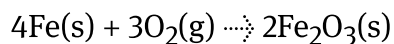
- (R3.2.1) Deduce the oxidation state of an atom in a compound or ion.
- (R3.2.1) Identify the oxidised and reduced species and the oxidising and reducing agents in a chemical reaction.
- (R3.2.2) Deduce redox half-equations and equations in acidic or neutral solutions.



Figure 1. A rusty bicycle.

Credit: fhm, Getty Images

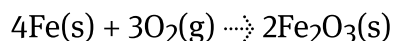
You might have noticed that certain metals rust with time. For example, the bicycle in **Figure 1** is covered in that characteristic reddish-brown colour of rust which indicates the presence of iron(III) oxide, produced when iron reacts with oxygen in the air:



When any iron material is left in contact with air, you will see it rust. You might have heard this described as the oxidation of iron.

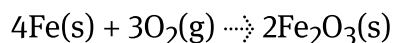
Defining oxidation and reduction

The rusting of iron can be represented by the equation:



Worked example 1

Describe what happens to iron in the equation:



Iron, Fe, reacts with oxygen to form a new compound, iron(III) oxide, Fe_2O_3 . It can be said that the iron gains oxygen, or becomes oxidised.

In short, iron undergoes oxidation.

Use **Interactive 1** to assign oxidation states (see [section S3.1.6](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-states-id-44696/>)) to all the species involved in the rusting of iron.

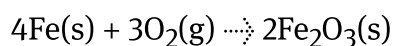
$$4\text{Fe(s)} + 3\text{O}_2\text{(g)} \rightarrow 2\text{Fe}_2\text{O}_3\text{(s)}$$

+7	+8	+9
+4	+5	+6
+1	+2	+3
0		
-1	-2	-3
-4	-5	-6
-7	-8	-9

Interactive 1. Assigning Oxidation States.

Worked example 2

For the rusting of iron:



describe what is happening to iron and to oxygen in terms of their oxidation states.

The oxidation state of iron is *increasing* from zero to +3. The oxidation state of oxygen is *decreasing* from zero to -2.

It can be said that:

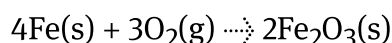
- Iron is increasing in oxidation state or becoming oxidised.
Iron is undergoing oxidation.

- Oxygen is decreasing in oxidation state or becoming reduced. Oxygen is undergoing reduction.

You can also consider what happens to iron and oxygen in terms of the number of electrons each possess, based on the changes to their oxidation states.

Worked example 3

For the rusting of iron:



describe what happens to iron and to oxygen in terms of the number of electrons that each species possesses.

Each iron atom loses three electrons and becomes a cation, Fe^{3+} .

Each oxygen atom gains two electrons and becomes an anion, O^{2-} .

It can be said that:

- Iron loses electrons or becomes oxidised. Iron undergoes oxidation.
- Oxygen gains electrons or becomes reduced. Oxygen undergoes reduction.

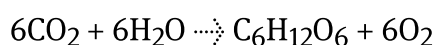
Analysing all the information above, you can see that oxygen was added to iron – this is called oxidation. Whenever one of the species in the chemical reaction is oxidised, another species is reduced. It is said that it undergoes reduction, which in this case could be defined as the removal of oxygen. The definition of oxidation as the addition of oxygen is historical, not all oxidation-reduction reactions involve the addition of oxygen.

Making connections

The surface oxidation of metals is often known as corrosion. What are some of the consequences of this process? (see subtopic S2.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43111/>)).

Rusting is an example of corrosion. Corrosion creates structural damage and reduces the functionality of the material. It can lead to increased maintenance costs, safety risks, economic impact and reputational damage. The use of preventive measures, such as proper material selection, protective coatings and regular maintenance, can help mitigate the consequences of corrosion.

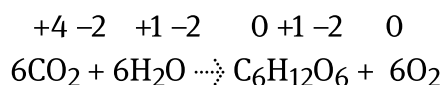
Photosynthesis is a reduction reaction in which plants use energy from the Sun to convert carbon dioxide and water into glucose (sugar) and oxygen, according to the equation:



Describe what happens to carbon dioxide during photosynthesis.

Carbon dioxide, CO_2 , reacts with water to form a new compound, glucose, $\text{C}_6\text{H}_{12}\text{O}_6$. It can be said that the carbon atom gains hydrogen, or becomes reduced. Carbon undergoes reduction.

Describe what happens to carbon and to oxygen in terms of their oxidation states during photosynthesis.



The oxidation state of carbon decreases from +4 to zero.

The oxidation state of oxygen increases from -2 to zero.

It can be said that:

- Carbon decreases in oxidation state or becomes reduced. Carbon undergoes reduction.

- Oxygen increases in oxidation state or becomes oxidised.
Oxygen undergoes oxidation.

Based on the changes to their oxidation states, describe what happens to carbon and oxygen in terms of the number of electrons they possess.

Each carbon atom gains four electrons and each oxygen atom loses two electrons. In this case, no ions are created as carbon and oxygen share their electrons in covalent bonds.

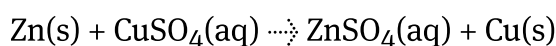
It can be said that:

- Carbon gains electrons or becomes reduced. Carbon undergoes reduction.
- Oxygen loses electrons or becomes oxidised. Oxygen undergoes oxidation.

You can see hydrogen atoms were removed from the water (oxidation) and added to the carbon dioxide (reduction) – this is another way to define redox reactions.

Recall that ions are formed when an atom loses or gains electrons (see [subtopic S2.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/)).

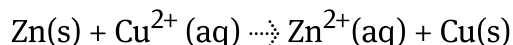
The reaction of zinc metal with copper(II) ions in a CuSO_4 solution has the equation:



Notice that zinc metal, Zn(s) , loses electrons (oxidation), becoming zinc ions, $\text{Zn}^{2+}\text{(aq)}$, while copper ions, Cu^{2+} , gain electrons (reduction) becoming copper metal, Cu(s) . The sulfate ions, SO_4^{2-} , remain unchanged – they are called spectator ions.

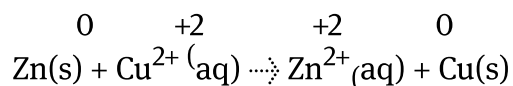
You can rewrite the equation, omitting the spectator ions as they are not involved in the reaction.

Rewrite the reaction of zinc and copper(II) sulfate, omitting the spectator ions.



The copper ions in the copper(II) sulfate compound have a charge of 2+. This means that the zinc ion in the zinc sulfate also has a charge of 2+.

Assign oxidation states to all of ions in the previous equation and determine which species were oxidised and which were reduced.



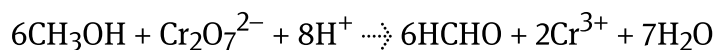
The oxidation state of zinc increases from zero to +2.

The oxidation state of copper decreases from +2 to zero.

It can be said that:

- Zinc increases in oxidation state or becomes oxidised. Zinc undergoes oxidation.
- Copper decreases in oxidation state or becomes reduced. Copper undergoes reduction.

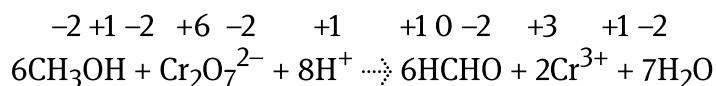
Look at the equation that represents the reaction that transforms methanol, CH_3OH , to methanal, HCHO :



One of the most common oxidising agents, potassium dichromate, $\text{K}_2\text{Cr}_2\text{O}_7$, is used in the reaction. You will learn more about oxidising agents later in this section. The reaction takes place in the presence of acid (H_2SO_4) and heat.

Worked example 4

Deduce the oxidation states of all atoms in the equation above and determine which species are oxidised and reduced.



The oxidation state of carbon increases from -2 to 0 .

The oxidation state of chromium decreases from $+6$ to $+3$.

It can be said that:

- Carbon increases in oxidation state or becomes oxidised.
Carbon undergoes oxidation.
- Chromium decreases in oxidation state or becomes reduced.
Chromium undergoes reduction.

Study skills

Include the use of oxidation numbers in the naming of compounds, as in the examples above for iron(III) oxide and copper(II) sulfate.

Note that there is no space between the name of the metal and the parentheses that include the Roman numeral for the oxidation state.

Recall that the terms 'oxidation number' and 'oxidation state' are often used interchangeably, and either term is acceptable in assessment (see [section S3.1.6](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-states-id-44696/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-states-id-44696/>)).

There are four ways to define oxidation and reduction. Complete the table in **Interactive 2** by dragging the correct sentences to the spaces to create correct comparisons.

	Oxidation	Reduction
Oxidation state		
Electron transfer		
Oxygen		
Hydrogen		

Added

Decreases

Removed

Gain

Increases

Removed

Loss

Added

☒ Check

Interactive 2. Summary Table for the Definitions of Oxidation and Reduction.

Making connections

What are the advantages and limitations of using oxidation states to track redox changes? (see [subtopic S3.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/)).

Oxidation states are a useful tool for tracking redox changes in many chemical reactions. They provide a simple and systematic way to track changes in electron transfer during redox reactions. Oxidation states follow specific rules and guidelines and can help students understand and visualise the changes in electron distribution during redox reactions, aiding in conceptual understanding.

However, they have limitations and should be used with caution, especially in complex chemical systems, such as enzymes, organometallic compounds and transition metal complexes. Oxidation states are an approximation of the electron

distribution in a compound or ion, and they may not always accurately reflect the true electron density and distribution. Careful consideration of the specific context and chemical environment is necessary for an accurate interpretation of oxidation states in redox reactions.

Oxidising and reducing agents

In redox reactions, the reduction of one species occurs only because there is oxidation of the other species. Therefore, we can say that:

- the species that causes oxidation is the oxidising agent and becomes reduced.
- the species that causes reduction is the reducing agent and becomes oxidised.

Figure 2 shows a representation of the roles of oxidising and reducing agents.

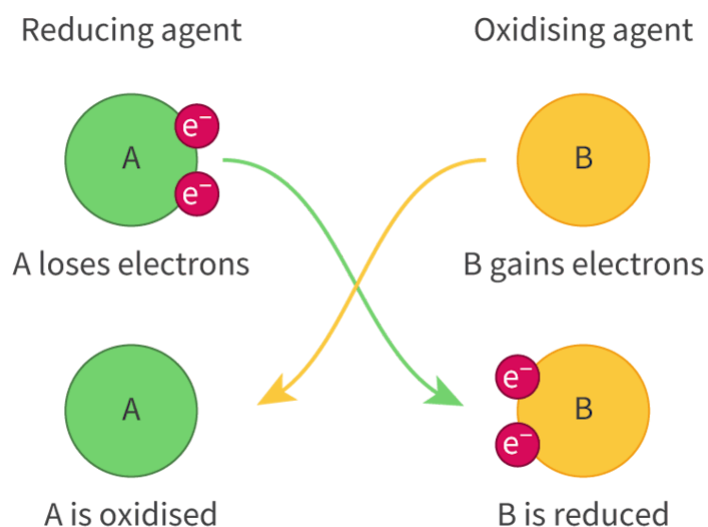


Figure 2. The different roles of oxidising and reducing agents in terms of electron transfer.

 More information for figure 2

The diagram illustrates the roles of oxidizing and reducing agents with electron transfer between two species labeled as A and B. On the left, a green circle labeled 'A' represents the reducing agent, which loses electrons (e^-) as indicated by two small red circles labeled with ' e^- '. The process is depicted by a green arrow pointing from A to the electrons. Below this, it states 'A is oxidised'.

On the right, a yellow circle labeled 'B' is shown as the oxidizing agent, which gains the electrons that A loses. The gained electrons are illustrated with red circles attached to B, and an orange arrow points towards it. Below this, the text reads 'B is reduced'.

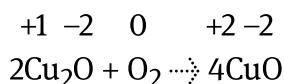
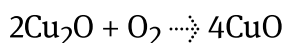
The diagram visually represents electron transfer where A loses and B gains, thereby showing A is oxidized and B is reduced.

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Let's apply this to the reactions below.

Worked example 5

Deduce the oxidation states of the atoms in the reaction below, when copper(I) oxide is heated in air or oxygen:

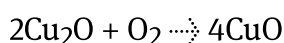


The copper ion attached to oxygen in the reactants has a charge of +1, given that there are two copper ions bonded to one oxygen ion with a charge of 2−.

The copper ion attached to oxygen in the products has a charge of +2, given that there is only one copper ion bonded to one oxygen ion with a charge of 2−.

Worked example 6

Based on the oxidation states found in the previous question, identify the oxidising and reducing agents when copper(I) oxide is heated in air or oxygen:



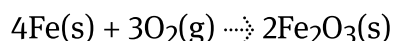
The oxidising agent in this reaction is oxygen.

The oxidising agent leads to the oxidation of the other species and becomes reduced (decreased in oxidation state).

The reducing agent in this reaction is copper.

The reducing agent leads to the reduction of the other species and becomes oxidised (increased in oxidation state).

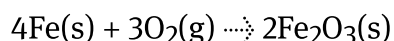
Identify the oxidising agent in the iron rusting reaction:



The oxidising agent in this reaction is the oxygen atom.

The oxidising agent leads to the oxidation of the other species and becomes reduced.

Identify the reducing agent in the iron rusting reaction:



The reducing agent in this reaction is the iron atom.

The reducing agent leads to the reduction of the other species and becomes oxidised.

In this reaction, iron increased in oxidation number from zero to +3, which means it was oxidised and, therefore, caused the reduction of oxygen (that decreases in oxidation number from zero to -2).

Writing redox equations

As described previously, if you place zinc metal, Zn(s) , in a solution of copper(II) sulfate, the zinc metal loses electrons, becoming zinc ions, $\text{Zn}^{2+}(\text{aq})$, while the copper ions, Cu^{2+} , gains electrons becoming copper metal, Cu(s) .

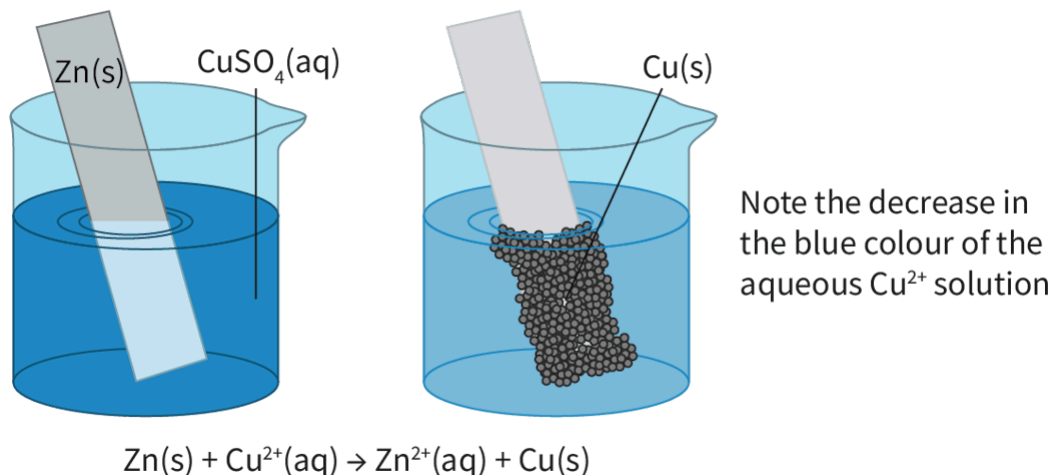


Figure 3. The reaction of zinc metal with copper(II) sulfate solution.

 More information for figure 3

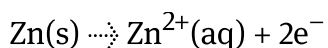
The image shows an illustration of a chemical reaction between zinc metal and copper(II) sulfate solution in two stages, represented by two beakers. The first beaker on the left contains a blue solution labeled " $\text{CuSO}_4(\text{aq})$," with a zinc strip labeled " Zn(s) " partially submerged. The second beaker on the right shows the same setup but with the zinc strip now covered in small black particles labeled " Cu(s) ," indicating copper metal formation. Below the beakers is the reaction equation: " $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$." Additional text notes a decrease in the blue color of the solution, which illustrates the reduction of copper ions to copper metal and the oxidation of zinc to zinc ions.

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How can we represent what happens in oxidation and reduction?

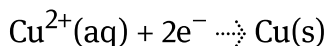
For any redox reaction, we can write separately what happens to each of the species in two half-reactions. Zinc metal, Zn(s) , lost electrons and became oxidised, which can be represented by:

Oxidation:



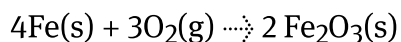
Note that each zinc atom loses two electrons (e^{-}) that are represented on the products side. Copper ions, Cu^{2+} , receive electrons and are reduced, becoming copper metal, Cu(s) .

Reduction:



Note that each copper ion gains two electrons (e^{-}) that are represented on the reactants side.

Now it is your turn to do the same for the iron rusting reaction:

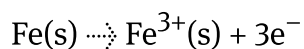


Start by identifying the two ions in iron(III) oxide and focus on describing what happens to each of them.

Worked example 7

State the oxidation half-reaction for the rusting of iron.

Oxidation:

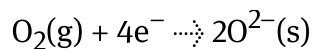


Each iron atom loses three electrons and forms an iron(III) ion.

Worked example 8

State the reduction half-reaction for the rusting of iron.

Reduction:

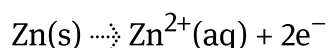


Each oxygen atom gains two electrons and forms an oxygen ion. Note that because there are two oxygen atoms in O_2 , four electrons are required.

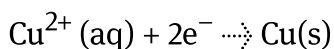
Balancing redox reactions

In the examples above, you might have noticed that the two half-equations for the reaction of zinc with copper(II) sulfate can be added up to get the overall equation:

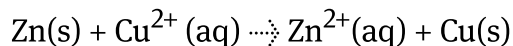
Oxidation:



Reduction:



Overall reaction:

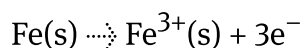


Recall that Hess's law involves adding the equations together and cancelling out terms that appear on both sides (see [section R1.2.2](#)

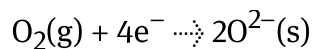
(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/hesss-law-id-46252/>). The same concept applies here because it all goes back to the law of conservation of mass/energy. Summation of equations, even half reaction equations, will equal the total in terms of mass and energy.

However, that is not always so simple – let's look at the rusting of iron. Consider the two half-reactions and compare them with the overall equation:

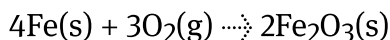
Oxidation:



Reduction:

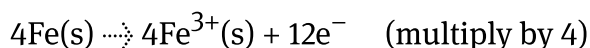


Overall reaction:

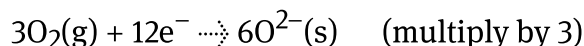


In the half-equations above, note that the number of electrons lost by one iron atom(three electrons) does not match the number of electrons received by the two oxygen atoms(four electrons total). Given the electrons gained by the oxygens need to be lost by iron atoms, you need to match the total number of electrons on both equations. To do that, you multiply each of the equations by a factor that will create the same number of total electrons. In this case:

Oxidation:

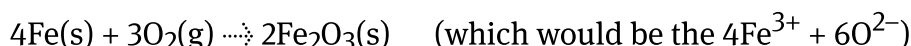


Reduction:



Now that the total number of electrons matches, you can cancel them out and add up all the reactants and products:

Overall reaction:

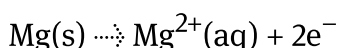


You have now learned how to balance redox reactions in neutral conditions. Try it again with the reaction between magnesium metal and hydrochloric acid.

Worked example 9

State the oxidation half-reaction for the reaction between magnesium metal and hydrochloric acid.

Oxidation:

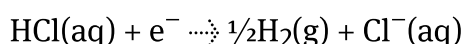


Each magnesium atom will lose two electrons and form a magnesium ion.

Worked example 10

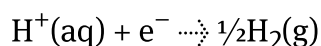
State the reduction half-reaction for the reaction between magnesium metal and hydrochloric acid.

Reduction:

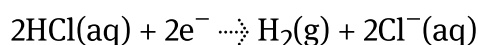


Each hydrogen atom will gain one electron and form half of a diatomic molecule of hydrogen gas.

Because the chloride ion is a spectator ion, you can leave it out of the equation:

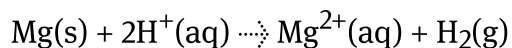


You can also write the equation without fractional quotients – both are correct:



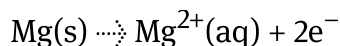
State the overall equation for the reaction between magnesium metal and hydrochloric acid.

Overall reaction:

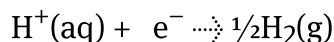


The half-equations are:

Oxidation:

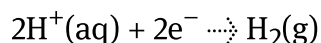


Reduction:



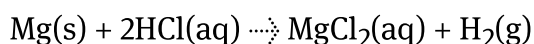
Because the number of electrons is not the same on both equations, you need to multiply the reduction equation by two:

Reduction:



Now you can cancel the electrons and add up the remaining equation.

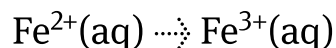
You can also include the spectator ion back, and write the products as $\text{Mg}^{2+}(\text{aq}) + 2\text{Cl}^{-}(\text{aq})$ or as $\text{MgCl}_2(\text{aq})$:



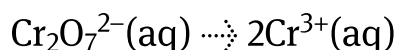
Not all reactions happen in neutral conditions. You will also learn to balance redox equations in acidic conditions. What is the overall equation for the oxidation of Fe^{2+} to Fe^{3+} by $\text{Cr}_2\text{O}_7^{2-}$ (giving Cr^{3+}) in an acid solution?

1. Start by writing the oxidation and reduction half-equations and balance all elements except H and O:

Oxidation:



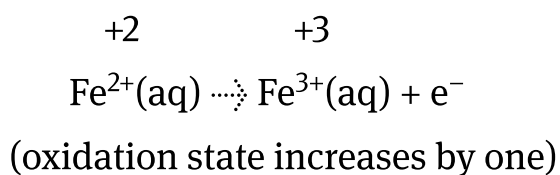
Reduction:



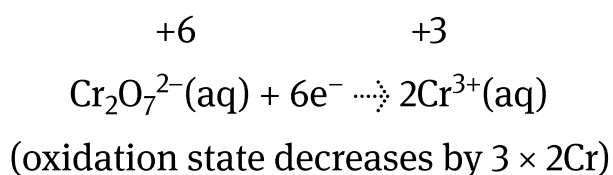
(note that there are two Cr that need to be balanced)

2. Assign oxidation states to add the total number of electrons lost/gained in each half-reaction:

Oxidation:

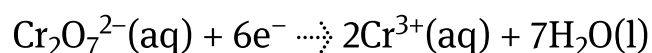


Reduction:



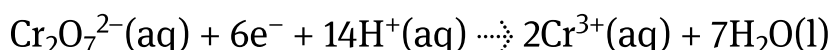
3. Balance any oxygen atoms by adding water:

Reduction:



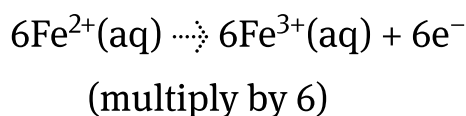
4. Balance any hydrogen atoms by adding hydrogen ions:

Reduction:

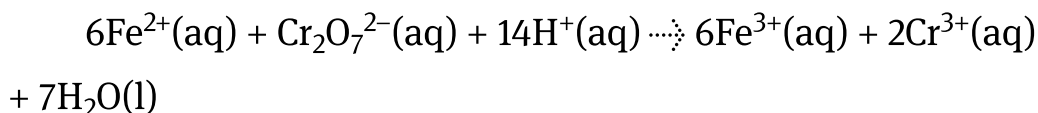


5. Multiply the equations as needed to match the number of electrons on both half-reactions:

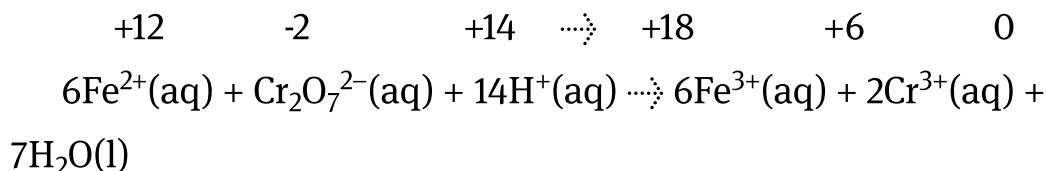
Oxidation:



6. Add both half-reactions:

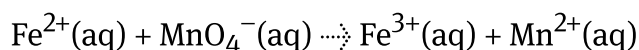


7. Check your work by counting all atoms on both sides and confirming the total oxidation state on both sides of the equation:

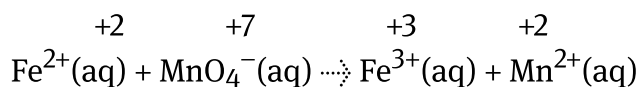


Worked example 11

Write the balanced equation for this unbalanced reaction:



1. Assign oxidation states to identify oxidised and reduced species:



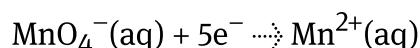
Iron(II) ions are oxidised and manganese is reduced.

2. Write the half-equations:

Oxidation:



Reduction:



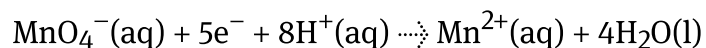
3. Balance any oxygen atoms by adding water:

Reduction:



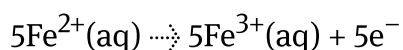
4. Balance any hydrogen atoms by adding hydrogen ions:

Reduction:

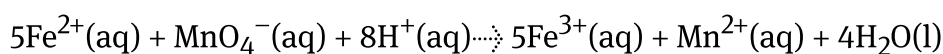


5. Because the number of electrons is not the same on both equations, you need to multiply the oxidation equation by five:

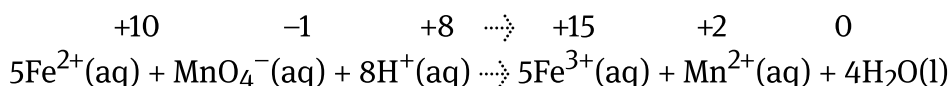
Oxidation:



6. Add both half-reactions:



7. Check your work by counting all atoms on both sides and confirming the total oxidation state on both sides of the equation:



Check your understanding with this question.



Exercise 1

Click a question to answer



1. Determine the correct equation for this reaction in acidic solution.



Check answers

Making connections

Why are some redox titrations described as 'self-indicating'?

Redox titrations, like acid-base reactions, can be studied through titration. Titration (see [section R3.1.8 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-for-strong-acid-base-neutralisation-reactions-id-46057/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-for-strong-acid-base-neutralisation-reactions-id-46057/)) is a laboratory technique that can be used for various purposes such as determining the concentration of a substance in a solution or the amount of a particular substance in a sample, or for measuring the purity of a compound. During a titration, a solution called an indicator is generally used to identify the endpoint of the reaction by a change of colour (see [section R3.1.4—5a \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)).

Some redox titrations are described as 'self-indicating' because the analyte itself acts as the indicator, without the need for an external indicator solution. In other words, the analyte undergoes a distinct and easily observable change in colour or some other physical property at the endpoint of the titration, indicating the completion of the reaction. One example is the bromine water test for alkenes (see [section R3.4.4—5 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-addition-reaction-id-46329/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-addition-reaction-id-46329/)).

Activity

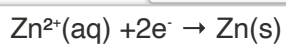
- **IB learner profile attribute:**
 - Open-minded
 - Thinker
- **Approaches to learning:**
 - Thinking skills — Applying key ideas and facts in new contexts
 - Communication skills — Using terminology, symbols, and communication conventions consistently and correctly
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual/pair activity

Now test your knowledge of oxidation and reduction by sorting the terms in **Interactive 3**.

Oxidation**Reduction**

Gain of electrons

Oxidation state decreases

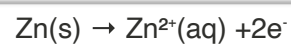


Oxygen added

Hydrogen added

Oxidation state increases

Loss of electrons



Oxygen removed

Hydrogen removed

 Check**Interactive 3. Sorting Game: Oxidation and Reduction.**

Ease of oxidation and reduction

Learning outcomes

By the end of this section you should be able to:

- (R3.2.3) Determine the relative ease of oxidation of different metals by analysing redox reactions between metals and aqueous metal ions.
- (R3.2.3) Predict the relative ease of oxidation of metals.
- (R3.2.3) Predict the relative ease of reduction of halogens.
- (R3.2.3) Interpret data regarding metal and metal ion reactions.
- (R3.2.4) Deduce equations for reactions of reactive metals with dilute HCl and H₂SO₄.

In the previous section, you saw a very rusty children's bicycle and you have learned that iron can undergo a redox reaction when in contact with air for a prolonged time. You may also know that nowadays most bicycle frames are made of aluminium. Why is that? Does aluminium react differently to iron when exposed to air?

You learned in the last section what happens when you place zinc metal in a solution of copper ions. But what would happen if you placed copper metal in a solution of zinc ions?

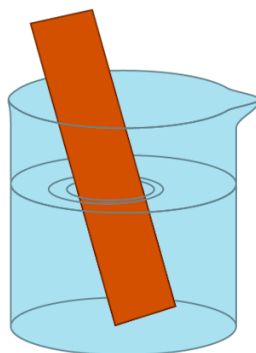


Figure 1. Copper metal in a solution of zinc ions.

No matter how long you leave the copper strip in the zinc ion solution, you will not see anything happen. Why?

Some reactions can proceed in both directions (see [subtopic R2.3 \(/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/\)](#)), allowing the system to reach an equilibrium. This does not happen in these

reactions as they are only evolving in one direction. Why is that?

Relative ease of oxidation of metals

When left in contact with the air, some metals undergo oxidation, which can lead to the formation of metal oxides or other compounds. You learned about the rusting of iron in the previous section and might also have encountered the corrosion of copper, where a greenish-blue colour develops, or the tarnishing of silver, when it loses its shiny appearance.



Credit: Philippe Gerber, Getty Images



Credit: Becart, Getty Images

Figure 2. A corroded copper pipe with its characteristic greenish-blue colour and tarnished silver jewellery, lacking its natural shine.

As seen in **Figure 1**, when a strip of copper metal is placed in a solution of zinc ions there are no visible changes in the solution. However, if you place zinc metal in a solution of copper ions a redox reaction will happen and you will see the blue colour of the solution fade and new copper form around the zinc (**Figure 3**). From these examples, we can deduce that some metals are more reactive than others.

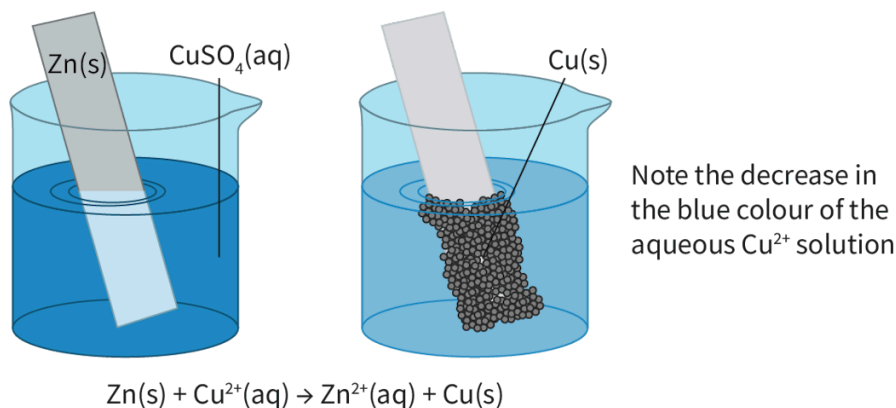
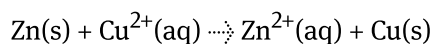


Figure 3. The reaction of zinc metal with copper(II) sulfate solution.

📖 More information for figure 3

The reactions discussed above are known as displacement reactions. In a displacement reaction the metal that is more reactive is able to displace, or replace, the metal ion in the solution.



All metals have a tendency to lose electrons and form positive ions (see [subtopic S2.1 \(/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/\)](#)). In the equation above, you can see that copper has already lost its electrons (it is a positive ion), but zinc still has them when in its metallic form. From the information provided, you can deduce that zinc has the ability to give its two valence electrons to the copper ion, which will be reduced to copper metal. This means that zinc is oxidised (loses its electrons) more easily than copper, making it a stronger reducing agent. You can also say that zinc is more reactive than copper.

Let's look at a few other reactions:

Table 1. Some displacement reactions.

Met als rea ctin g	Equ atio n	Qualitative observations
Zin c (Zn) wit h lea d (Pb)	$\begin{aligned} &\text{Zn(s)} \\ &+ \\ &\text{Pb}^{2+} \\ &+ \\ &\text{(aq)} \\ &\rightarrow \\ &\text{Zn}^{2+} \\ &\text{(aq)} \\ &+ \\ &\text{Pb(s)} \\ & \\ &\text{Pb(s)} \\ &+ \\ &\text{Zn}^{2+} \\ &\text{(aq)} \\ &\rightarrow \\ &\text{no} \\ &\text{reac} \\ &\text{tion} \end{aligned}$	Zinc Reacts with Lead (II) Nitrate to Produce Lead and Zinc Ni  Video 1. Zinc Reacts with Lead (II) Nitrate to Produce Lead and Zinc Nit 🕒 More information for vic
Lea d (Pb) wit h cop per (Cu)	$\begin{aligned} &\text{Pb(s)} \\ &+ \\ &\text{Cu}^{2+} \\ &+ \\ &\text{(aq)} \\ &\rightarrow \\ &\text{Pb}^{2+} \\ &+ \\ &\text{(aq)} \\ &+ \\ &\text{Cu(s)} \\ & \\ &\text{Cu(s)} \\ &+ \\ &\text{Pb}^{2+} \\ &+ \\ &\text{(aq)} \\ &\rightarrow \\ &\text{no} \\ &\text{reac} \\ &\text{tion} \end{aligned}$	Lead and Copper II Chloride  Video 2. Lead and Copper II Chloride 🕒 More information for vic

Metals reacting	Equation	Qualitative observations
Copper (Cu) with silver (Ag)	$\text{Cu(s)} + \text{Ag}^+(\text{aq}) \rightarrow \text{Cu}^+(\text{aq}) + \text{Ag(s)}$ $\text{Ag(s)} + \text{Cu}^+(\text{aq}) \rightarrow \text{no reaction}$	Chemistry Experiment - Copper and Silver Nitrate Redox React 

Based on the information provided in **Table 1**, order these metals by increasing reactivity.

Increasing Reactivity →

< < < <

Ease of oxidation →

Zn Ca Cu Ag Pb

✓ Check

Interactive 1. Reactivity Series of Elements.

🔗 More information for interactive 1

Study skills

The relative R of metals observed in metal/metal ion displacement reactions does not need to be learned; appropriate data will be supplied in examination questions.

You learned previously (see [subtopic S3.1 \(/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/\)\)](#) that the periodic table is a useful resource to help us predict some properties of elements. Ease of oxidation is one such property.

Worked example 1

Use the periodic table to determine whether sodium (Na) or potassium (K) shows a more vigorous reaction with water.

Moving down the group, the valence electrons are further away from the nucleus (higher energy level and more shielding), which makes them easier to remove, which leads to increased ease of oxidation. Generally, metals at the bottom of a group are more reactive than metals at the top of that same group.

Worked example 2

Use the periodic table to determine which metals are least reactive. Metals from Groups 1, 2 or 3?

The least reactive metals would be from Group 3 as alkali metals (Group 1) and alkaline earth metals (Group 2) are highly reactive metals that readily oxidise due to their low ionisation energies.

Moving down a group, metals have similar chemical properties and in general the ease of oxidation tends to increase.

Study skills

Lithium is an exception to the general trend of reactivity due to its density and ionisation energy (see [section S3.1.3a \(/study/app/preview-p/sid-426-cid-753301/book/trends-in-atomic-and-ionic-radius-id-43570/\)\)](#)).

Worked example 3

Use the periodic table to determine which metal is more reactive. Sodium (Na) or aluminium (Al).

Across a period, the valence electrons are positioned in the same energy level (same shielding), which makes them more attracted to the nucleus due to an increase in effective nuclear charge. These outer electrons are harder to remove, which leads to decreased reactivity and ease of oxidation.

Worked example 4

Use the periodic table to determine which metals are least reactive. Transition metals or metals from Groups 1 and 2?

Transition metals (Groups 3–12) are generally less reactive compared to alkali metals and alkaline earth metals due to their higher ionisation energies and lower reactivity with oxygen.

Metals in the same period may exhibit varying levels of ease of oxidation. In general, metals towards the left side of the periodic table tend to be more easily oxidised than metals towards the right side.

Making connections

The observations may vary depending on the specific metals, halogens, and halide ions involved, as well as the conditions of the reaction, such as concentration, temperature, and presence of catalysts.

Metals mixed with aqueous metal ions

When a more reactive metal is mixed with a solution containing ions of a less reactive metal, a redox reaction occurs, resulting in the more reactive metal displacing the ions of the less reactive metal. This can sometimes be observed as a colour change, such as the change in colour of aqueous copper(II) sulfate when solid zinc is added. This results in a colour change from blue to colourless. Some metals can react with metal ions to produce gas, which can be observed as effervescence or bubbling.

Halogens mixed with aqueous halide ions

If a more reactive halogen is mixed with a solution containing halide ions of a less reactive halogen, a redox reaction occurs, resulting in the more reactive halogen displacing the less reactive halogen. This can result in the change of colour of the solution, the appearance of a precipitate or gas formation. Chlorine solutions are usually a pale green, and bromine solutions are more orange in colour depending on the concentration. Iodine solutions are brown while fluorine solutions are a pale yellow to colourless depending on the concentration.

Relative ease of reduction of halogens

Halogens have a tendency to gain one electron to complete their octet (see [subtopic S2.1 \(/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/\)\)](#) (Table 2).

Table 2. Displacement reactions involving the halogens.

Reaction	Equations	Qualitative observations
Fluorine with chloride ions Chlorine with fluoride ions	$\text{F}_2(\text{aq}) + 2\text{Cl}^-(\text{aq}) \rightarrow 2\text{F}^-(\text{aq}) + \text{Cl}_2(\text{aq})$ $\text{Cl}_2(\text{aq}) + \text{F}^-(\text{aq}) \rightarrow \text{no reaction}$	The colourless solution turns into a pale yellow colour due to the formation of chlorine.
Chlorine with bromide ions Bromine with chloride ions	$\text{Cl}_2(\text{aq}) + 2\text{Br}^-(\text{aq}) \rightarrow 2\text{Cl}^-(\text{aq}) + \text{Br}_2(\text{aq})$ $\text{Br}_2(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{no reaction}$	Solution changes from pale yellow to orange/brown colour due to the formation of bromine. Chlorine reaction KBr 

Reaction	Equations	Qualitative observations
Bromine with iodide ions	$\text{Br}_2(\text{aq}) + 2\text{I}^-(\text{aq}) \rightarrow 2\text{Br}^-(\text{aq}) + \text{I}_2(\text{aq})$	Solution changes from orange/brown to da colour due to the formation of iodine.
Iodine with bromide ions	$\text{I}_2(\text{aq}) + \text{Br}^-(\text{aq}) \rightarrow \text{no reaction}$	Bromine water reacting with ioc 

Table 2. Displacement reactions involving the halogens.

Use the information provided in **Table 2** to order these metals by increasing reactivity.

The diagram consists of two horizontal arrows pointing to the right. The top arrow is labeled "Increasing Reactivity". Below it, three less-than signs (<) are spaced out. The bottom arrow is labeled "Ease of reduction". Below this arrow, four boxes containing the chemical symbols Cl, I, F, and Br are arranged from left to right. At the bottom left of the diagram area is a "Check" button with a checkmark icon.

Interactive 2. Reactivity Series of Halogens.

For the halogens, moving down the group, there is an increase in atomic radius due to valence electrons being placed further away from the nucleus. This reduces their pull to the nucleus, which also decreases the electrostatic force between the nucleus and the electrons. Hence the ease of reduction is very easy to follow in the periodic table; it decreases moving down the group.

Making connections

Why does metal reactivity increase, and non-metal reactivity decrease, moving down the main groups of the periodic table? (see [subtopic S3.1 \(/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/\)](#)).

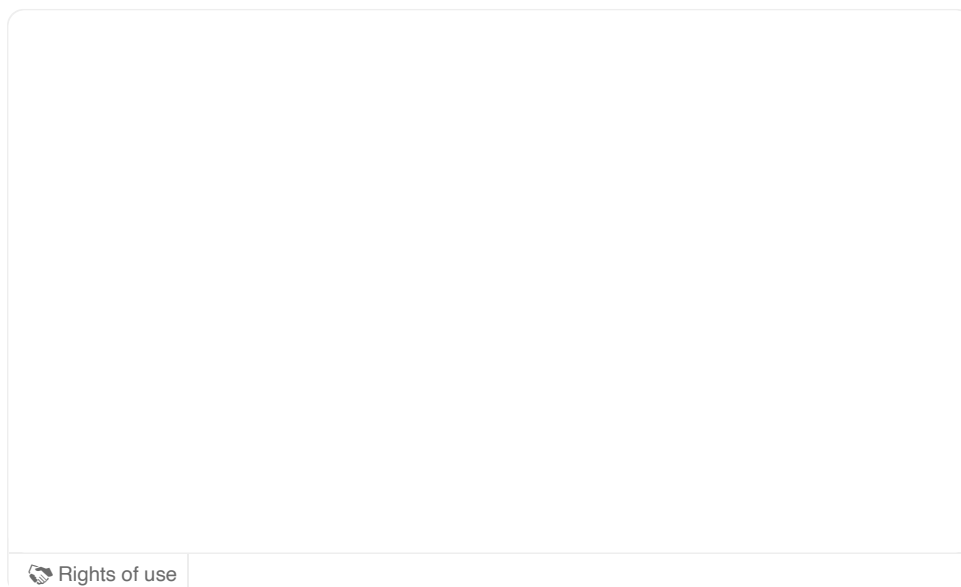
These two trends were discussed in the examples above.

Moving down a group, the valence electrons are placed at higher energy levels, which means there is more shielding between the nucleus and the outer electrons. This decreases the effective nuclear charge, which means the nucleus has less strength to attract the outer electrons. Because of this lower electrostatic force, less energy is required to remove the outer electrons from the elements at the bottom of the group.

For the metals, this means they are more easily oxidised — they can lose their outer electrons more easily. For the non-metals, this means they have less pull on the electrons that they receive to complete their octet — so they are less easily reduced.

Reaction of dilute acids with metals

Now let's look at a reaction between a metal, like magnesium with hydrochloric acid. What do you observe? What do you think is happening?

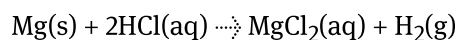


 Rights of use

Video 1. Magnesium Ribbon Placed in Hydrochloric Acid.

 More information for video 1

The equation for the reaction is:



The bubbles that you see in the clip above are the hydrogen gas produced by this displacement reaction. When acids react with certain metals, such as zinc, aluminium and iron, hydrogen gas is produced. In this reaction, the magnesium is oxidised (loses electrons) to form Mg^{2+} ions, and the hydrogen ions (H^+) in the acid are reduced (gain electrons) to form hydrogen gas (H_2). You can test the gas produced by using a lit splint or a flame; if hydrogen gas is present, it will combust and produce the characteristic pop sound.

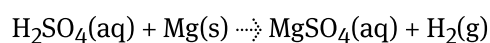
Test for hydrogen gas



Video 2. The test for hydrogen gas. Magnesium can also react with sulfuric acid.

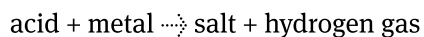
Worked example 5

State the balanced equation for the reaction between magnesium and sulfuric acid.



The hydrogen ion from the acid is displaced by the magnesium which loses two electrons to become Mg^{2+} and each hydrogen ion gains one electron to form hydrogen gas (H_2).

The general chemical equation for the reaction between hydrochloric or sulfuric acid and a reactive metal is:

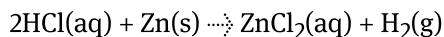


Some common examples of metals that react with acids are:

- Zinc (Zn)
- Iron (Fe)
- Magnesium (Mg)
- Aluminium (Al)
- Tin (Sn)
- Lead (Pb)
- Nickel (Ni)
- Copper (Cu)

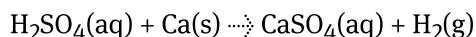
Worked example 6

State the balanced equation for the reaction between hydrochloric acid and zinc.



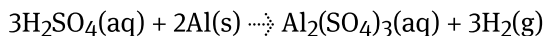
The hydrogen ion from the acid is displaced by the zinc which loses two electrons to become Zn^{2+} and each hydrogen ion gains one electron to form hydrogen gas (H_2).

State the balanced equation for the reaction between sulfuric acid and calcium.



The hydrogen ion from the acid is displaced by the calcium which loses two electrons to become Ca^{2+} and each hydrogen ion gains one electron to form hydrogen gas (H_2).

State the balanced equation for the reaction between sulfuric acid and aluminium.



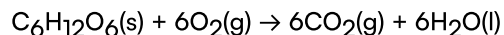
The hydrogen ion from the acid is displaced by the aluminium which loses three electrons to become Al^{3+} and each hydrogen ion gains one electron to form hydrogen gas (H_2).

Aspect: Global impact of science

In the search for renewable, non-pollutant, and sustainable sources of energy, fuel cells were created. You may recall that in fuel cells hydrogen is used as a power source (see [section R1.3.5 \(/study/app/preview-p/sid-426-cid-753301/book/fuel-cells-id-44783/\)](#)). Hydrogen needs to be produced industrially when acids react with metals, such as magnesium, zinc or iron. The use of these redox reactions has allowed the mass production of hydrogen gas, but not without challenges and risks associated with its production and use (see [section R3.2.6—7 \(/study/app/preview-p/sid-426-cid-753301/book/voltaic-cells-and-rechargeable-batteries-id-46490/\)](#)).

Redox reactions are also essential in ecosystems as they are involved in the cycling of various nutrients. For example, dead organic matter in the soil (or water) will be decomposed by microorganisms through a reaction called cellular

respiration:



This leads to the oxidation of the organic compounds that made up the organism and returns simpler inorganic and/or organic compounds to the ecosystem to be reused.

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 30 minutes
- **Activity type:** Pair activity

You have studied various displacement reactions in this section. Use this [simulation](https://teachchemistry.org/classroom-resources/metals-in-aqueous-solutions-simulation)  (<https://teachchemistry.org/classroom-resources/metals-in-aqueous-solutions-simulation>) to review and observe what happens when you place some metals in:

- a metal ion solution (activities one, two, three)
- hydrochloric acid (activity four).

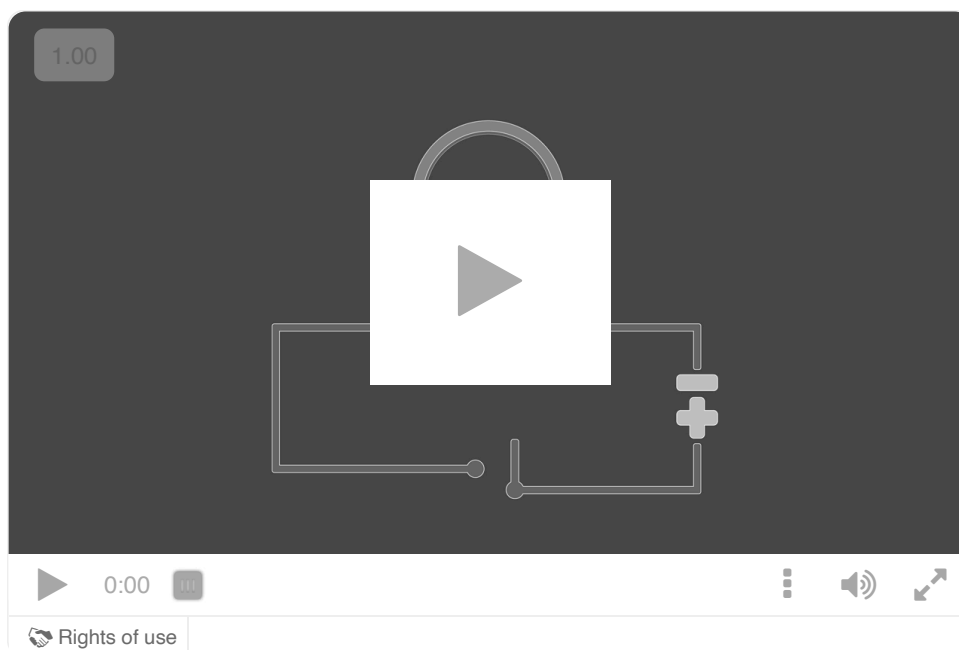
Create a table where you can write down your results and finish by ordering the metals by increased reactivity.

Electrodes in electrochemical cell

Learning outcomes

By the end of this section you should be able to label electrodes as anode and cathode and identify their polarities in voltaic and electrolytic cells, based on the type of reaction occurring at the electrode.

Batteries are used to power devices, like in the example of the circuit in **Video 1** where a battery is lighting a lamp.



Video 1. The Battery Supplies Energy to the Lamp.

 More information for video 1

A video illustrates a simple electrical circuit with a light bulb, a battery, and a switch. When the circuit is open, the bulb remains off. As the switch is closed, completing the circuit, the bulb lights up, indicating the flow of electricity. This continues in a loop as the video progresses.

An animation video illustrates a simple electrical circuit with a light bulb, a battery, and a switch. When the circuit is open, the bulb remains off. As the switch is closed, completing the circuit, the bulb lights up, indicating the flow of electricity. This continues in a loop as the animation progresses.

In **Video 1** the battery is represented by the positive and negative signs. Have you noticed when using a battery that there are different ends? Batteries need to be inserted in the correct orientation in order to work. Why do you think this is?

Batteries are used to supply electricity; the movement of electrons through a conducting material. What does **Video 1** tell you about the 'flow' of electrons? Electrons flow in one direction, like water flowing in a pipe. How might that relate to the orientation of a battery?

Electrochemical cells

Redox reactions happen in systems that are called electrochemical cells. The term 'cell' was used for the first time by an Italian physicist Alessandro Volta in 1800 when he invented the first electrochemical cell. He named his battery a 'voltaic pile'. Perhaps you can see why in **Figure 1**?

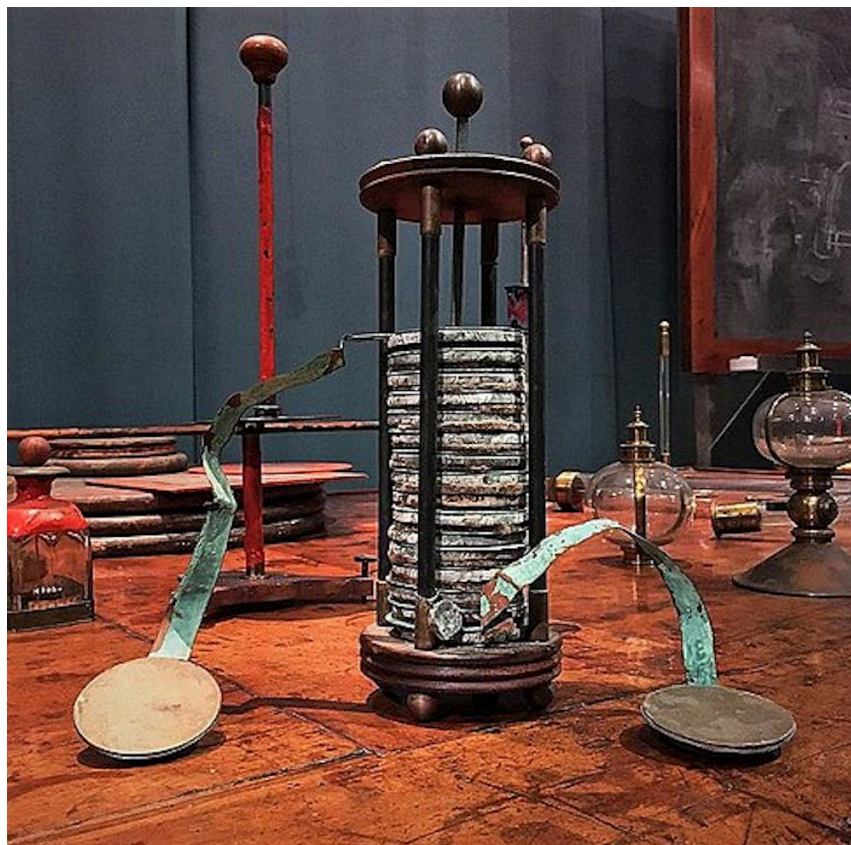


Figure 1. A voltaic pile, the first electrochemical cell.

Source: "Pila di volta

(https://commons.wikimedia.org/wiki/File:Pila_di_volta.jpg)" by Markkv is

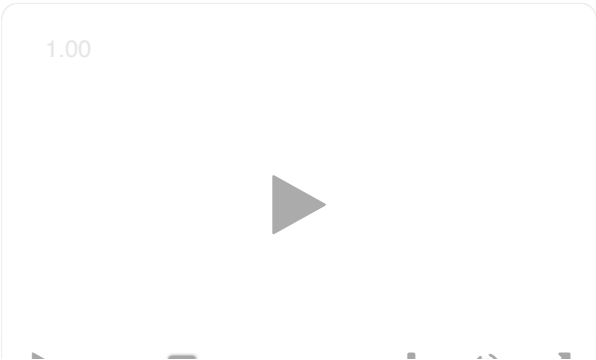
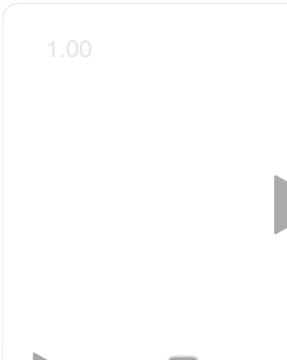
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Because there is a transfer of electrons due to chemical reactions happening inside cells, they are called electrochemical cells. For the cells to work, the redox reaction happening needs to be split into the two half-reactions, but they need to be connected by a conducting wire.

There are two types of electrochemical cells, and they are called voltaic cells and electrolytic cells. Both involve conversions between chemical and electrical energy but in opposite directions. Check **Table 1** for a comparison between the two types

of cells.

Table 1. A comparison between voltaic (also called galvanic cells) and electrolytic cells.

Voltaic cells	Electrolytic cells
Convert chemical energy to electrical energy	Convert electrical energy to chemical energy
Energy from spontaneous exothermic chemical reactions is converted to electrical energy	Energy is provided by an external source to drive a non-spontaneous chemical reaction
Energy is released, as you learned in <u>subtopic R2.1</u> (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43479/).  Interactive 1. Voltaic Cell.  More information for interactive 1 The animation video demonstrates the concept of voltaic cells using the movement of a ball. The ball initially moves along a small horizontal path and then rolls downhill, representing the release of energy. After that, the ball continues in a horizontal direction, forming a stepped-line pattern. This video highlights how a spontaneous reaction occurs without any external force and how a voltaic cell naturally converts chemical energy into electrical energy without external input.	Energy is required, as you learned in <u>subtopic R2.1</u> (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43479/).  Interactive 2. Electrolytic Cell.  More information for interactive 2 The animation video illustrates the concept of electrolytic cells using the movement of a ball. The ball initially moves in a small horizontal path and then moves uphill, indicating that an external force is required. After reaching the top, the ball continues in a horizontal direction, forming a stepped-line pattern. This video highlights how a non-spontaneous reaction requires an external energy source to proceed.

For either of the cells to work, two electrodes are required, each containing one of the half-reactions. Electrodes are conductors of electricity used to make contact with a non-metallic part of a circuit, such as a solution in a cell, called the electrolyte. Generally, the electrodes are metal, but they can also be other materials that conduct electricity, such as graphite.

In **Figure 2**, you can see a general diagram of a voltaic and an electrolytic cell for comparison.

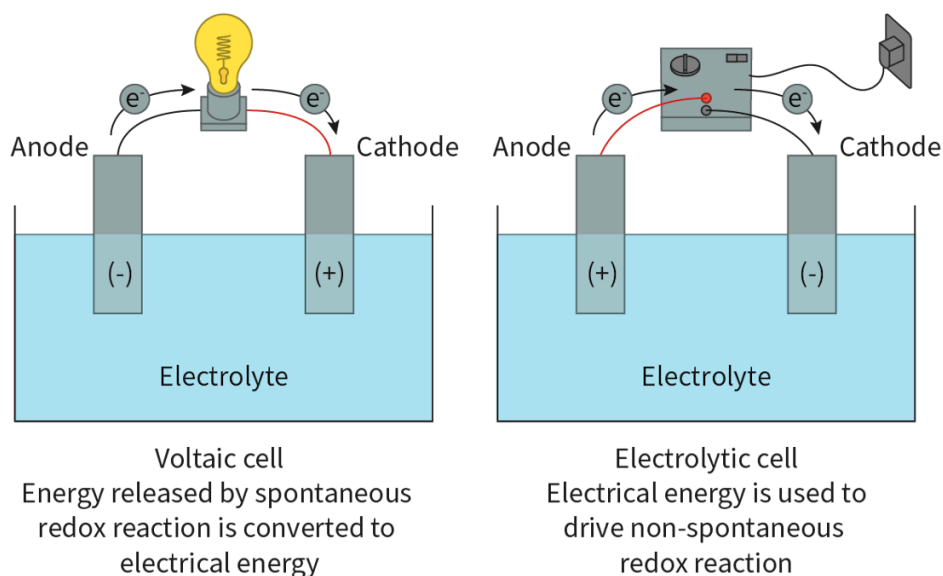


Figure 2. A voltaic and an electrolytic cell for comparison.

 More information for figure 2

The image is a diagram comparing a voltaic cell and an electrolytic cell. On the left side, the voltaic cell is shown with one anode and one cathode submerged in an electrolyte solution. The anode is marked with a negative sign (-), while the cathode is marked with a positive sign (+). Above the electrolyte, there is a wire connecting the two electrodes with a light bulb, indicating the flow of electrons from the anode to the cathode, generating electrical energy.

On the right side, the electrolytic cell is similarly structured with an anode and a cathode, marked with a positive sign (+) and a negative sign (-) respectively. Here, the flow of electrons is driven by an external power source displayed between the electrodes, emphasizing that electrical energy is used to drive a non-spontaneous redox reaction.

Text beneath the voltaic cell reads: "Voltaic cell: Energy released by spontaneous redox reaction is converted to electrical energy." Beneath the electrolytic cell, the text reads: "Electrolytic cell: Electrical energy is used to drive non-spontaneous redox reaction."

[Generated by AI]

As seen in **Figure 2**, electrochemical cells contains two electrodes:

- An anode: where oxidation takes place and electrons are lost by the species being oxidised.
- A cathode: where reduction occurs and electrons are gained by the species being reduced.

You can come up with your own mnemonic to remember this information, but here is one to get you started:

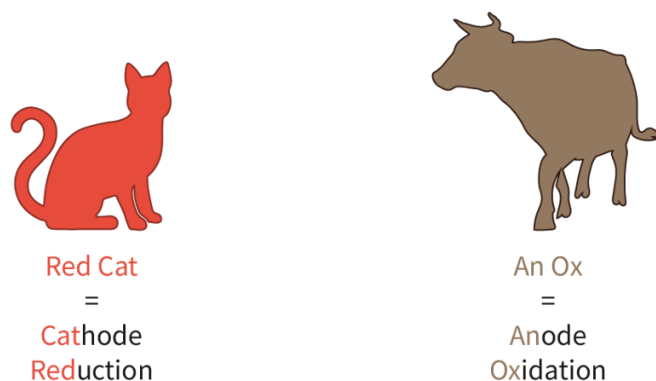


Figure 3. A mnemonic to remember what happens at each electrode.

You may have noticed in **Figure 2**, that depending on the electrochemical cell, the electrodes will have different polarities. This polarity refers to the relative charge of the two electrodes in the cell – where electrons are being released, there is a more negative charge (negative pole). In voltaic cells, the electrons released are dependent on the species undergoing oxidation. In electrolytic cells, electrons are initially provided by the battery and they enter the cell at the cathode. However, note that electrons always flow from the anode to the cathode in both types of cells. This is why you need to place batteries with the correct orientation in a device in order for it to work.

Table 2. A summary of the polarity of electrodes on electrochemical cells.

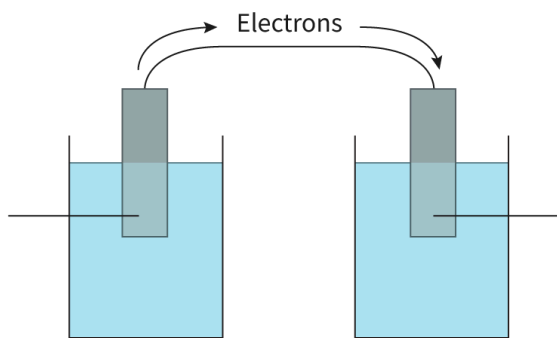
Electrode	Reaction happening	Voltaic cells	Electrolytic cell
Anode	Oxidation	Negative	Positive
Cathode	Reduction	Positive	Negative

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Communication skills — Using terminology, symbols
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual/pair activity

Use **Interactive 3** to correctly position the labels on the diagrams. Click on the arrow at the bottom right corner for the second part the interactive.

Voltaic cells – where energy is produced

☒ Check

Interactive 3. A Voltaic and an Electrolytic Cell for Comparison.

Voltaic cells and rechargeable batteries

Learning outcomes

By the end of this section you should be able to:

- (R3.2.6) Explain the direction of electron flow from anode to cathode in the external circuit, and ion movement across the salt bridge.
- (R3.2.7) Deduce the reactions of the charging process from given electrode reactions for discharge, and vice versa.

Batteries are, in essence, very simple devices. Look at the simple aluminium/air battery made of activated charcoal, aluminium foil and a salt solution in **Figure 1**.

Aluminum Air Battery - Exploratorium Science Snack



Video 1. An aluminium/air battery powering up a lamp.

Eventually batteries ‘die’ and need to be replaced – what do you think is happening in terms of the chemical reaction inside?

Creativity, activity, service

Strand: Service

Learning outcomes:

- Demonstrate engagement with issues of global significance
- Recognise and consider the ethics of choices and actions

The use of electric cars is likely to increase over the next decade as governments aim to phase out the production 

(<https://www.statista.com/chart/29359/official-targets-phase-out-sale-gasoline-cars/>) of automobiles that run on petrol (gasoline) or diesel. At first, this might seem to be an environmental benefit due to the reduction in carbon dioxide emissions. However, an increase in the production of the batteries required for electric cars means that not all countries will feel this benefit. The materials required for the production of these batteries are often found in developing countries and according to the UNCTAD, it is these countries that may end up paying the greatest human and environmental cost 
(<https://unctad.org/news/developing-countries-pay-environmental-cost-electric-car-batteries>) of switching over to electric cars.

As an IB student, what can you do to raise awareness of such issues? Should we consider all aspects of our behaviour as consumers even if the repercussions do not directly affect us?

Voltaic cells (primary batteries)

In a voltaic cell, electricity is produced as a redox reaction proceeds. This means at the anode, electrons will be lost, and they will be gained at the cathode. If the transfer of electrons is done through a wire connecting both electrodes, the electricity created can be harnessed. To allow electrons to move in a wire, the two half-reactions need to be separated into two half-cells.

In metal/metal-ion electrode batteries, each half-cell consists of a bar of metal dipped into a solution containing cations of the same metal, like the one in **Figure 1**.

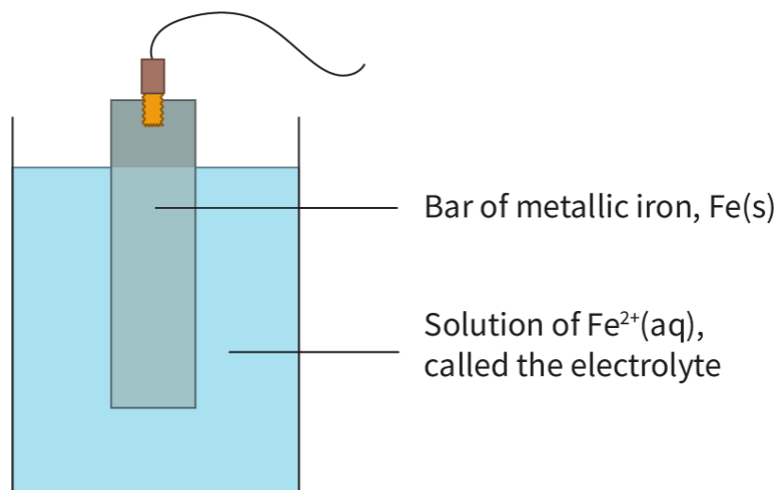


Figure 1. An example of a half-cell of Fe(s) with its metal ion in solution.

 More information for figure 1

The image is a diagram depicting a half-cell used in metal/metal-ion electrode batteries. In the diagram, there is a beaker filled with a light blue liquid, labeled as a 'Solution of Fe²⁺(aq), called the electrolyte.' Inside the beaker, a vertical bar labeled 'Bar of metallic iron, Fe(s)' is partially submerged in the solution. The bar is connected to a wire at the top, representing an electrode system. The diagram illustrates the setup of the iron in the electrolyte solution, which is typical for demonstrating electrochemical reactions in such half-cells.

[Generated by AI]

You see an electrode of iron, Fe(s), in the electrolyte, a solution of iron ions, Fe²⁺(aq). The half-cell can be represented as:



The straight line between the metal and metal ion represents the phase boundary or junction, the surface of the metal in contact with the solution. In the half-cell, the atoms of the metal will form ions by releasing electrons. So, the metal surface is negative compared to the solution. This charge separation is called the electrode potential. The electrode potential denotes an equilibrium between the metal and its ions in the electrolyte.

You can have an electrode of any metal and its metal ion. Practice with these two examples that represent half-cells in voltaic cells.

In each voltaic cell there are two half-cells connected through the wire attached to the metal bars. In that wire, you can find a lamp or a voltmeter represented but they are not required for the running of the battery. A lamp, or any other device, can be placed on the wire to harness the electricity being produced by the electron flow, while a voltmeter is a device that simply reads the electrical potential created by the movement of electrons from the zinc to the copper half-cells. Voltmeters are used to indicate if the battery is working and how much energy is being produced. They should give a positive value as energy is being produced. If the value is negative it means the wires were attached to the incorrect poles. To fix this, you just need to reverse the cables attached to the anode and cathode.

Once the half-cells are connected by the external wire, the reaction can proceed, but will immediately run into a problem. As electrons flow from the anode to the cathode, there will be a build-up of negative charge at the cathode and positive charge at the anode, which makes the cell stop. In order to prevent this from happening, another component is required; the salt bridge. The salt bridge will allow the continuous flow of electrons on the external wire because it provides a liquid junction between the two half-cells with free to move ions. A salt bridge is essential for the functioning of a battery because it:

- physically separates the cathode and anode as there is no mixing of the two solutions.
- provides an electrical path for the movement of the cations and anions in the cell.
- reduces the liquid-junction potential (voltage generated when two different solutions come into contact with each other, which occurs due to unequal cation and anion movement across the junction).

The cell in **Interactive 1** needs your help to function. Despite the fact that the electrodes and wire are in place, the electrons are not flowing in the wire. You need to balance the charges accumulating at each half-cell with the ions from the salt bridge. Drag the correct boxes to the spaces available on the voltaic cell to restore the production of electricity in the battery.

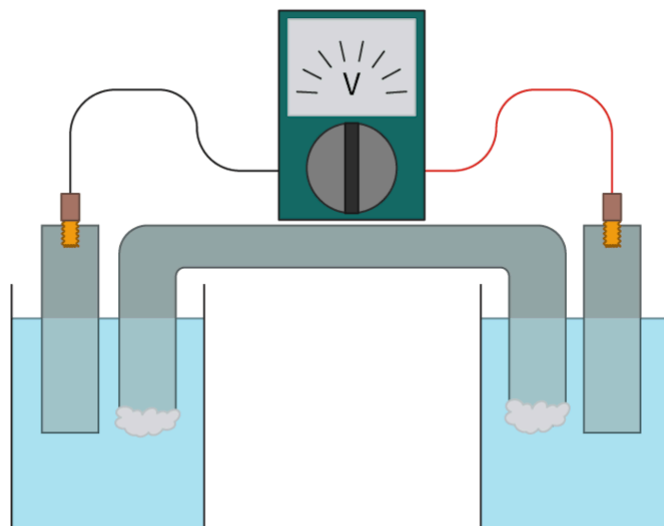
Interactive 1. A Voltaic Cell and the Movement of Ions on the Salt Bridge.

Interactive 1. A voltaic cell and the movement of ions on the salt bridge.

As you can see, the salt bridge contains a concentrated solution of a strong electrolyte which allows ions to diffuse out of it. The ions should be inert (not react with the other ions in the solution). Two common salt solutions in salt bridges are sodium sulfate (Na_2SO_4) and potassium chloride (KCl). At the openings of a salt bridge, there is generally cotton wool that prevents the solution from leaving the tube but still allows the movements of ions in and out.

In a salt bridge, the ions will move to the half-cell that is accumulating an opposite charge. In the example in **Figure 2** the positive ion in the salt bridge (K^+) is moving to the cathode as there has been an accumulation of negative charge (electrons arriving from the anode). The anion, Cl^- , moves to the anode as there has been an accumulation of the iron ions, Fe^{2+} .

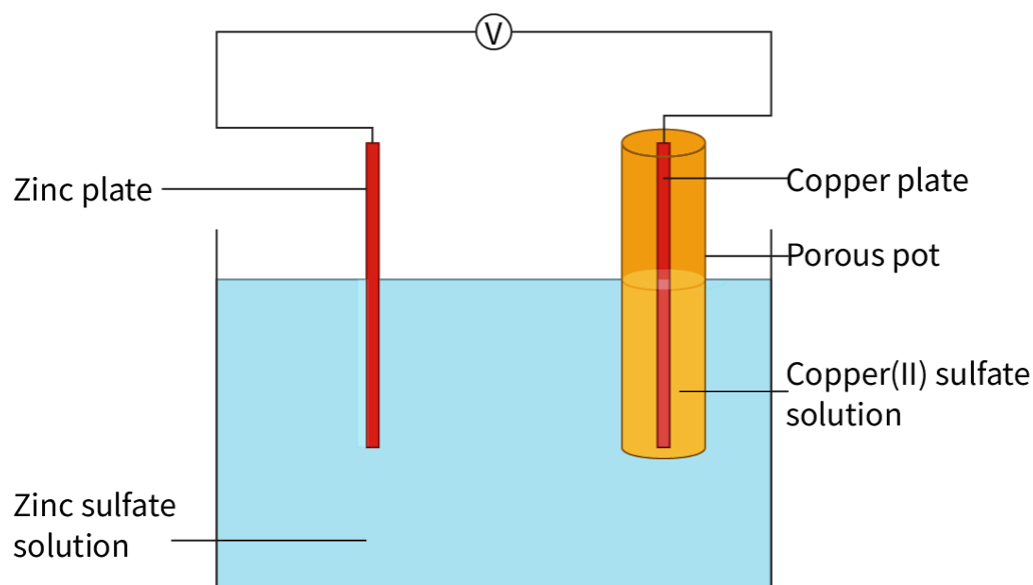
In a school setting, sometimes a porous cup or paper soaked in a salt solution is used as a simplified version of a salt bridge as in **Figure 2**.



 More information

The image shows a diagram of an electrochemical cell setup. It includes two containers filled with a solution connected by a salt bridge. Each container has an electrode submerged in the solution. The electrodes are connected by wires to a voltmeter placed on top of the setup. The two half-cells are connected by a salt bridge.

[Generated by AI]



 More information

The image is a diagram of a galvanic cell, depicting a simple electrochemical setup. On the left side, there is a zinc plate labeled "Zinc plate," immersed in a "Zinc sulfate solution." On the right side, there is a "Copper plate" placed inside a "Porous pot," which is immersed in a "Copper(II) sulfate solution." The zinc and copper plates are connected by a wire with a voltmeter symbolized by a circle marked with a "V." The wire indicates a flow of electrons from the zinc electrode to the copper electrode. The diagram illustrates the different components of the galvanic cell and their interactions, showing how chemical energy is converted into electrical energy.

[Generated by AI]

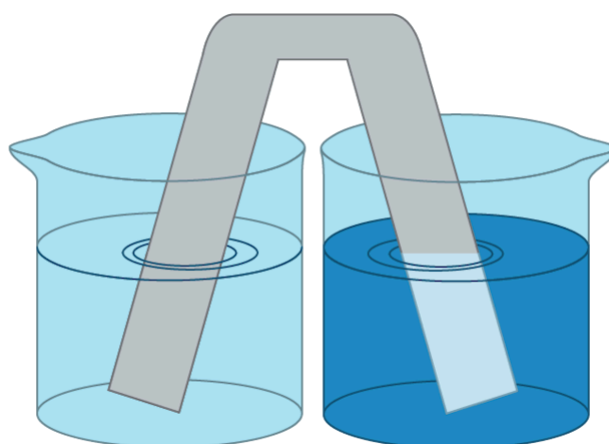


Figure 2. Examples of salt bridges used in a school laboratory setting.

 More information for figure 2

The image shows an illustration of a laboratory setup involving two beakers, each partially filled with liquid. The beaker on the left contains a light blue liquid, while the beaker on the right holds a darker blue liquid. A gray, inverted U-shaped tube, representing a salt bridge, connects the two beakers. This setup is commonly used to demonstrate electrochemical cells where ions can pass through the salt bridge to complete the electrical circuit.

[Generated by AI]

In 1836, a British chemist and meteorologist called John Frederic Daniell invented a type of electrochemical cell that became known as the Daniell cell. In it, he improved the voltaic pile created by Alessandro Volta by separating the half-cells of copper and zinc by a porous ceramic cup, known as a 'semi-permeable diaphragm.' The electrodes used in the Daniell cell can be represented as:



Figure 3. A Daniell cell recreated in a school laboratory.

Source: "Daniell cell demonstration"

(https://commons.wikimedia.org/wiki/File:Daniell_cell_demonstration.jpg)”

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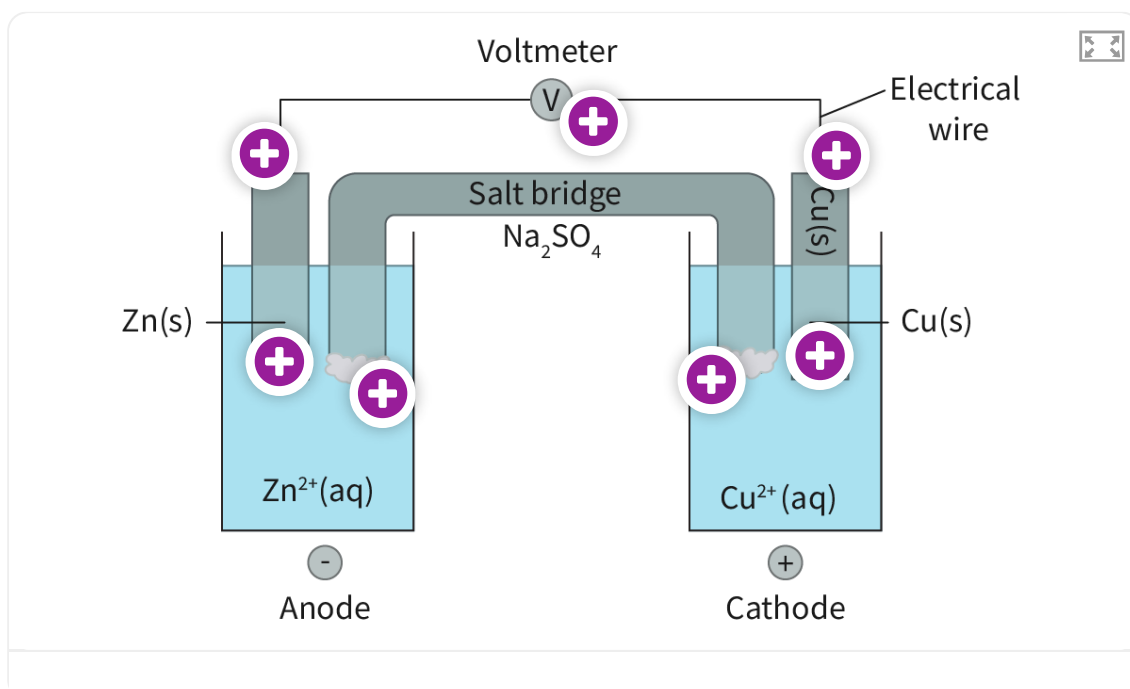
 More information for figure 3

The image shows a recreated Daniell cell in a laboratory setting. There are two clear beakers. The left beaker is labeled "ZnSO₄" and contains a clear liquid with a metallic strip connected to a wire. The right beaker is labeled "CuSO₄" and is filled with a blue liquid, also with a metallic strip connected to a wire. These wires lead to a digital multimeter on the left, displaying a reading of 1.07 volts. The setup demonstrates the basic principles of electrochemistry with zinc and copper electrodes in their respective sulfate solutions, connected to measure voltage.

[Generated by AI]

In the Daniell cell in **Figure 3**, zinc is more easily oxidised than copper and undergoes oxidation. The equilibrium between metal and metal ions on the copper electrode is more towards the metal (less tendency to lose electrons) so copper has a higher electrode potential than zinc. This means that the zinc atoms are more likely to release electrons; the zinc half-cell is more negative. The copper ions are more likely to accept electrons; the copper half-cell is more positive. This results in the flow of electrons from the zinc to the copper half-cell.

Use the hotspots in **Interactive 2** to learn more about what is happening at each electrode, the electrical wire, and the salt bridge.



Interactive 2. What Happens Inside a Daniell Cell?

 More information for interactive 2

An interactive displays a diagram of a galvanic cell with two half-cells connected by a salt bridge containing Na_2SO_4 . The left half-cell contains a zinc electrode labeled Zn solid immersed in an aqueous solution of Zn^{2+} . The right half-cell contains a copper electrode labeled Cu solid immersed in an aqueous solution of Cu^{2+} . The zinc electrode is labeled anode with a negative sign, and the copper electrode is labeled cathode with a positive sign. A voltmeter is connected between the two electrodes through an electrical wire. White clouds are near the salt bridge openings in both half-cells.

There are 7 Hotspots represented by plus signs at various places. Hotspot 1 above the zinc electrode. Hotspot 2 near the voltmeter. Hotspot 3 above the copper electrode. Hotspot 4 on the zinc electrode. Hotspot 5 on the aqueous solution of zinc. Hotspot 5 on the aqueous solution of copper. Hotspot 7 on the copper electrode. Clicking on the hotspots reveals what is happening in that place.

The following items are revealed at respective hotspots:

Hotspot 1: The electrons lost by zinc are carried by the electrical wire in the direction of the cathode.

Hotspot 2: The voltmeter is the device that reads the electrical potential created by the movement of electrons from the zinc to the copper half-cells.

Hotspot 3: The electrons lost by zinc arrive at the cathode and will be received by the copper ions in the solution.

Hotspot 4: In the anode, you will see the zinc metal breaking apart as the zinc atoms will be losing electrons to the external wire. The zinc ions are added to the electrolyte in the zinc half-cell. The zinc bar gets thinner.

Hotspot 5: Due to the increase of positive charges in the zinc half-cell by the addition of zinc ions, the negative ions in the salt bridge (SO_4^{2-}) move to the solution to balance out the charges and allow the reaction to continue.

Hotspot 6: Due to the decrease of positive charges in the copper half-cell by the removal of copper ions, the positive ions in the salt bridge (Na^+) move to the solution to balance out the charges and allow the reaction to continue.

Hotspot 7: The copper ions approach the metal and receive the electrons that arrived on the electrical wire. Copper metal is created, and the copper bar increases. The blue colour of the copper ion solution fades due to the removal of the copper ions.

Nature of Science

Aspect: Experiments

Alessandro Volta, an Italian scientist, is credited with the invention of the first true voltaic cell, known as the 'voltaic pile,' in 1800. Volta conducted a series of experiments to investigate the effects of different metals and electrolytes on the generation of electricity. He stacked alternating layers of zinc and copper discs separated by pieces of cardboard soaked in salt water, which served as the electrolyte. However, the voltaic pile had limitations such as low voltage output, short lifespan, and the production of corrosive chemicals.

John Daniell, a contemporary of Volta, sought to improve upon these limitations. The Daniell cell used a porous ceramic cup, known as a 'semipermeable diaphragm,' to separate the copper sulfate solution from the zinc sulfate solution. The Daniell cell had several advantages: it had a more stable and longer-lasting output of electrical energy, a higher voltage output and produced fewer corrosive chemicals.

Over the years, extensive experimentation has been conducted in an attempt to develop more modern and efficient voltaic cells. For example, researchers have experimented with different electrode materials, electrolyte compositions, and cell configurations to optimise cell performance, durability, and safety. These experiments have led to the development of various types of voltaic cells, which are used in everyday devices, such as batteries, flashlights, and calculators and also more modern designs like fuel cells (see [section R1.3.5](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fuel-cells-id-44783/>)) and solar cells that are used in renewable energy technologies today. Despite the benefits of producing cleaner renewable energy and increased efficiency, both fuel and solar cells are not only expensive solutions but have other drawbacks.

All these experiments show the tentative and dynamic nature of scientific knowledge, as new discoveries and technologies can challenge or modify existing theories and models and help solve new and old problems. They also reflect the influence of financial, social, and cultural factors on scientific practice and knowledge production.

A voltaic cell can be represented using a cell diagram as in **Figure 4**. The convention is to draw the anode on the left and the cathode on the right. Note that the phase boundary between a solid and an aqueous solution is represented by a single vertical line and the salt bridge is represented by the two parallel vertical lines between the anode and the cathode.

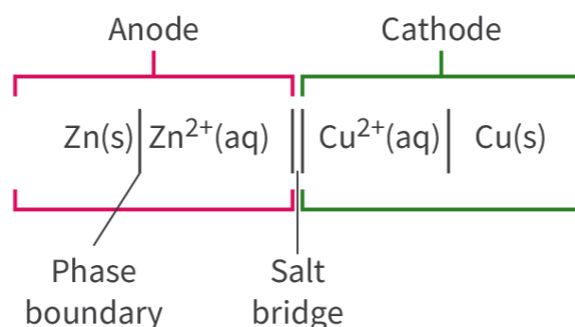


Figure 4. The cell diagram convention for a voltaic cell.
 More information for figure 4

This diagram represents a voltaic cell with an anode on the left and a cathode on the right. The anode side features zinc in solid form (Zn(s)) transitioning to zinc ions in aqueous solution (Zn²⁺(aq)), indicated by a single vertical line representing the phase boundary. The cathode portion shows copper ions in aqueous solution (Cu²⁺(aq)) transitioning to copper in solid form (Cu(s)), separated by a phase boundary. Between these two sections is a salt bridge, symbolized by two parallel vertical lines. Labels indicate 'Anode' above the left section and 'Cathode' above the right section, with 'Phase boundary' and 'Salt bridge' labeled under their respective lines.

[Generated by AI]

In **Interactive 1**, write the cell diagram for a voltaic cell with electrodes of nickel and lead by dragging the boxes to the correct locations below.



Pb(s)

Ni(s)

Pb²⁺(aq)

Ni²⁺(aq)

|

||

☒ Check

Interactive 3. A voltaic cell with electrodes of nickel and lead.

Study skills

Construction of primary cells should include: half-cells containing metal/metal ion, anode, cathode, electric circuit, and salt bridge, as in **Figure 5**.

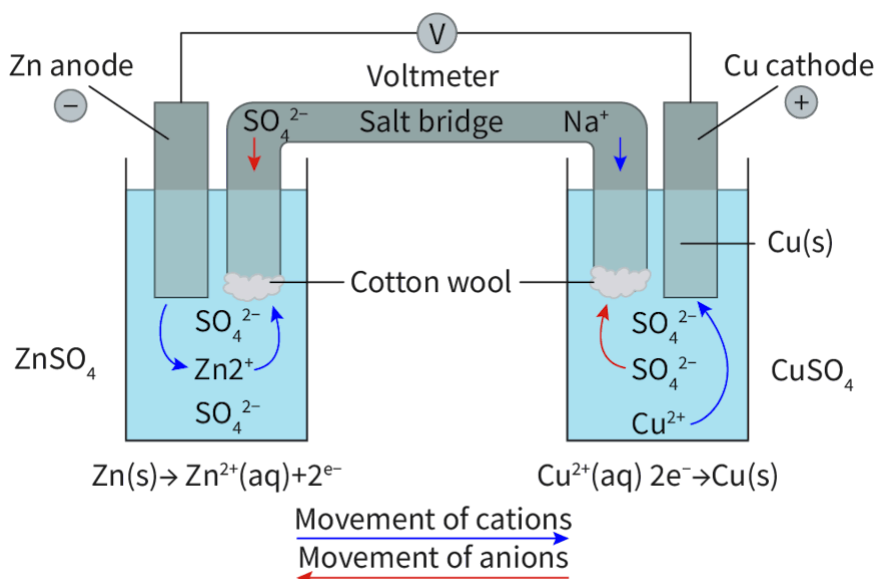


Figure 5. A representation of a voltaic cell.

 More information for figure 5

This diagram illustrates a voltaic cell consisting of two half-cells connected by a salt bridge. On the left, there is a zinc (Zn) anode submerged in a ZnSO_4 solution. The zinc undergoes oxidation, releasing electrons and forming Zn^{2+} ions. The movement of cations is indicated by a blue arrow heading towards the salt bridge. Below the anode, the reaction is represented as $\text{Zn(s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^{-}$.

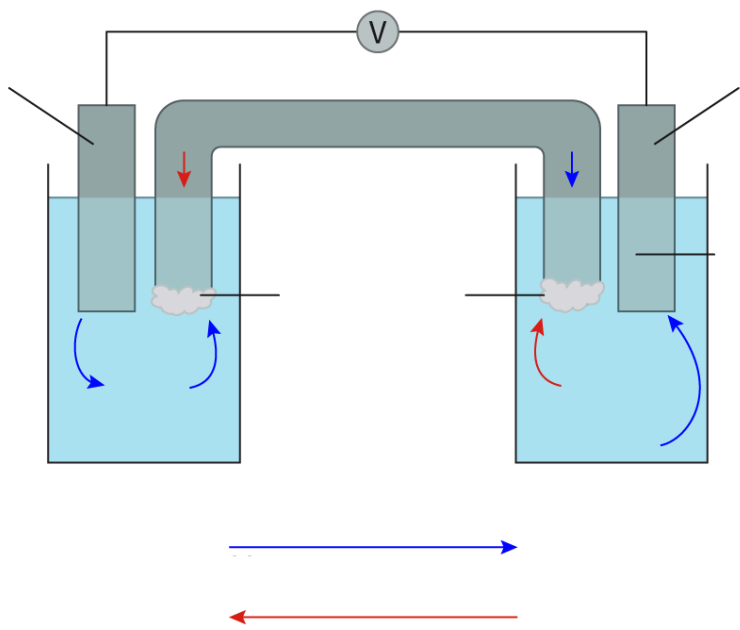
On the right, a copper (Cu) cathode is submerged in a CuSO_4 solution. Here, Cu^{2+} ions gain electrons to form copper metal. The movement of anions, indicated by red arrows, also heads towards the salt bridge. The cathodic reaction is depicted as $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Cu(s)}$.

A salt bridge, filled with Na^{+} and SO_4^{2-} ions, allows for the movement of ions between the two half-cells, maintaining electrical neutrality. Cotton wool appears to be used as a medium within the salt bridge. A voltmeter is connected between the anode and cathode to measure the potential difference.

Overall, the diagram depicts the flow of electrons from the anode to the cathode through an external circuit, while ions travel through the solutions and salt bridge to balance the charge.

[Generated by AI]

Use **Interactive 4** to practice labelling a voltaic cell.



Movement of cations	Ni anode	Cotton wool	Na ⁺	Ni ²⁺	+
Movement of anions	$\text{Pb}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Pb}(\text{s})$	Pb cathode	Pb ²⁺	PbSO ₄	-
Salt bridge	$\text{Ni}(\text{s}) \rightarrow \text{Ni}^{2+}(\text{aq}) + 2\text{e}^{-}$	Voltmeter	SO ₄ ²⁻	NiSO ₄	Pb(s)

✓ Check

Interactive 4. Label the voltaic cell.

Rechargeable batteries (secondary batteries)

Unlike the single use batteries studied in the previous section, cell phones, computers and cars use rechargeable batteries. Many electric or electronic devices can be powered by rechargeable batteries.

Video 2. Many Devices Use Rechargeable Batteries.

 More information for video 2

A video shows a close-up of a battery charger holding four cylindrical rechargeable batteries. The screen on the charger is illuminated with a bright blue backlight, displaying four battery icons. Each icon represents a battery's charging status, with three of them showing low charge levels and one showing a slightly higher charge, indicated by partially filled bars. The screen also displays the text CHG at the top left, signifying that the batteries are currently in the process of charging. The background is slightly blurred, focusing attention on the charging display.

How do these batteries compare to the ones you already studied?

How do rechargeable batteries work? Can you guess how redox reactions might occur in rechargeable batteries?

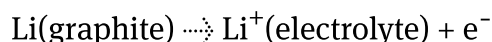
Rechargeable batteries use chemical reactions to provide energy to power devices, just like primary cells, but can be used more than once, if they are recharged. To recharge, the chemical reaction is reversed by the use of an electrical current to drive the reaction back to its original state. The current used needs to be carefully controlled to prevent damage to the battery. That is why generally there are specific chargers for specific batteries, as they monitor the current and temperature of the battery to extend battery life. Battery life is a measure of the

time that the battery will continue to provide energy. This means that even with proper care and maintenance, rechargeable batteries will eventually need to be replaced.

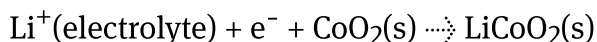
The most common types of rechargeable batteries for daily use include nickel-metal hydride (NiMH), typical batteries like AA or AAA, and lithium-ion (Li-ion) batteries like the ones used in cell phones and laptops.

Let's look at what happens inside a lithium-ion battery made up of lithium cobalt oxide (LiCoO₂). In this type of cell, the cathode is made of a lithium metal oxide such as lithium cobalt oxide (LiCoO₂) and the anode is composed of lithium atoms absorbed onto a lattice of graphite. The electrolyte is a non-aqueous solution containing lithium ions, which flow between the cathode and anode during charging and discharging.

When being discharged, the following half equations take place. At the anode, the lithium atoms undergo oxidation to form lithium ions.



The electrons flow in the external circuit from anode to cathode. The lithium ions flow through the electrolyte to the cathode where they undergo reduction.



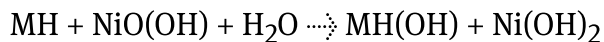
Note that when being recharged, the two half equations above are reversed. During charging, lithium ions flow from the cathode to the anode through the electrolyte. This process stores energy in the battery. During discharging, the lithium ions flow back from the anode to the cathode through the electrolyte, generating an electric current that can be used to power devices.

Note that these reactions are simplified representations of the actual reactions that occur in rechargeable batteries. The actual reactions involve multiple steps and intermediate compounds.

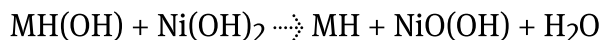
Learn more about other types of rechargeable batteries and predict their discharge or charge reactions with the examples below.

Worked example 1

A nickel-metal hydride (NiMH) battery is a very common AA or AAA battery. This is the reaction that happens during discharge:



State the charging reaction.



To write the reverse equation, simply swap the reactants and products.

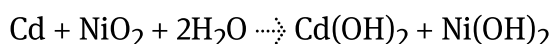
The reaction that occurs during discharge is the oxidation of the metal hydride and the reduction of nickel oxide hydroxide.

During charging, the reaction is reversed, and nickel hydroxide is oxidised while the metal hydride is reduced.

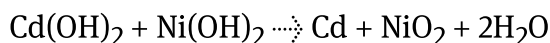
Worked example 2

Nickel-cadmium (NiCd) batteries were commonly used in cordless power tools such as drills, saws, and screwdrivers, but have mostly now been replaced.

The reaction that happens during discharge is:



State the charging reaction.



To write the reverse equation, simply swap the reactants and products.

The reaction that occurs during discharge is the oxidation of the cadmium and the reduction of nickel oxide.

During charging, the reaction is reversed, and nickel hydroxide is oxidised while the cadmium hydroxide is reduced.

Making connections

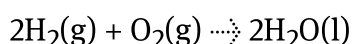
Secondary cells rely on electrode reactions that are reversible. What are the common features of these reactions? (see [subtopic R2.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>)).

Reversible reactions allow the combination of the products to regenerate the original reactants. In batteries, this allows the creation of stored potential in a redox cell that will be released and harnessed to power a device. During discharge (when the battery is supplying electrical energy), the anode (negative electrode) undergoes oxidation (loses electrons) while the cathode (positive electrode) undergoes reduction (gains electrons). During charging (when the battery is being recharged), the electrodes are reversed to allow the restoration of the original components of the cell.

Fuel cells

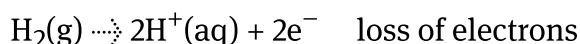
Fuel cells (see [section R1.3.5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fuel-cells-id-44783/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fuel-cells-id-44783/>)) generate electrical energy directly from a redox reaction. The fuel, such as hydrogen or methanol, is oxidised at the anode, generating electrons and cations.

For the hydrogen fuel cells in acidic conditions, the net reaction is:

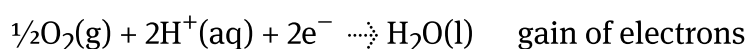


and can be divided into two half-reactions.

Oxidation:



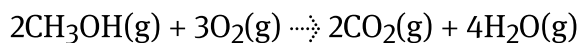
Reduction:



The electrons pass through an external circuit, producing an electrical current, while the cations move through the electrolyte to the cathode. At the cathode, the electrons combine with oxygen, or another oxidising agent, to form water or

another byproduct. The continuous flow of electrons through the external wire generates a steady supply of electrical energy.

For the methanol fuel cells, the combustion of methanol in the presence of oxygen gas produces carbon dioxide and water vapour, according to the equation:

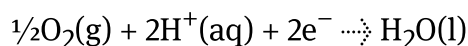


A methanol fuel cell works with the same chemical reaction but separates the process into two half-reactions.

Oxidation:



Reduction:



Nature of Science

Aspect: Science as a shared endeavour

The development of hydrogen fuel cells may be the solution to the use of fossil fuels in motor vehicles. How can scientists work together to overcome global problems such as these

Advantages and disadvantages of the different cells

It is now time to consider the advantages and disadvantages of the different types of cells you have studied. These are summarised in **Table 1**.

Table 1. Advantages and disadvantages of the different typ

	Primary cells	Secondary cells	
Distinctive feature	Single-use battery Irreversible reaction	Rechargeable battery Reversible reaction	I
Advantages	Longer shelf life Lower cost Greater reliability	Higher energy density Longer lifespan Lower environmental impact	I
Disadvantages	Lower energy densities Generate waste when discarded Environmental impact	Lead-acid batteries — heavy and require regular maintenance Nickel-cadmium (Ni-Cd) batteries — contain toxic cadmium, harmful to the environment Nickel-metal hydride (Ni-MH) batteries — less durable and have a shorter lifespan Lithium-ion (Li-ion) batteries — can be expensive and require careful handling to avoid safety risks	

	Primary cells	Secondary cells	
Uses	Medical devices Emergency equipment Remote sensors	Lead-acid batteries — commonly used in automobiles Nickel-cadmium (Ni-Cd) batteries — from portable electronics to aircraft Nickel-metal hydride (Ni-MH) batteries — consumer electronics and hybrid electric vehicles Lithium-ion (Li-ion) batteries — portable electronics, electric vehicles, and renewable energy systems	1

Use **Interactive 5** to identify some advantages of different cells.

Use	Desirable property	Type of cell
	Reliability	Primary cells
Electric vehicles	High energy density	
	Continuous reaction	Fuel cells

Back-up power generation

Secondary cells

Fire alarm

☒ Check

Interactive 5. Advantages of different cells.

Making connections

Electrical energy can be derived from the combustion of fossil fuels or from electrochemical reactions. What are the similarities and differences in these reactions? (see [subtopic R1.3 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43475/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43475/))

Similarities

Electrical energy is an essential part of modern life. To produce it, we use fossil fuels, such as coal, natural gas, oil and electrochemical reactions, such as hydrogen fuel cells and lithium-ion batteries. Both the combustion of fossil fuels and electrochemical reactions convert chemical energy into electrical energy. To harness or use the electricity produced, both methods require the use of an external circuit to allow for the flow of electrons, which generates an electrical current.

Differences

Fossil fuel combustion is a non-renewable process, while some electrochemical reactions in batteries are a renewable process, as they can be recharged and reused.

Electrochemical reactions in batteries are often used for smaller-scale applications, such as portable electronics and electric vehicles, while fossil fuel combustion typically involves a large-scale power generation process.

The combustion of fossil fuels releases harmful emissions into the environment, including carbon dioxide, sulfur dioxide, and nitrogen oxides, which contribute to climate change and air pollution, while electrochemical reactions in batteries do not produce any harmful emissions.

Fossil fuels have historically been cheaper to obtain than the production of batteries, although the cost of batteries is decreasing as technology improves and demand increases.

Electrochemical reactions in batteries are more energy efficient processes, as some of the energy is lost as heat during the combustion process of fossil fuels.

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Social skills — Assigning and accepting specific roles during group activities

- **Time required to complete activity:** 30 minutes
- **Activity type:** Pair/group activity

1. Use this [simulation](https://teachchemistry.org/classroom-resources/galvanic-voltaic-cells-2)  (<https://teachchemistry.org/classroom-resources/galvanic-voltaic-cells-2>) to create your own primary cells by selecting the electrode pairs and their molarities, and answer the following questions:
 - Find the combination of electrodes and their molarities that produces the highest electrical output.
 - Predict what will happen at each electrode and the salt bridge and click the 'see molecular scale buttons' to check if you were correct.
 - Write the half-reaction happening at each electrode.
2. With your pair or group, research other types of rechargeable batteries and write their charging and discharging equations. In your research, include common uses and the benefits/drawback of using this type of battery.

Electrolytic cells

Learning outcomes

By the end of this section you should be able to:

- Explain how current is conducted in an electrolytic cell.
- Deduce the products of the electrolysis of a molten salt.

You have surely held an aluminium can in your hand once or twice. Aluminium cans are used daily to hold a wide range of products, from carbonated soft drinks to canned fruits and vegetables, soups, aerosol sprays and even some pharmaceutical products.



Credit: Thomas-Soellner, Getty Images



Credit: Aleksandr Durnov, Getty Images

Figure 1. A variety of products are kept in aluminium cans.

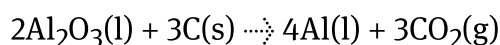
Have you ever stopped to consider where all the aluminium for aluminium cans comes from?

Electrolytic cells

Many elements are not found in nature in their elemental form. They are found as ores, where they occur as ions bonded with other ions (see [subtopic S2.1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/>)). In order to change from the ion to the atom, there must be electron transfer (see [subtopic R3.1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/>)).

Aluminium, as well as other chemicals needed by humans, can be produced by using electricity in electrolytic cells. You may recall that electrolytic cells use electrical energy to produce chemical compounds (see [section R3.2.5](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrodes-in-electrochemical-cell-id-46489/>)). The process uses electrical energy to drive a non-spontaneous chemical reaction and is called electrolysis.

In the previous example, electrolytic cells can be used to extract aluminium metal from its ores by reducing the aluminium ions in molten aluminium oxide, Al_2O_3 (also known as alumina), using graphite as electrodes. The reaction to produce pure aluminium metal and carbon dioxide gas according to the reaction:



This is a non-spontaneous reaction which means energy needs to be provided for the reaction to occur. That energy comes from the use of electricity in the form of a battery or a power generator.

All electrolytic cells consist of:

- a single container
- two inert electrodes (generally graphite is used)
- a solution (the electrolyte)
- a battery which can be considered an electron pump (DC power source).

The first step in creating an electrolytic cell is to melt the salt that contains the ions that will be reduced and oxidised. Remember that solid ionic salts are bad conductors of electricity (see [subtopic S2.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/>)), which means they cannot be electrolysed.

Making connections

Under what conditions can ionic compounds act as electrolytes? (see [subtopic S2.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/>)).

Ionic compounds will act as electrolytes when subjected to high temperatures and an electric current so that they become molten. In the liquid state, the lattice structure is disrupted and the cations and anions are free to move and will be able to reach the electrodes to undergo either oxidation or reduction, respectively.

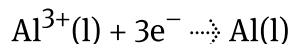
Ionic compounds will also be able to act as electrolytes when they are dissolved in a polar solvent, such as water, and dissociate into their constituent ions. In a polar solvent, the positive and negative poles of the solvent molecules interact with the positively and negatively charged ions, causing them to separate and become surrounded by solvent molecules in a process called solvation or hydration.

In the example of aluminium oxide, Al_2O_3 , the salt needs to be exposed to a high temperature and an electric current to undergo electrolysis. The melting point of aluminium oxide is 2000°C however, cryolite is added to lower the melting point. Cryolite is sodium aluminium fluoride (Na_3AlF_6), a colourless to white halide mineral. Its addition to the electrolyte helps to reduce the cost of the process, as much less energy is required to heat the ore to a temperature of around $960\text{--}980^\circ\text{C}$, the temperature at which the aluminium oxide melts, allowing the ions to move freely in the liquid. The electric current applied to the electrodes inserted into the molten aluminium oxide then separates the aluminium and oxygen ions as each species travels to one of the electrodes.

The electrodes used in this type of cell are inert, as oxidation and reduction should occur only with the ions produced by the molten salt. Platinum (Pt) is a commonly used material for electrodes. In a school lab setting, generally, graphite is used as it is a cheaper option and a good conductor of electricity (see [subtopic S2.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/>)) but relatively inert, which allows the electrical current to pass to the electrolyte without interfering with the reactions happening at each electrode.

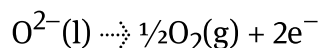
The battery supplies electrons to the cathode, where reduction occurs. From the ions available in the electrolyte, Al^{3+} and O^{2-} , the most likely to receive electrons and become reduced is the aluminium ion, which leads to the following half-reaction:

Cathode(reduction):



Because the cathode is gaining electrons from the power source, it has a more negative charge than the anode. This makes the anion in the electrolyte, O^{2-} , travel to the anode to lose electrons and become oxidised, according to the half-reaction:

Anode(oxidation):



Notice that the product here is oxygen gas and will be visible as bubbles being produced at the anode. The oxygen that is produced reacts with the carbon of the positive electrodes, forming carbon dioxide, so they gradually burn away, which adds to the cost of the process. As a result, the positive electrodes have to be replaced frequently.

Watch the video below about the electrolysis of alumina.

How To Extract Aluminium By Electrolysis | Environmental C...

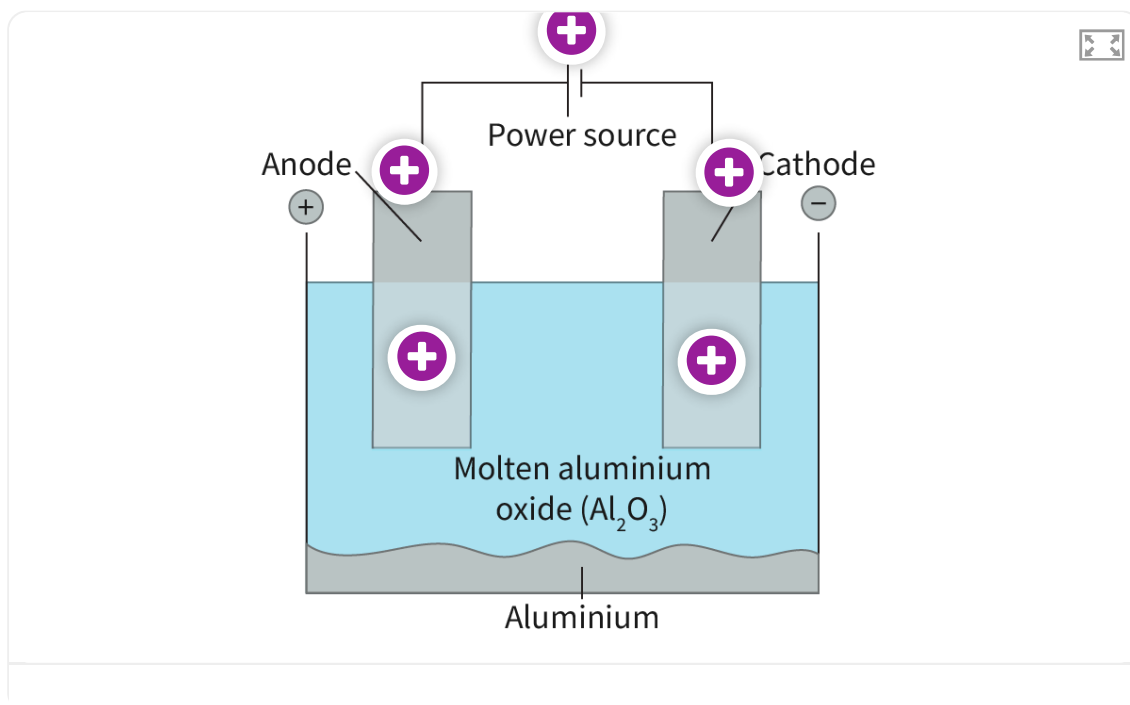


Video 1. The electrolysis of aluminium oxide.

At the cathode, liquid aluminium will be collected and can then be separated from the electrolyte due to its density difference.

In the external wire connecting both electrodes, there is a current of electrons moving from the anode to the cathode, just like in the voltaic cells, but the electrodes have reversed potentials (the anode is positive and the cathode is negative). The circuit is closed by the movement of the ions in the electrolyte (cations to the cathode and anions to the anode).

Use **Interactive 1** to investigate what is happening at each location in the electrolysis of aluminium oxide.

**Interactive 1.** Electrolysis of Molten Aluminium Oxide.

 More information for interactive 1

An interactive diagram displays the electrolysis of molten aluminum oxide (Al_2O_3). Two vertical electrodes are immersed in the molten aluminum oxide. The left electrode is labeled anode and has a positive sign, while the right electrode is labeled cathode and has a negative sign. A power source is connected to both electrodes. The bottom of the container is labeled aluminum, representing the deposited metal.

There are 5 Hotspots represented by plus signs at various places. Hotspot 1 is next to the label anode. Hotspot 2 is above the power source. Hotspot 3 is next to the label cathode. Hotspot 4 on the anode. Hotspot 5 on the cathode. Clicking on the hotspots reveals what is happening in each of the places.

The following items are revealed at respective hotspots:

Hotspot 1: It displays electron flow exiting from the anode and entering the positive terminal of the power supply.

Hotspot 2: It displays the electron flow from the negative terminal to the positive terminal.

Hotspot 3: It displays the electron flow from the negative terminal entering the cathode.

Hotspot 4: It displays an O^{2-} moving towards the anode and liberating two electrons and an O_2 molecule released from the electrolyte.

Hotspot 5: It displays 3 electrons moving down the cathode and an Al^{3+} ion moving towards the cathode inside the electrolyte and releasing Al ion from the cathode.

Study skills

Remember that the construction of electrolytic cells should include a DC power source connected to anode and cathode and an electrolyte.

Look back at **Figure 2** in section R3.2.5

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrodes-in-electrochemical-cell-id-46489/>) for all the details required on a diagram of an electrolytic cell.

To work out what happens during the electrolysis of molten salt, follow these steps:

1. Identify all species present (the anions and cations that can be released when the salt melts).
2. Identify which species are attracted to the anode (positive electrode) and which species are attracted to the cathode (negative electrode).
3. Deduce the two half-reactions, knowing the oxidation happens at the anode and reduction happens at the cathode.
4. Determine the direction for the flow of electrons in the external wire and of the ions in the electrolyte.

5. State what will be observed at each electrode based on the products of the two half-reactions.

Worked example 1

Consider the electrolysis of a molten salt such as lead(II) bromide.

State the ions that will be present in the electrolyte.

Lead(II) bromide, PbBr_2 , when molten, will release the cation Pb^{2+} and the anion Br^- into the electrolyte.

Worked example 2

Consider the electrolysis of a molten salt such as lead(II) bromide.

State the half-reaction happening at the anode and deduce the product formed and what you will be able to observe.

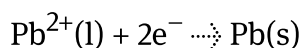


At the anode, oxidation will occur. Two anions of Br^- will move to the anode and will lose two electrons to become bromine gas, $\text{Br}_2(\text{g})$. A brown gas will be visible, bubbling around the anode.

Worked example 3

Consider the electrolysis of a molten salt such as lead(II) bromide.

State half-reaction happening at the cathode and deduce the product formed and what you will be able to observe.



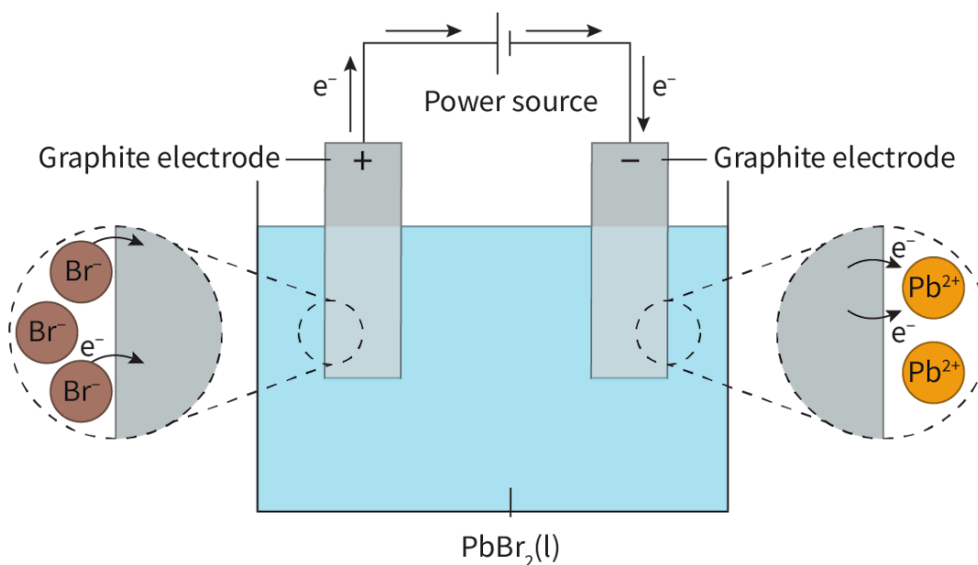
At the cathode, reduction will occur. Each cation Pb^{2+} will move to the cathode and will gain two electrons to become lead metal, Pb(l) . The lead will be visible around the cathode.

Worked example 4

Consider the electrolysis of a molten salt such as lead(II) bromide.

Draw and annotate a diagram of the electrolytic cell, including the movement of the ions and electrons, as well as the charges for each electrode.

The diagram should look similar to this one.



Electrolysis of molten lead(II) bromide.



Worked example 5

Consider the electrolysis of a molten salt such as lead(II) bromide.

Discuss why lead(II) bromide molten conducts electricity while it does not when in a solid form.

Molten lead(II) bromide has free ions that can move. Electricity is passed when there are charged particles able to move freely. In the solid form, all the ions that make up lead(II) bromide are fixed into a lattice structure and, therefore cannot freely move. This means they cannot conduct electricity.

Aspect: Experiments

The concept of electrolysis and its uses were developed by several scientists over time, but the modern understanding of the process can be traced back to the work of Michael Faraday in the early 19th century. Faraday's experiments on electrolysis showed that an electric current could cause chemical reactions to occur in a solution or molten material.

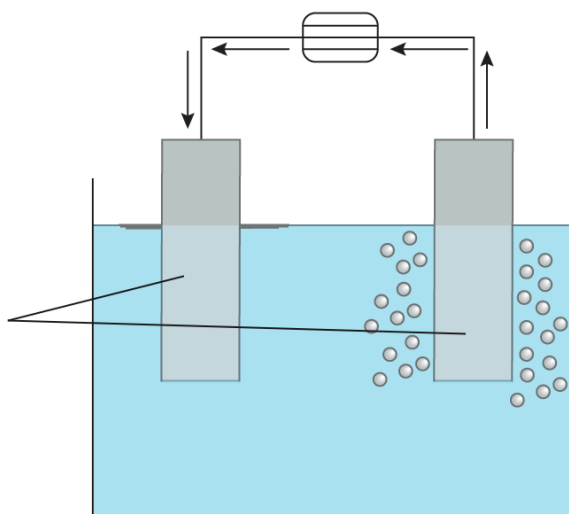
Nowadays, electrolysis is used in various industries and applications, like the production of aluminium from alumina discussed at the beginning of this section. Other reactions include:

- Electrolysis of water, used to produce hydrogen gas for fuel cells and other applications.
- Electroplating, used to deposit a thin layer of metal onto a surface using an electric current; it improves the appearance, corrosion resistance, or wear resistance of various metal objects, such as jewellery, car parts, and electronics.
- Electrolytic refining, used to purify metals such as copper, nickel, and zinc, which are used in various industries, such as electronics and telecommunications.

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Communication skills — Using terminology, symbols, and communication conventions consistently and correctly
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual/pair

The electrolysis of calcium chloride (CaCl_2) is used on a small scale to produce pure calcium metal for laboratory use. Drag all the labels to the correct spaces on the image that illustrates the electrolysis of calcium chloride.



Molten calcium
chloride CaCl_2

Anode

Graphite

Battery

Cathode

-

+

$\text{Cl}_2(\text{g})$

$\text{Ca}(\text{l})$

e^-

e^-

✓ Check

Interactive 2. Drag the labels to the correct spaces.

Oxidation of organic compounds

Learning outcomes

By the end of this section you should be able to deduce equations to show changes in the functional groups during the oxidation of primary and secondary alcohols, including the two-step reaction in the oxidation of primary alcohol.

The environmental impact of plastics is an increasing concern and a threat to ecosystems around the globe. Scientists are studying the creation of products that could replace plastics and also ways to reduce the plastic that we already have introduced into the environment.

Carboxylic acids can be used to make biodegradable polymers, such as polylactic acid (PLA), which is a biodegradable polymer made from lactic acid, a carboxylic acid derived from renewable sources such as corn starch. Ketones and carboxylic acids can also be used as solvents or reactants in the degradation of pollutants. For instance, carboxylic acids can be used to extract heavy metals from contaminated soils, while ketones can be used to break down organic pollutants, such as polycyclic aromatic hydrocarbons (PAHs) or bisphenol A (BPA), which are commonly found in industrial and household waste.

How can we have access to more of these useful ketones and carboxylic acids?

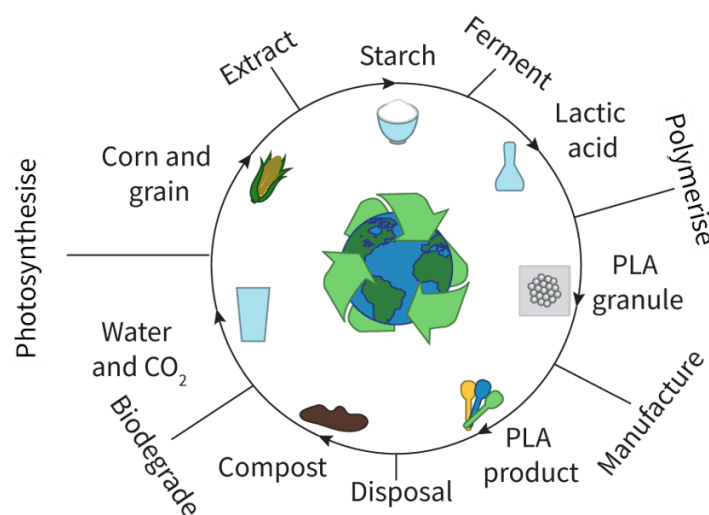


Figure 1. PLA production and use.

 More information for figure 1

The circular diagram illustrates the production and lifecycle of Poly(lactic acid) (PLA). At the center is an image of the Earth surrounded by a recycling symbol, indicating the eco-friendly nature of the process. The cycle begins with 'Photosynthesis', where corn and grain are grown, labeled as 'Corn and grain'. From there, 'Extract' leads to 'Starch' and then 'Ferment', producing 'Lactic acid'. The process continues with 'Polymerise' to form 'PLA granule', which is then used in 'Manufacture' to create a 'PLA product'. The 'PLA product' can be 'Disposal'd or 'Biodegrade'd into 'Compost', which is then used in 'Photosynthesis' to grow 'Corn and grain' again, completing the cycle.

where 'PLA granule' is formed. 'Manufacture' creates 'PLA product'. After use, the product is 'Disposed' and goes through 'Compost'. Finally, it 'Biodegrades' to produce 'Water and CO₂', which then returns to 'Photosynthesis', completing the cycle.

[Generated by AI]

Alcohol structure

You might recall that the alcohols (see [subtopic S3.2](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/>)) have a hydroxyl (–OH) group, the ketones have a carbonyl (=O) group and the carboxylic acids have a carboxyl (–COOH) group.

Table 1. Alcohols, carboxylic acids and ketones.

C o m p o u n d	Functional group	Example
Al co ho l	$\text{R} - \text{OH}$ Hydroxyl	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{O}$  More information The image depicts the chemical structure formula of Butanol. It is shown as a molecule chain: CH₃CH₂CH₂CH₂OH. The letters represent the atoms in the molecule: C for carbon, H for hydrogen, and OH for the hydroxyl group. The chain of letters is displayed horizontally, indicating the linear structure of the Butanol molecule. [Generated by AI] Butanol
Ca rb		

C o m p o u n d	Functional group	Example
ox y l i c a c i d	More information The image is a diagram showing the chemical structure of a carboxyl group. It consists of a carbon (C) atom double-bonded to an oxygen (O) atom, and single-bonded to a hydroxyl group (OH). Additionally, the carbon atom is single-bonded to a variable group represented as R. This structural formula represents the functional group typical in carboxylic acids. [Generated by AI] Carboxyl	CH_3COOH Ethanoic acid
Ke to ne	More information The image is a chemical structure diagram of a ketone molecule. It shows a carbon (C) atom double-bonded to an oxygen (O), labeled as "O=C". Attached to the carbon are two groups, indicated as "R" and "R'", which represent the rest of the molecule and can vary. These groups form single bonds with the carbon atom to complete the angular structure typical of simple ketones. [Generated by AI]	CH_3COCH_3 Propanone

C o m p o u n d	Functional group	Example
	Ketone	

Earlier in your studies you learned that alcohols can be classified as primary, secondary or tertiary (see section S3.2.6 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structural-isomerism-id-45665/>)), depending on the number of carbon atoms bonded to the carbon atom that is directly bonded to the functional group. **Figure 2** shows some examples of primary, secondary and tertiary alcohols.

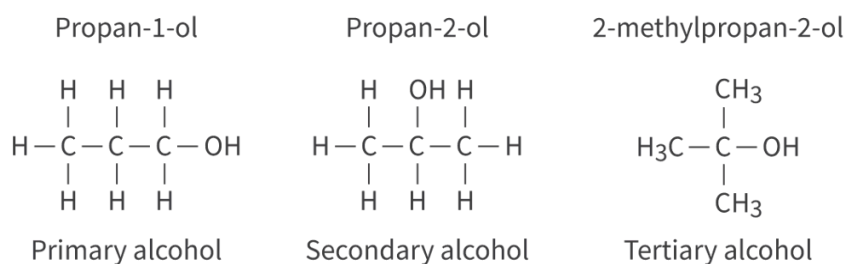


Figure 2. Some examples of primary, secondary and tertiary alcohols.

 More information for figure 2

The image depicts molecular structures of three types of alcohols, highlighting their classification as primary, secondary, and tertiary. The first structure is for Propan-1-ol, a primary alcohol, shown with three carbon atoms in a chain. Each carbon is bonded with hydrogen atoms, and the functional group (-OH) is attached to the end carbon. The second structure is for Propan-2-ol, a secondary alcohol, with the -OH group attached to the middle carbon of the three-carbon chain. Hydrogen atoms are bonded to the other available sites on the carbon atoms. The third structure is for 2-methylpropan-2-ol, a tertiary alcohol, where the -OH group is attached to a central carbon atom that is also bonded to a methyl group (CH₃) in addition to other carbon atoms. This diagram visually differentiates the alcohols by illustrating their different functional group attachments to the carbon chain.

[Generated by AI]

Check your understanding of the classification of alcohols by answering these questions.



Exercises 1 and 2

Click a question to answer



1. What type of alcohol is 2-methylpropan-2-ol?



2. Butan-2-ol is classified as a _____ alcohol.

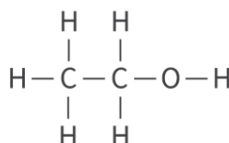
✓ Check answers

You also learned previously that oxidation can be defined as the addition of oxygen or the removal of hydrogen (see [section R3.2.1–2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/>)).

Figure 3. A comparison of oxidation and reduction.

Now, let's compare ethanol, ethanal and ethanoic acid in terms of their oxidation.

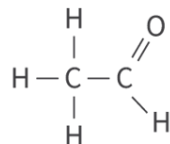
Which one is the most oxidised or reduced?



More information

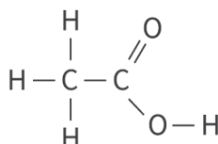
The image shows the structural formula of an ethanol molecule. It is represented by a linear arrangement of carbon (C), hydrogen (H), and oxygen (O) atoms connected by single bonds. The formula follows this sequence: a central carbon atom (C) is bonded to two hydrogen atoms (H) on the left and right sides. The carbon atom is also bonded upwards to another carbon atom, which is connected to two hydrogen atoms above and below. This second carbon atom is attached to an oxygen atom (O) on its right, which in turn is bonded to a hydrogen atom. This sequence forms the molecular configuration CH₃CH₂OH, representing ethanol.

[Generated by AI]

[More information](#)

The image is a chemical structure diagram representing the acetaldehyde molecule. It consists of two carbon atoms bonded together, labeled as 'C'. One of the carbon atoms is single-bonded to three hydrogen atoms labeled 'H'. The other carbon atom is double-bonded to an oxygen atom labeled 'O' and single-bonded to a hydrogen atom. This structure shows the typical arrangement of atoms in an acetaldehyde molecule, with the oxygen atom forming a double bond with one of the carbon atoms to create the carbonyl group.

[Generated by AI]

[More information](#)

The image shows the chemical structure of ethanol, a primary alcohol. It is depicted with a central carbon (C) atom bonded to three hydrogen (H) atoms and one hydroxyl (OH) group. The carbon is also bonded to another carbon which is double-bonded to an oxygen (O) atom. The hydroxyl group is bonded to the second carbon atom. This structure is a typical representation of ethanol, commonly found in alcoholic beverages. The schematic highlights the arrangement of atoms, demonstrating typical bond angles and orientations associated with organic chemical structures.

[Generated by AI]

Ethanol
(primary alcohol)

Ethanal
(aldehyde)

Ethanoic acid
(carboxylic acid)

Figure 4. Ethanol, ethanal and ethanoic acid.

You might be able to identify that the aldehyde has two fewer hydrogens than the alcohol, which means it is more oxidised than the alcohol. It might be also clear than in comparison, the carboxylic acid has one more oxygen than the aldehyde, meaning it is more oxidised.

Oxidation of primary alcohols

Primary alcohols can be oxidised into aldehydes and carboxylic acids when in the presence of an oxidising agent [O] according to the reactions in **Figure 5**. Notice that the oxidation number of the carbon in the alcohol will increase and either an aldehyde or a carboxylic acid will be produced.

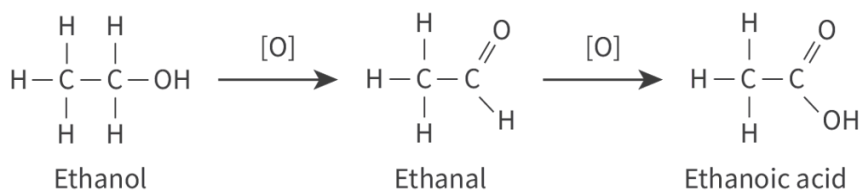


Figure 5. The oxidation of alcohols into aldehydes and carboxylic acids.

 More information for figure 5

The image is a chemical diagram illustrating the oxidation process of ethanol into ethanal and further into ethanoic acid. The structures are depicted as follows:

1. Ethanol (C₂H₅OH): Two carbon atoms are shown, with the first carbon bonded to three hydrogen atoms and the second carbon bonded to one hydrogen and a hydroxyl group (OH).
2. An arrow marked with "[O]" indicates the presence of an oxidizing agent that facilitates the reaction to form ethanal.
3. Ethanal (C₂H₄O): The first carbon remains bonded to three hydrogen atoms. The second carbon now forms a double bond with an oxygen atom and is also bonded to one hydrogen atom, indicating the formation of an aldehyde.
4. Another arrow marked with "[O]" shows further oxidation to ethanoic acid.
5. Ethanoic acid (C₂H₄O₂): The structure has the first carbon bonded to three hydrogen atoms. The second carbon is bonded to a hydroxyl group (OH) and double-bonded to an oxygen atom, representing the carboxylic acid formation.

[Generated by AI]

For example, glycerol (C₃H₈O₃) is a naturally occurring alcohol that can be used as a sweetener for low-fat food products, and it can be oxidised into glyceraldehyde or glyceric acid in the presence of an oxidising agent (**Figure 6**).

In the human body, glycerol is involved in the balance of cellular redox, crucial for the proper cellular functioning and the prevention of oxidative stress that can damage cellular components.

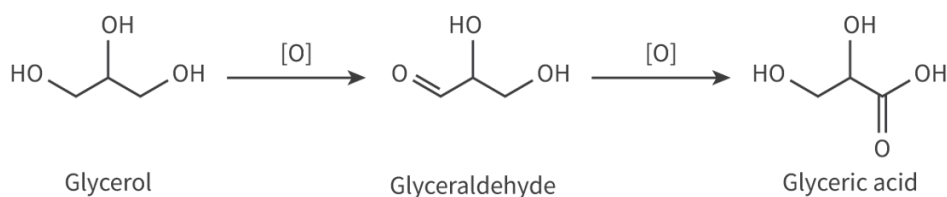


Figure 6. The oxidation of glycerol to glyceraldehyde and glyceric acid.

 More information for figure 6

The image is a chemical reaction diagram illustrating the oxidation process of glycerol to form glyceraldehyde and subsequently glyceric acid. The diagram is composed of three main chemical structures connected by arrows indicating the flow of reaction. On the left, the structure of glycerol is depicted, which consists of a three-carbon chain with each carbon (C) attached to a hydroxyl group (OH). An arrow labeled with [O] points to the next structure, glyceraldehyde, a three-carbon aldehyde with a double-bonded oxygen on the first carbon. A similar arrow labeled with [O] leads to glyceric acid, which has a similar three-carbon structure but features both a double-bonded oxygen and an additional hydroxyl group.

[Generated by AI]

The presence of the hydroxyl functional group makes alcohols susceptible to oxidation in the presence of a suitable oxidising agent. You do not need to know the names of suitable oxidising agents, but an understanding of their function is important. During the reaction, the oxidising agent removes electrons from the alcohol (or aldehyde) and becomes reduced. In the process, the oxidising agent changes colour, as shown in **Figure 7** and **Figure 8**.

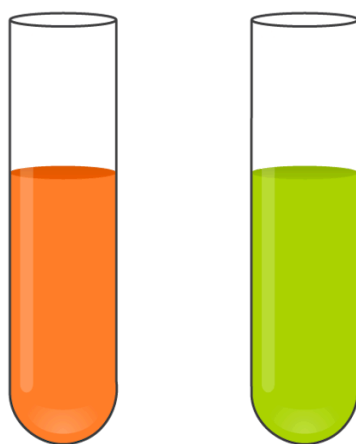


Figure 7. The colour change from orange to green when an alcohol undergoes oxidation with acidified potassium dichromate(VI).

Making connections

Why is there a colour change when an alcohol is oxidised by a transition element compound? (see [subtopic S3.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/)).

Transition element compounds have different colours depending on the oxidation state of the transition element. For example, the dichromate ion ($\text{Cr}_2\text{O}_7^{2-}$) with Cr^{6+} ions forms an orange solution which changes to a green colour as it is reduced to Cr^{3+} (Figure 7).

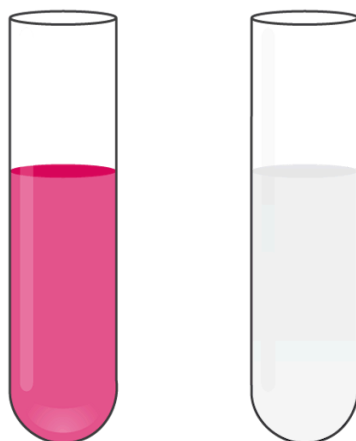
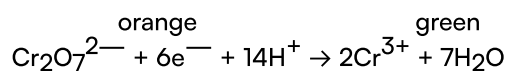


Figure 8. The colour changes from purple to colourless when an alcohol undergoes oxidation with acidified potassium manganate(VII) solution.

Study skills

Names and formulas of specific oxidising agents and the mechanism will not be assessed.

Primary alcohols, such as ethanol, produce aldehydes or carboxylic acids, depending on the degree of oxidation (partial or complete). The partial oxidation of a primary alcohol produces an aldehyde. The reaction is carried out with an excess of alcohol as in **Figure 9**. Note the use of [O] to represent the oxidising agent in the reaction.

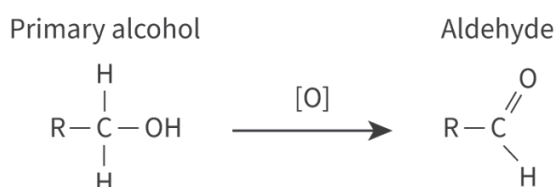


Figure 9. The partial oxidation of a primary alcohol to an aldehyde.

 More information for figure 9

The image is a diagram illustrating the partial oxidation of a primary alcohol to an aldehyde. On the left, there is a structural representation of a primary alcohol with the generic formula $R-C-OH$, where the carbon (C) is bonded to two hydrogen atoms (H) and a hydroxyl group (OH). To the right of this structure, an arrow points towards another structure labeled 'Aldehyde'. Above the arrow, the symbol $[O]$ is used to represent the oxidizing agent involved in the reaction. The structure on the right represents an aldehyde with the formula $R-C-H$, where the carbon is double bonded to an oxygen atom (O) and single bonded to a hydrogen atom (H). This diagram communicates the chemical transformation from a primary alcohol to an aldehyde through partial oxidation.

[Generated by AI]

Figure 10 shows the partial oxidation of the primary alcohol ethanol to the aldehyde ethanal.

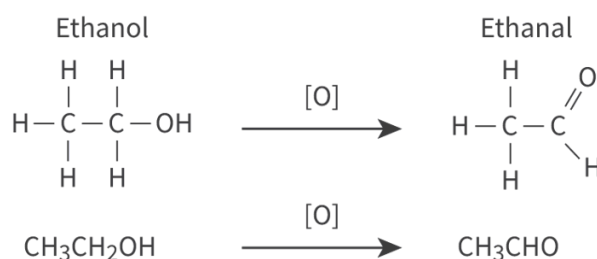


Figure 10. The partial oxidation of ethanol to ethanal.

 More information for figure 10

The image depicts a chemical diagram that illustrates the partial oxidation of ethanol to ethanal. On the left, the ethanol molecule is shown with the chemical formula C_2H_5OH , consisting of a hydroxyl group (OH) bonded to a 2-carbon chain, with each carbon atom bonded to hydrogen atoms (H). To the right, after the oxidation process indicated by an arrow and $[O]$ placeholder, the ethanol molecule is transformed into ethanal. The ethanal molecule is represented by the chemical formula CH_3CHO and includes an aldehyde group (CHO) attached to the carbon chain, indicating the replacement of one hydrogen atom with an oxygen atom double-bonded to a carbon atom.

[Generated by AI]

Distillation apparatus

If the aldehyde is the desired product, it can be removed as it forms by distillation to prevent further oxidation (**Figure 11**). This is possible because the boiling point of the aldehyde is lower than that of both the alcohol and the carboxylic acid. As the aldehyde evaporates and rises up the distillation column, it passes through the condenser where it condenses back to a liquid and runs down into the flask.

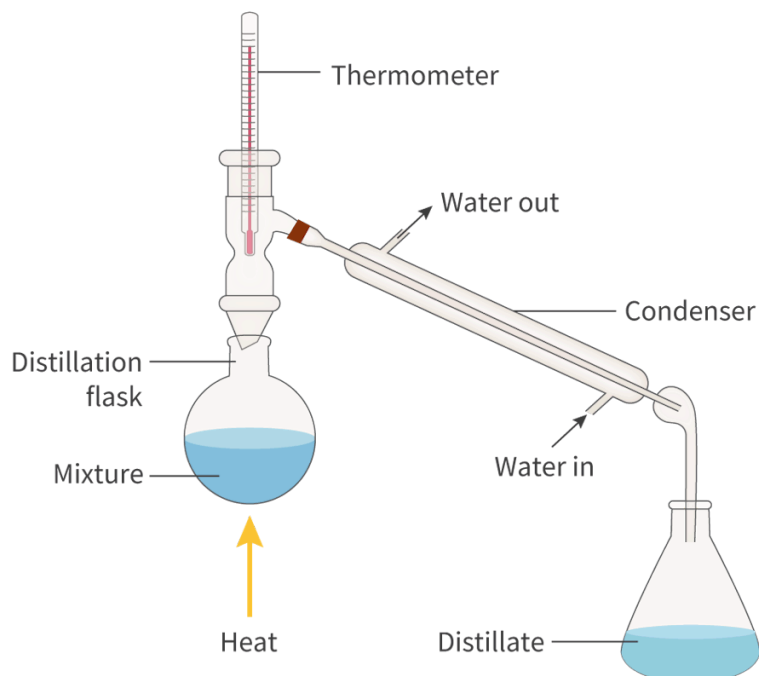


Figure 11. Distillation apparatus used to separate components of a mixture with different boiling points.

 More information for figure11

The image is a labeled diagram of a distillation apparatus used for separating components of a mixture based on their boiling points. The apparatus consists of several key components:

1. A distillation flask containing the mixture, which is heated.
2. A thermometer is inserted into the apparatus to measure the temperature.
3. The vapor rises and moves through the condenser.
4. The condenser has labels for 'Water in' at the lower end and 'Water out' at the upper end to control the condensation process.
5. The vapor condenses back into liquid in the condenser tube.
6. The liquid, known as distillate, is collected in a receiving flask.

The diagram shows the process of heating the mixture to facilitate evaporation of the component with a lower boiling point, which then condenses and is collected separately as a distillate.

[Generated by AI]

Making connections

How does the nature of the functional group in a molecule affect its physical properties, such as boiling point? (see [subtopic S3.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/>)).

The nature of the functional group in a molecule can significantly affect its physical properties, including the boiling point, due to the intermolecular forces present between the particles of the substance.

Boiling point is a measure of the temperature at which a liquid changes state from liquid to gas. The stronger the intermolecular forces between the molecules, the higher the boiling point of the substance.

In general, we can say that alkanes have lower boiling points as they can only form London forces between their molecules. Alcohols contain a hydroxyl (—OH) functional group, which can participate in hydrogen bonding with other molecules, therefore alcohols generally have higher boiling points than similar-sized hydrocarbons. Aldehydes and ketones contain a carbonyl (C=O) functional group, which can participate in dipole-dipole interactions which are weaker than hydrogen bonding, so aldehydes and ketones generally have lower boiling points than alcohols of similar size. Finally, carboxylic acids contain a carboxyl (—COOH) functional group, which can participate in hydrogen bonding and dipole-dipole with other molecules, which generally gives them the highest boiling points.

Reflux apparatus

The complete oxidation of a primary alcohol produces a carboxylic acid as shown in **Figure 12**. The conditions used are heat under reflux with an excess of the oxidising agent.

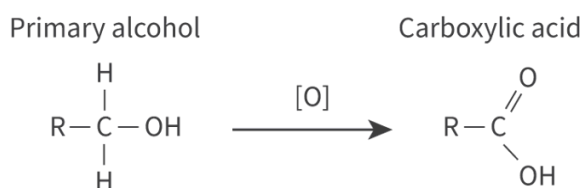


Figure 12. The complete oxidation of a primary alcohol produces a carboxylic acid.

 More information for figure 12

The image is a chemical reaction diagram illustrating the oxidation process of a primary alcohol to form a carboxylic acid. The diagram shows a structural formula of a primary alcohol on the left side, labeled as 'Primary alcohol,' with a generic structure: $\text{R-CH}_2\text{OH}$. An arrow labeled with "[O]" points towards the right side, indicating the addition of an oxidizing agent. On the right, the resultant carboxylic acid is represented, labeled as 'Carboxylic acid,' with the structure: R-COOH . The arrow signifies the conversion process facilitated by oxidation.

[Generated by AI]

The complete oxidation of ethanol to ethanoic acid is shown in **Figure 13**.

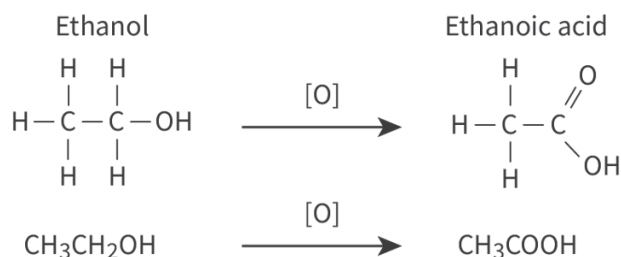


Figure 13. The complete oxidation of ethanol to produce ethanoic acid.

 More information for figure 13

The image displays a chemical reaction showing the complete oxidation of ethanol to ethanoic acid. The structural formula for ethanol is shown on the left with the atoms and bonds depicted as: two carbon atoms with three hydrogen atoms bonded to the first carbon and one hydroxyl group (OH) bonded to the second carbon. Below the structural formula is the molecular formula 'CH₃CH₂OH.' An arrow points to the right, indicating the reaction process with an '[O]' above and below the arrow, representing oxidation. On the right, the structural formula for ethanoic acid is depicted with a carboxyl group (COOH) bonded to a carbon atom, which is also connected to two other hydrogen atoms. Below the structural formula, the molecular formula 'CH₃COOH' is shown. This visual representation illustrates the conversion of ethanol to ethanoic acid through an oxidation process.

[Generated by AI]

If the carboxylic acid is the required product, the mixture containing the alcohol and oxidising agent is heated under reflux (**Figure 14**). As the mixture is heated, the vapours rise up the condenser, cool down and condense back into a liquid. This allows the alcohol and the oxidising agent to remain in contact for a prolonged period of time and results in complete oxidation.

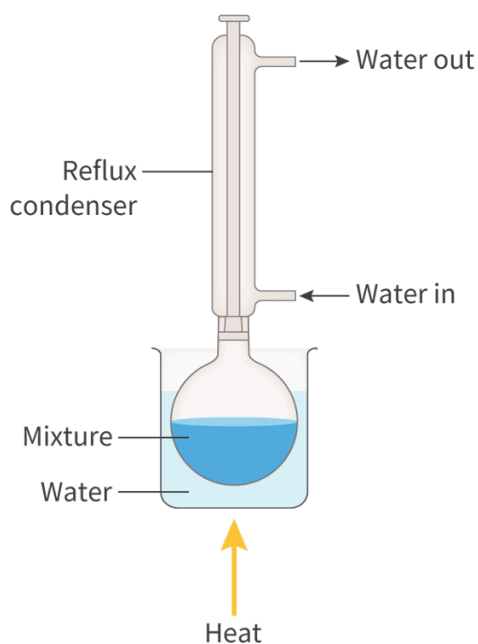


Figure 14. Apparatus for heating under reflux. More information for figure 14

The image shows a diagram of an apparatus for heating under reflux. It consists of a round-bottom flask containing a blue liquid labeled as the mixture. The flask sits in a larger container filled with water. Underneath this container is an arrow pointing upward labeled "Heat," indicating the application of heat. Above the flask is a vertical tube labeled "Reflux condenser" with arrows and labels showing "Water in" on one side and "Water out" on the other. This setup allows vapors to rise, condense in the condenser, and return to the flask, maintaining the reaction mixture in a closed system.

[Generated by AI]

Aldehyde has a lower boiling point than carboxylic acid due to the differences in the intermolecular forces:

- Aldehyde – weaker dipole–dipole intermolecular forces
- Carboxylic acid – stronger hydrogen bonds intermolecular forces

Worked example 1

Draw and name the product in the oxidation of butan-1-ol under reflux.

Under reflux, the primary alcohol will oxidise completely into a carboxylic acid, in this case butanoic acid.

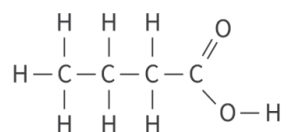


Figure 15. Butanoic acid.

**Worked example 2**

Draw and name the product of the oxidation of hexan-1-ol under distillation.

Under distillation, the primary alcohol will oxidise partially into an aldehyde, in this case hexanal.

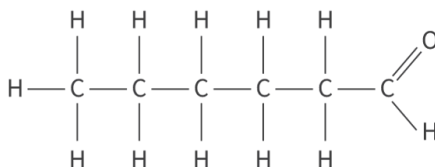


Figure 16. The full structural formula of hexanal.



Oxidation of secondary alcohols

The products of the oxidation reaction depend on whether the alcohol is primary or secondary, and on the reaction conditions (note that tertiary alcohols do not undergo oxidation).

What would oxidation look like for a secondary alcohol? Where will hydrogen atoms be removed or oxygen atoms added? How far can the oxidation occur?

The oxidation of a secondary alcohol produces only one product – a ketone as in **Figure 17**. The required reaction conditions are heat under reflux with a suitable oxidising agent such as acidified potassium dichromate(VI).

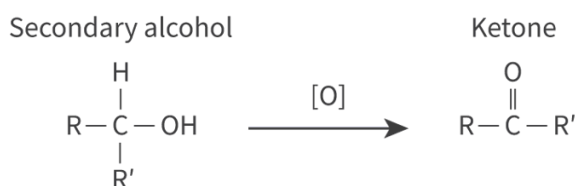


Figure 17. Secondary alcohols are oxidised to form ketones.

More information for figure 17

The image shows a chemical reaction equation where a secondary alcohol is oxidized to form a ketone. On the left side of the equation, there's a structural formula for secondary alcohol labeled "Secondary alcohol," with the structure: $\text{R}_1\text{-C}(\text{-OH})(\text{-H})\text{-R}_2$. An arrow labeled "[O]" points from the secondary alcohol to the ketone on the right side. The ketone is labeled "Ketone," with the structure: $\text{R}_1\text{-C}(=\text{O})\text{-R}_2$. This illustration depicts the oxidation process where the hydrogen bonded to the carbon in the alcohol is replaced by an oxygen to form the carbonyl group of a ketone.

[Generated by AI]

Figure 18 shows the oxidation of the secondary alcohol, propan-2-ol, to produce propanone.

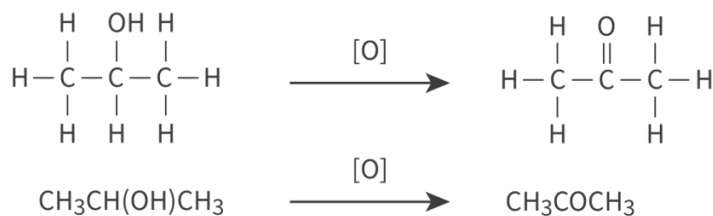


Figure 18. The oxidation of propan-2-ol to propanone.

 More information for figure 18

The image is a chemical diagram illustrating the oxidation process of propan-2-ol to propanone. On the left side, the structure of propan-2-ol is shown, represented by a chain of three carbon atoms. The first carbon atom is bonded to three hydrogen atoms, the second carbon is bonded to one hydrogen atom and one hydroxyl group (OH), and the third carbon is bonded to three hydrogen atoms. Below the structure is the molecular formula $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$. An arrow labeled with $[\text{O}]$ points from the propan-2-ol structure to the right side, indicating an oxidation reaction. On the right side, the structure of propanone is shown, represented by a similar chain of three carbon atoms. The first carbon atom is bonded to three hydrogen atoms, the second carbon is double-bonded to an oxygen atom and bonded to another carbon atom, and the third carbon is bonded to three hydrogen atoms. Below this structure is the molecular formula CH_3COCH_3 . The diagram effectively shows the transformation of the secondary alcohol to a ketone through the oxidation process.

[Generated by AI]

Worked example 3

Draw and name the product of the oxidation of butan-2-ol under reflux.

Under reflux, the secondary alcohol will oxidise completely into a ketone, in this case butanone.

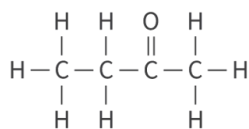


Figure 19. Butanone.



Making connections

What is the difference between the combustion and oxidation of an alcohol? (see subtopic R1.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43475/>)).

Alcohols combust in the presence of oxygen and produce carbon dioxide and water, as well as energy in the form of heat and light. The combustion of an alcohol is a redox reaction in which the alcohol is oxidised and the oxygen is reduced.

Alcohols can also be oxidised through the addition of an oxidising agent. This process can be used to convert alcohols to other functional groups, such as aldehydes or carboxylic acids.

The key difference between the combustion and oxidation of alcohols is the reaction rate and the nature of the products. Combustion is a rapid process that typically results in the complete oxidation of the alcohol to carbon dioxide and water, while the oxidation of alcohols is a slower process that can be controlled to produce specific products, such as aldehydes or carboxylic acids, depending on the choice of oxidising agent and reaction conditions, like a distillation or reflux apparatus.

Activity

- **IB learner profile attribute:**
 - Inquirer
 - Thinker
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual/pair

Real-world applications of the oxidation of alcohols

Research how primary alcohols are oxidised to aldehydes and carboxylic acids, and how these reactions are relevant to biological and industrial processes.

- Compare the oxidation of different types of alcohols (primary, secondary and tertiary) using acidified sodium dichromate as an oxidising agent.
- Explore real-world applications of alcohol oxidation, such as the biological metabolism of ethanol and methanol, the reflux and distillation setups, the commercial production of various types of vinegar or kombucha using acetic acid bacteria, or the production of acetic acid from ethanol.
- You might consider investigating the common uses and commercial production of ethanoic acid (acetic acid) or methanoic acid (formic acid) — two very important compounds.

Share your report with a partner and reflect on what you have learned.

Reduction of organic compounds

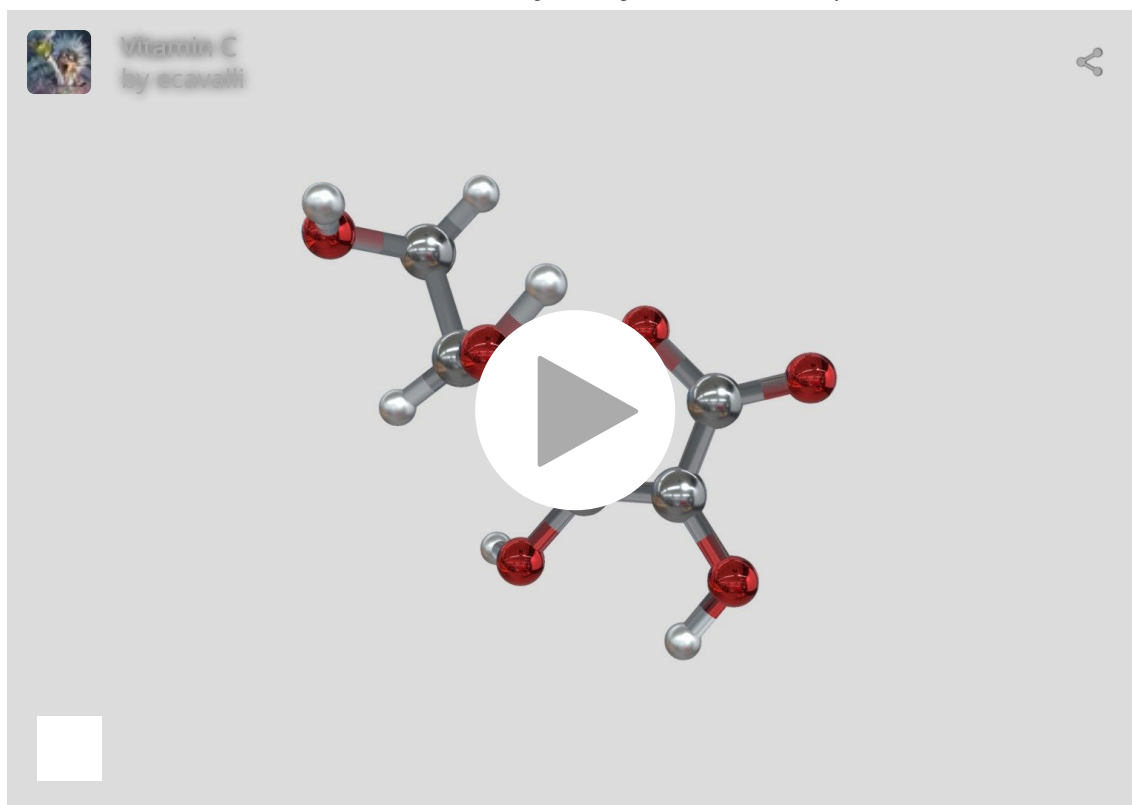
Learning outcomes

By the end of this section you should be able to deduce equations to show the reduction of carboxylic acids to primary alcohols via the aldehyde, and the reduction of ketones to secondary alcohols.

Vitamin C, also known as ascorbic acid, is a water-soluble vitamin that plays a crucial role in various biological functions. It is an essential nutrient, which means that it cannot be synthesised by the human body and must be obtained from the diet. Vitamin C is found in many fruits and vegetables, including citrus fruits, strawberries, kiwi, broccoli, and peppers.

Vitamin C is a six-carbon compound with the chemical formula $C_6H_8O_6$, belonging to the family of organic compounds known as polyols or sugar alcohols. Ascorbic acid contains both hydroxyl ($-OH$) and carbonyl ($C=O$) functional groups as shown in **Interactive 1**, which give it its unique chemical and biological properties.

It can be produced industrially by the reduction of carboxylic acids.



Interactive 1. Ascorbic acid, also known as vitamin C.

 More information for interactive 1

An interactive simulation features a three-dimensional ball-and-stick model of ascorbic acid. Users can rotate and manipulate this model in any direction. The carbon, oxygen, and hydrogen atoms are represented by spheres, and the connecting bonds are represented by thin rods.

The ascorbic acid structure features a pentagonal structure with an oxygen atom at the top, a carbon at the top right, a carbon at the bottom right, a carbon at the bottom left, and a CH group at the top left. All are connected by single bonds, except the carbon atoms on the bottom, which are connected to each other by a double bond.

The carbon atoms at the bottom are each connected to an OH group, and the carbon at the top right is double-bonded to an oxygen atom. The CH group on the top left is connected to another CH group, which is further bonded to an OH group and a CH₂ group. This CH₂ is bonded to an OH group.

This model helps users understand the structure, bonding, and geometry of ascorbic acid.

In the early 20th century, it was discovered that vitamin C deficiency was the cause of scurvy, and that scurvy could be prevented by consuming foods containing vitamin C. This led to the development of vitamin C supplements, which have been widely used since then as a dietary supplement. The industrial production of vitamin C is a complex process, but it highlights the conversion of carboxylic acids into aldehydes, ketones or alcohols through reduction reactions.

How is the reduction of carboxylic acids used to produce aldehydes, ketones or alcohols?

Reduction of organic compounds

As mentioned, carboxylic acids can be reduced back to aldehyde and primary alcohol in the presence of a reducing agent.

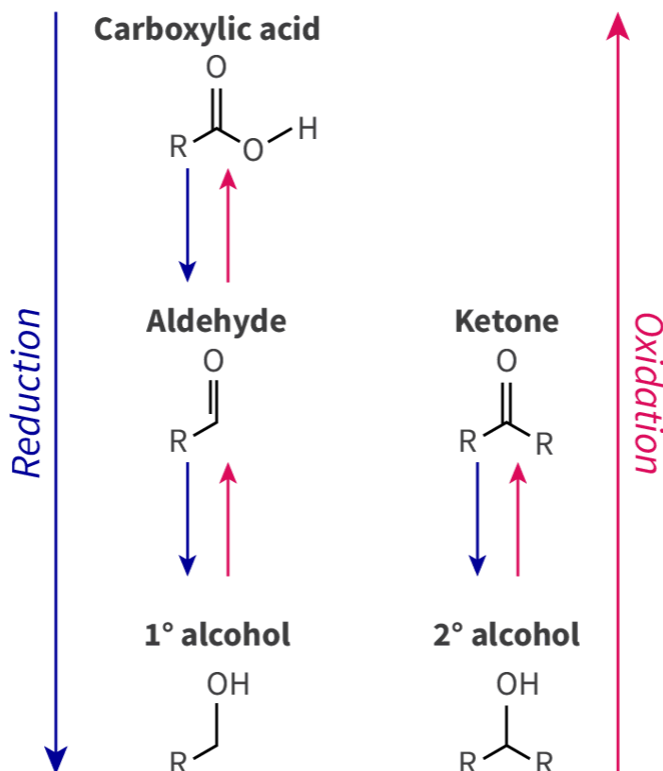


Figure 1. Reaction pathways between carboxylic acids, aldehydes, alcohols and ketones.

 More information for figure 1

The diagram presents the reaction pathways between carboxylic acids, aldehydes, alcohols, and ketones, illustrating flows of oxidation and reduction.

On the left, labeled 'Reduction,' the diagram begins with carboxylic acid, represented by a structural formula ($\text{R}-\text{C}(\text{O})\text{OH}$). Below it is 'Aldehyde' ($\text{R}-\text{C}(\text{O})\text{H}$) with an arrow pointing downward, indicating reduction. Further down is ' 1° alcohol' ($\text{R}-\text{CH}_2\text{OH}$), also connected via a downward arrow.

On the right, labeled 'Oxidation,' the pathway illustrates 'Ketone' ($\text{R}_2\text{C}=\text{O}$) converting back to ' 2° alcohol' (R_2CHOH), both connected with arrows indicating the direction of reactions.

Overall, the diagram emphasizes the reversible nature of these chemical transformations via redox processes, underscoring how carboxylic acids can be reduced to aldehydes and then to primary alcohols, while ketones can be oxidized and reduced to secondary alcohols.

[Generated by AI]

Recalling that reduction can involve the removal of oxygen or the addition of hydrogen, what element must the reducing agent have in order to perform its function?

The reducing agent needs to have hydrogen atoms to donate to the carboxylic acid, in order for reduction to occur. That can be in the form of hydride ions (H^-) or hydrogen ions (H^+).

Theory of Knowledge

The commercial synthesis of natural substances, such as vitamins, are chemically identical to those synthesized by plants. To what extent do chemists rely on nature for knowledge?

These compounds can be reduced back to the respective alcohol in reduction reactions using a suitable reducing agent. The partial positive charge on the carbon atom makes this group vulnerable to attack from a nucleophile, which is the hydride ion (H^-). The sources of the hydride ions are the reducing agents sodium borohydride (NaBH_4) and lithium aluminium hydride (LiAlH_4).

Study skills

Remember that names and formulas of specific reducing agents and the mechanisms will not be assessed.

Figure 2 shows the reduction of an aldehyde, propanal, to a primary alcohol, propan-1-ol. The reducing agent is sodium borohydride (lithium aluminium hydride can also be used) followed by the addition of an acidic solution (H^+).

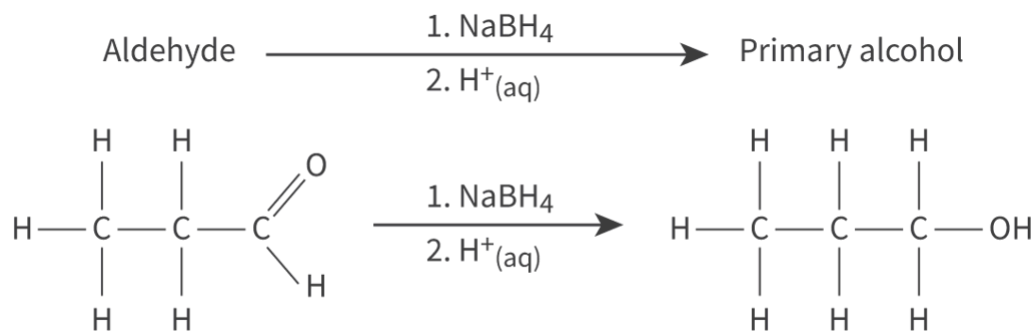


Figure 2. The reduction of the aldehyde propanal to the primary alcohol propan-1-ol.

 More information for figure 2

The image is a chemical reaction diagram illustrating the reduction of the aldehyde propanal to the primary alcohol, propan-1-ol. On the left, the structure of propanal is shown with a carbon chain: three carbon atoms with hydrogen atoms attached, and a double bonded oxygen atom at the end, indicating the aldehyde group. An arrow points to the right, labeled '1. NaBH₄, 2. H⁺(aq)', representing the reduction process. On the right, the structure of propan-1-ol is displayed, where the double-bonded oxygen has been replaced by an OH group, indicating the transformation into a primary alcohol.

[Generated by AI]

Figure 3 shows the reduction of a ketone, propanone, to a secondary alcohol, propan-2-ol. The reducing agent is sodium borohydride followed by the addition of an acidic solution. As with the reduction of aldehydes, it is also possible to use lithium aluminium hydride as the reducing agent.

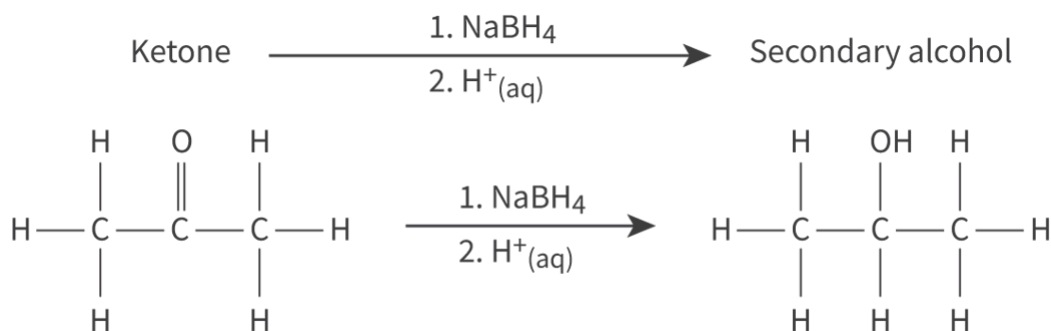


Figure 3. The reduction of the ketone propanone to the secondary alcohol propan-2-ol.

 More information for figure 3

The image is a chemical diagram depicting the reduction of propanone (a ketone) to propan-2-ol (a secondary alcohol). The diagram contains two chemical structures and a reaction arrow. On the left, the structure of propanone is shown, with a central carbon double-bonded to an oxygen and single-bonded to two methyl groups, forming the ketone group. An arrow labeled "1. NaBH₄ 2. H⁺(aq)" indicates the conversion process using sodium borohydride followed by an acid solution, leading to the structure on the right. The right structure represents propan-2-ol, where the central carbon is single-bonded to an -OH group and two methyl groups, indicating the transformation to an alcohol.

[Generated by AI]

Figure 4 shows the reduction of propanoic acid to the primary alcohol propan-1-ol.

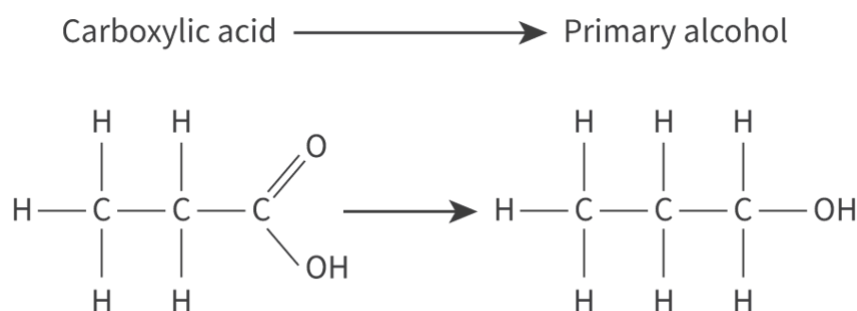


Figure 4. The reduction of propanoic acid to the primary alcohol propan-1-ol.

 More information for figure 4

The image is a chemical diagram depicting the reduction reaction of propanoic acid to propan-1-ol. On the left, there's a structural formula for propanoic acid, which includes three carbon atoms. The first two carbon atoms are each bonded to three hydrogen atoms. The third carbon atom is double-bonded to an oxygen atom and single-bonded to a hydroxyl group (-OH), indicating it is a carboxylic acid. An arrow points to the right, labeled from carboxylic acid to primary alcohol, indicating the reduction process.

On the right side, there's the structural formula for the primary alcohol propan-1-ol. It consists of three carbon atoms, with the first two carbon atoms each bonded to three hydrogen atoms. The third carbon atom is bonded to two hydrogen atoms and one hydroxyl group (-OH), showing that the acid has been reduced to an alcohol with the loss of one oxygen atom and gain of two hydrogen atoms.

[Generated by AI]

Worked example 1

Deduce the final product in the reduction of methanal.

Methanal is an aldehyde and therefore it will produce a primary alcohol through reduction so the final product will be methanol.

Worked example 2

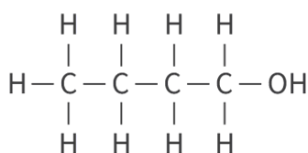
Deduce the final product in the reduction of hexanoic acid.

Hexanoic acid is a carboxylic acid and therefore it will produce a primary alcohol through reduction so the final product will be hexan-1-ol.

Worked example 3

Draw and name the final product in the reduction of butanal.

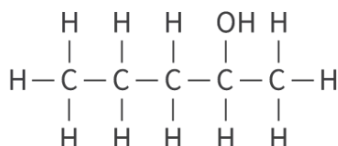
Butanal is an aldehyde and therefore it will produce a primary alcohol through reduction so the final product will be butan-1-ol.



Worked example 4

Draw and name the final product in the reduction of pentan-2-one.

Pentan-2-one is a ketone and therefore it will produce a secondary alcohol through reduction so the final product will be pentan-2-ol.



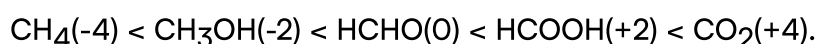
Making connections

How can oxidation states be used to show that the following molecules are given in increasing order of oxidation: CH_4 , CH_3OH , HCHO , HCOOH , CO_2 ? (see subtopic S3.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>)).

Oxidation state is a measure of the degree of oxidation of an atom in a molecule. Knowing that each molecule has an overall oxidation state of zero, we can assign the oxidation states of each atom in the given molecules:

- CH_4 : Carbon has an oxidation state of -4 as each hydrogen has an oxidation state of +1.
- CH_3OH : Carbon has an oxidation state of -2, as the oxygen has an oxidation state of -2, and each hydrogen has an oxidation state of +1.
- HCHO : Carbon has an oxidation state of 0, as the oxygen has an oxidation state of -2, and each hydrogen has an oxidation state of +1.
- HCOOH : Carbon has an oxidation state of +2, as each oxygen has an oxidation state of -2, and each hydrogen has an oxidation state of +1.
- CO_2 : Carbon has an oxidation state of +4 and each oxygen has an oxidation state of -2.

From these oxidation state assignments, we can see that the given molecules are ordered from increasing oxidation state as:



Activity

- **IB learner profile attribute:** Thinkers
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Pair/group activity

Research examples of pharmaceuticals that are synthesised using reduction reactions, such as ibuprofen and naproxen. Research about the synthesis of these drugs and focus on the role of reduction reactions in their synthesis.

Share your findings with a partner and reflect on what you have learned.

Reduction of unsaturated compounds

Learning outcomes

By the end of this section you should be able to deduce the products of the reactions of hydrogen with alkenes and alkynes.

Having a balanced diet is essential for good health. Humans need a diet filled with the main groups of biomolecules (carbohydrates, lipids, and proteins), and essential nutrients, vitamins and minerals. Nowadays, there are a wide variety of oils and fats collected or produced for human consumption. For a few decades in some countries, margarine has been used as a mainstream alternative to butter, as it is low cost and is marketed as a healthier fat and a vegetarian/vegan option or suitable for those with dairy allergies.

Margarine is a solid fat produced industrially from vegetable oils that are liquid at room temperature, however, margarine is solid. What distinguishes margarine from other plant oils and what allows it to be solid at room temperature?



Figure 1. Margarine is a solid substance at room temperature.

Credit: Science Photo Library, Getty Images

Reduction of unsaturated compounds

Recall the structure of unsaturated compounds (see subtopic S3.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/>)). Unsaturated compounds are hydrocarbons that possess at least one double or triple carbon-carbon bond in their main chain. Compounds with double bonds are called alkenes and compounds with triple bonds are called alkynes.

To identify an unsaturated compound, an unsaturation test can be performed that consists of adding bromine water and checking for a colour change. This reaction is classified as a halogenation, as the double bond in an alkene (or the triple bond in an alkyne) will react with the bromine, $\text{Br}_2(\text{l})$, and immediately change its colour from brown to colourless. When bromine is mixed with an alkane, there is no colour change.

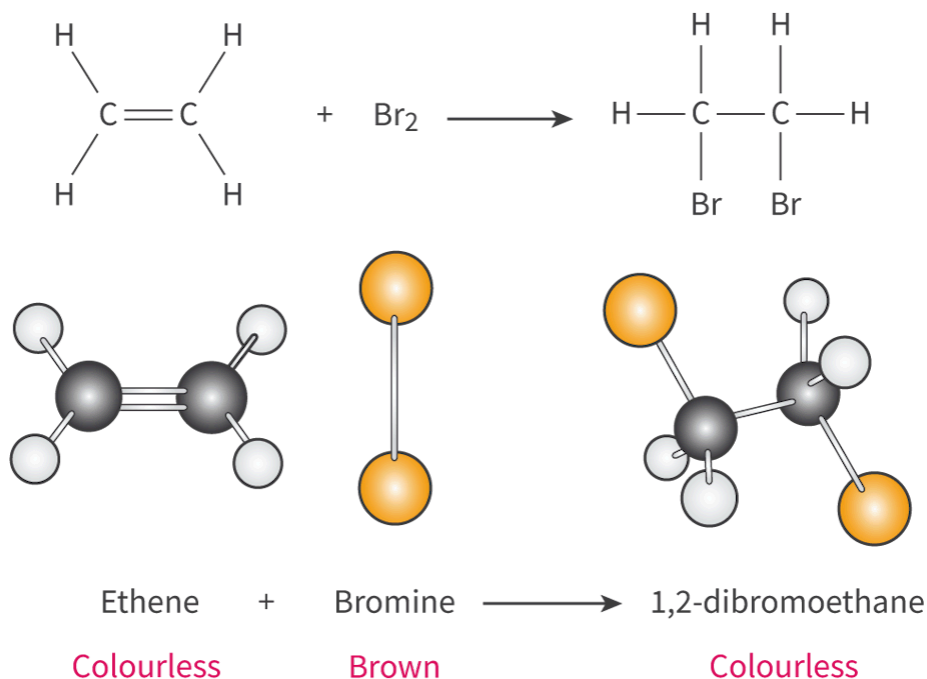


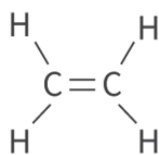
Figure 2. The halogenation of ethene.

 More information for figure 2

The image is a chemical diagram illustrating the halogenation reaction of ethene with bromine to form 1,2-dibromoethane. At the top, there is a balanced chemical equation: $(\text{C}_2\text{H}_4 + \text{Br}_2 \rightarrow \text{C}_2\text{H}_4\text{Br}_2)$. Below the equation, there is a structural representation of ethene on the left, depicted as a double-bonded carbon chain with hydrogen atoms attached, labeled "Colourless" underneath. In the center, there is a model of bromine as a diatomic molecule represented by two orange spheres, labeled "Brown." On the right, there is a newly formed molecule of 1,2-dibromoethane shown with the addition of two bromine atoms to the original ethene structure, labeled "Colourless." These visual representations are connected by an arrow, indicating the chemical reaction.

[Generated by AI]

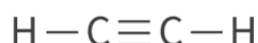
Look at the structure of the alkenes and alkynes in terms of potential oxidation. Which reactions are possible?



 More information

The image is a structural diagram of an ethylene molecule. It consists of two carbon (C) atoms connected by a double bond, represented by two parallel lines between the carbon atoms. Each carbon atom is also bonded to two hydrogen (H) atoms. The hydrogen atoms are positioned in a manner that creates a linear and symmetrical arrangement around the carbon atoms, forming a planar geometry. The bonds between carbon and hydrogen are represented by lines extending outward from each carbon atom towards the hydrogen atoms, creating a structural formula C_2H_4 .

[Generated by AI]



 More information

This image shows the structural formula of Ethyne (alkyne), also known as acetylene. It consists of two carbon atoms (C) connected by a triple bond, represented by three parallel lines. Each carbon atom is additionally bonded to a hydrogen atom (H) on the outer sides with single bonds. The overall structure is linear, with $H-C \equiv C-H$, indicating a bond between hydrogen and carbon atoms on both ends of the carbon triple bond.

[Generated by AI]

Ethene
(alkene)

Ethyne
(alkyne)

Figure 3. Alkenes and alkynes.

You might have thought that both alkenes and alkynes can be easily reduced by the addition of hydrogen atoms, when the double or triple bonds are broken. The process of adding hydrogens is called hydrogenation. This reaction decreases the degree of unsaturation of a compound. The degree of unsaturation is a measurement of the number of molecules of hydrogen (H_2) needed to convert the organic compound to the corresponding saturated, non-cyclic molecule.

In the hydrogenation of an alkene, gaseous hydrogen (H_2) is added across the carbon-carbon double bond. The product of the reaction is an alkane, a saturated hydrocarbon. Note that the conditions for this reaction are high pressure and temperature with a nickel (Ni) catalyst.

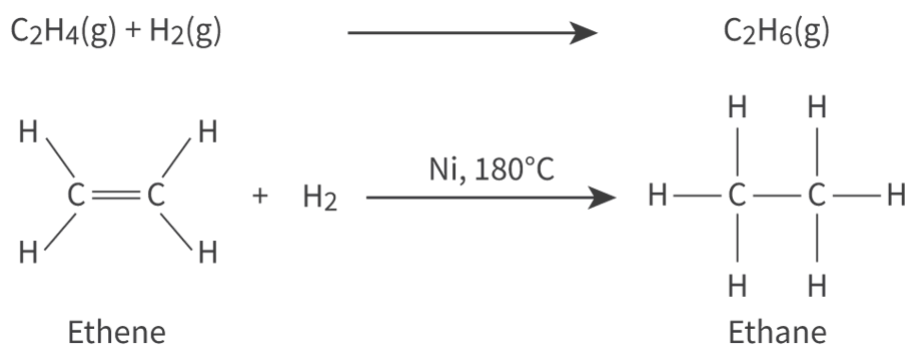


Figure 4. The hydrogenation of ethene.

 More information for figure 4

The image depicts a chemical equation representing the hydrogenation of ethene (C_2H_4) to ethane (C_2H_6). On the left side of the equation, the structure of ethene is shown with two carbon atoms double-bonded to each other and also bonded to two hydrogen atoms each. Gaseous hydrogen (H_2) is added to this structure. The direction of the reaction is indicated by an arrow pointing to the right. Above the arrow, it mentions that a nickel catalyst and a temperature of 180°C are required for this reaction. On the right side, the structure of ethane is displayed, consisting of two carbon atoms single-bonded to each other and each carbon atom bonded to three hydrogen atoms. This transformation results in a saturated hydrocarbon, as indicated by the change from the double bond in ethene to a single bond in ethane.

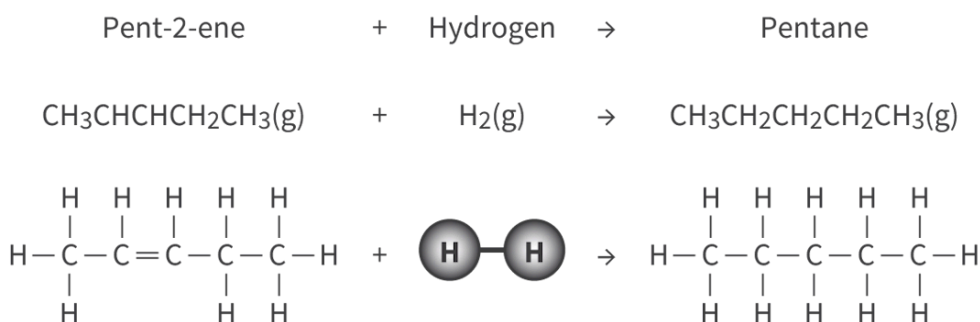
[Generated by AI]

Unsaturated compounds are bent due to the double and triple bonds in their chains. They are not easily stacked up when close to each other, which means they have reduced London forces when compared to the saturated hydrocarbon. Weaker London forces means a lower melting and boiling point. Vegetable oils, being composed of unsaturated lipids, have lower melting points and are, therefore, liquid at room temperature. In the production of margarine, the unsaturated compounds undergo a reduction reaction and become saturated, which means they will no longer be bent. Becoming straight chained, the compounds have stronger London forces between molecules, which leads to a higher melting point. This allows margarine to be solid at room temperature.

Worked example 1

Draw the overall reaction and name the product of the hydrogenation of pent-2-ene.

In the hydrogenation of pent-2-ene, the double bond is broken and each hydrogen takes one of the unpaired electrons from each of the carbons 2 and carbon 3 atoms, to produce pentane.



The hydrogenation of alkynes involves the addition of hydrogen gas (H_2) to a carbon-carbon triple bond, resulting in the formation of a carbon-carbon double bond or a single bond, depending on the reaction conditions.

For example, in the hydrogenation of ethyne, ethyne is converted into ethene by the addition of hydrogen gas. The reaction is typically carried out in the presence of a metal catalyst, such as palladium, platinum, or nickel.

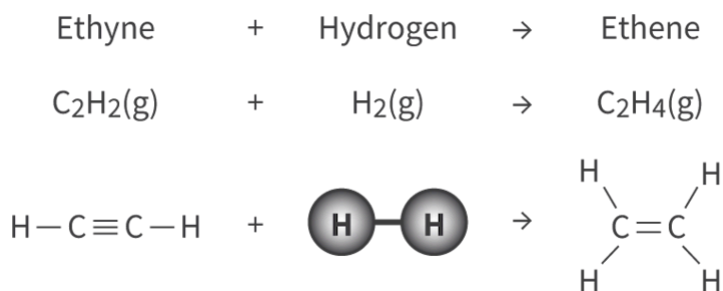


Figure 5. The hydrogenation of ethyne to ethene.

 More information for figure 5

The image is a diagram illustrating the chemical process of the hydrogenation of ethyne into ethene. At the top, it shows the words: "Ethyne + Hydrogen → Ethene." Below it, the chemical formulas are displayed: " $\text{C}_2\text{H}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{C}_2\text{H}_4(\text{g})$." The image visually represents the structural formulas of the molecules involved. It features a linear ethyne molecule with a triple bond between carbon atoms, represented as " $\text{H}-\text{C}\equiv\text{C}-\text{H}$." This molecule is on the left side of the equation. It then shows a diatomic hydrogen molecule (H_2), represented as two connected spheres, indicating a single bond, located in the middle of the equation. On the right side, the ethene molecule is shown, with a double bond between the carbon atoms: " $\text{C}=\text{C}$," with two hydrogen atoms bonded to each carbon atom, resulting in the formula " $\text{H}_2\text{C}=\text{CH}_2$." This structural transformation visually emphasizes the addition of hydrogen to the ethyne molecule to form ethene, illustrating the change from a triple to a double bond between carbon atoms.

[Generated by AI]

Alkynes can also be reduced to alkanes. In **Figure 6**, two molecules of hydrogen gas (H_2) are added to one molecule of ethyne (C_2H_2) to form one molecule of ethane (C_2H_6). The reaction is typically carried out in the presence of a metal catalyst, such as palladium or platinum, at high temperature and pressure.

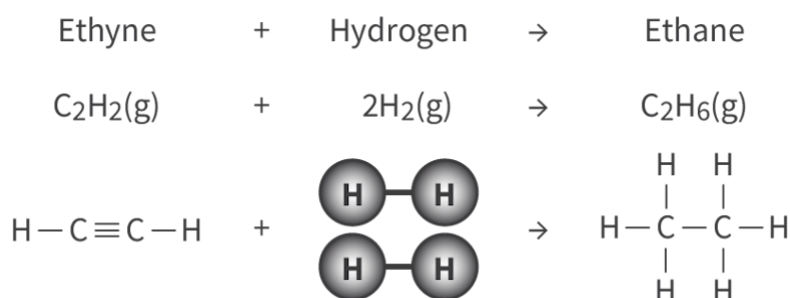


Figure 6. The hydrogenation of ethyne to ethane.

 More information for figure 6

The image is a chemical reaction diagram illustrating the hydrogenation of ethyne into ethane. It shows the chemical equation as follows: Ethyne (C_2H_2) plus two Hydrogen (H_2) molecules yield Ethane (C_2H_6).

On the left, the diagram displays a structural formula for ethyne, written as $\text{H}-\text{C}\equiv\text{C}-\text{H}$, indicating a triple bond between the carbon atoms. Next to it are two separate H_2 molecules represented by two pairs of ball-and-stick models, where each pair of balls represents a hydrogen molecule.

An arrow points to the products on the right side of the diagram, symbolizing the chemical reaction. Here, the ethane molecule is displayed as a structural formula: H-C-C-H, with single hydrogen atoms bonded to each carbon, forming a complete ethane structure with a single bond between the carbon atoms.

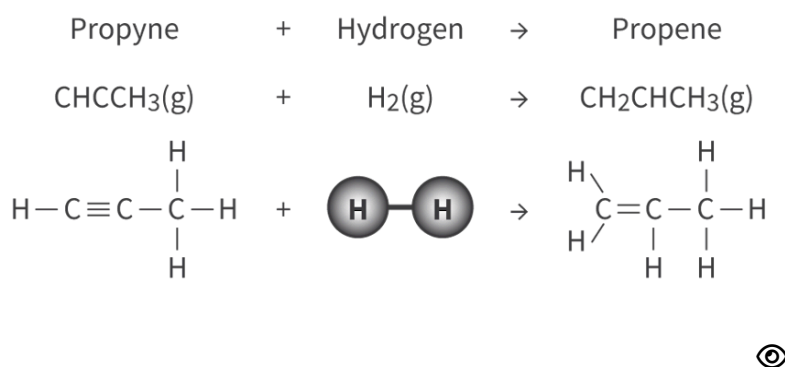
The diagram visually depicts the transformation involving the breaking of the triple bond and the formation of a single bond with the addition of hydrogen atoms, showing the transition from an alkyne to an alkane.

[Generated by AI]

Worked example 2

Draw the overall reaction and name the product of the hydrogenation of propyne (note: the final product has a C=C bond).

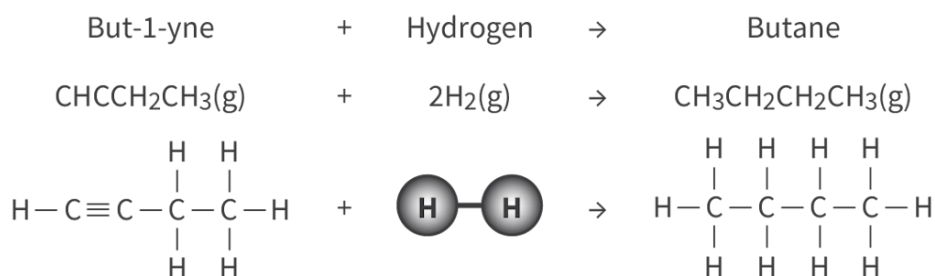
In the hydrogenation of propyne, the triple bond is broken and each hydrogen takes one of the unpaired electrons of carbons 1 and 2, producing propene. A double bond is still left.



Worked example 3

Draw the overall reaction and name the product of the hydrogenation of but-1-yne (note: the final product has only C-C bonds).

In the hydrogenation of but-1-yne, the triple bond is broken and each hydrogen takes one of the unpaired electrons of the carbon 1 and carbon 2 atoms, to produce butane.



Making connections

Why are some reactions of alkenes classified as reduction reactions while others are classified as electrophilic addition reactions? (see [subtopic R3.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43118/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43118/>))

The distinction between a reduction and an electrophilic addition reaction of alkenes depends on the molecule reacting with the alkene and the reaction mechanism.

In a reduction reaction, like the ones just studied, the alkene reacts with a molecule of hydrogen that can donate hydrogens (reduction), resulting in the decrease of the oxidation state of the carbons involved with the double bond, and the production of an alkane.

In an electrophilic addition (see [section R3.4.4—5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-addition-reaction-id-46329/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-addition-reaction-id-46329/>)), an electrophile forms a bond to the nucleophile (in this case the C=C) by accepting both bonding electrons from the carbon in the C=C. Common electrophiles include hydrogen halides (H—X), halogens (X₂), and polar molecules such as water (H₂O) and alcohols (—OH). Electrophilic addition reactions result in the formation of a single bond between the two carbon atoms and a new functional group on the molecule.

Activity

- **IB learner profile attribute:** Thinkers
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual/pair activity

Reduction of unsaturated compounds by hydrogenation

Materials

Paper, pencil, molecular model kit (optional)

Your task

1. Review the definition and examples of unsaturated compounds and degree of unsaturation. Recall that unsaturated compounds contain double or triple bonds between carbon atoms, and that the degree of unsaturation is the measurement of the number of hydrogen molecules (H_2) needed to convert the organic compound to the corresponding saturated, non-cyclic molecule.
2. On a piece of paper, draw the structures of three unsaturated compounds with different degrees of unsaturation: ethene (C_2H_4), propyne (C_3H_4) and cyclohexene (C_6H_{10}).
3. For each compound, write the balanced equation for its hydrogenation reaction. Draw the structure of the product.
4. Compare the degrees of unsaturation of the reactants and products. How does hydrogenation affect the degree of unsaturation? Explain your answer using chemical concepts and terminology.
5. Choose one of the compounds and research its uses and properties. How would hydrogenation change its uses and properties?
6. Write a paragraph summarising your findings and citing your sources.
7. Share your paragraph with a classmate and discuss your findings.

Standard electrode potentials (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to interpret standard electrode potential data in terms of ease of oxidation/reduction.

In your studies so far, you have been studying the reduction and oxidation of different species in a qualitative manner. You previously studied the ease with which some metals are oxidised compared to others and organised this information into a reactivity series (see [section R3.2.1–2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/>)). Though this allows you to compare the reactivity of one metal to another, it is important to be able to quantify how much more reactive one element is than another in a redox reaction.

Take for example a common non-rechargeable battery (see [section R3.2.3–4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ease-of-oxidation-and-reduction-id-46488/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ease-of-oxidation-and-reduction-id-46488/>)). These types of batteries are composed of a variety of different metals including cadmium, lead, zinc, manganese, nickel, silver, mercury and lithium. Why are these metals chosen to be used in batteries? How can you quantify the amount of electricity that could be generated when certain metals are combined in a voltaic cell?

In this section you will study the concept of standard electrode potentials and use these to determine the standard cell potential for a voltaic reaction such as those that occur in many primary cells.

Using the hydrogen half-cell potential (SHE)

In any situation where a measurement is recorded you must have an appropriate measuring device with an appropriate unit of measurement assigned to it.

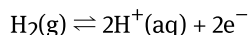
For example, temperature can be measured using various techniques and units. One scale commonly used is Celsius or centigrade which is based around the temperature at which water solidifies and evaporates. These two reference points were arbitrarily set at 0.0 °C and 100.0 °C. This notion of setting arbitrary standard values as reference points is one that is essential to measurements in electrochemistry.

In order to quantify whether one element will more readily undergo reduction or oxidation compared to another element, we need a standard value around which to base our measurements. This comes in the form of the standard hydrogen electrode (SHE).

The standard hydrogen electrode is made of a platinum electrode submerged in a solution of 1.00 mol dm⁻³ H⁺(aq) ions ([H⁺(aq)] = 1.00 mol dm⁻³). Hydrogen gas at 100 kPa is then bubbled over the platinum electrode and a redox reaction occurs. Platinum is used as it is inert and will not preferentially undergo reduction over hydrogen gas.

In the reaction, hydrogen ions are reduced to form hydrogen gas and hydrogen gas is oxidised to produce hydrogen ions in solution. When we combine the two half-reactions we get a net reaction that appears to be reversible. This is one of the reasons that hydrogen was chosen as the standard upon which to base all other values.

The hydrogen half-cell



is assigned an arbitrary standard electrode potential ($E^\ominus$) of zero volts (0V) by convention.

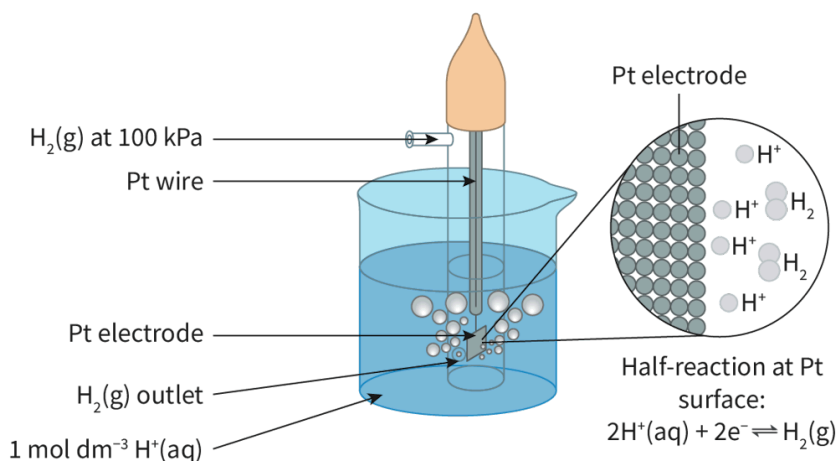


Figure 1. A standard hydrogen electrode.

 More information for figure 1

The diagram depicts a standard hydrogen electrode. A platinum electrode is submerged in an aqueous solution containing $1 \text{ mol dm}^{-3} \text{H}^+$ ions. Hydrogen gas (H_2) at 100 kPa is bubbled over the electrode. The diagram shows the gas entering through an outlet and a platinum wire connecting to the electrode. On the right, an enlarged section illustrates the half-reaction at the platinum surface: $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g})$. The components are labeled, including Pt electrode and Pt wire.

[Generated by AI]

Theory of Knowledge

A platinum electrode is a key requirement of the standard hydrogen electrode. Platinum is an incredibly hard and expensive metal with a high melting point (1775°C). Ancient societies in present-day Colombia and Ecuador worked with platinum to make jewellery, masks and vessels, yet platinum was largely unknown in Europe until the mid-sixteenth century. These ancient metallurgists were capable of making objects and purifying this unreactive metal without a furnace capable of reaching the melting point of platinum. To what extent does creativity and persistence play in the pursuit of knowledge?

So now that you are aware of this arbitrarily assigned standard value around which all standard electrode potentials ($E^\ominus$) are derived, how can you use this information?

As you have seen previously electrochemical reactions can be described in terms of their half-reactions (see [section R3.2.1–2](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/>)). These half-reactions are each associated standard electrode potentials ($E^\ominus$), measured in volts (V), at standard ambient temperature and pressure (SATP). It is not possible to obtain the standard electrode potential of a single half-cell reaction as voltages are measurements of potential difference between two locations, the anode and the cathode. Because of this issue, the SHE is used in conjunction with a metal electrode to determine the relative reducing power of different half-cells compared to hydrogen. Depending on the reducing power of the metal electrode, the SHE is then forced to act as either a cathode or an anode when connected to the other half-cell.

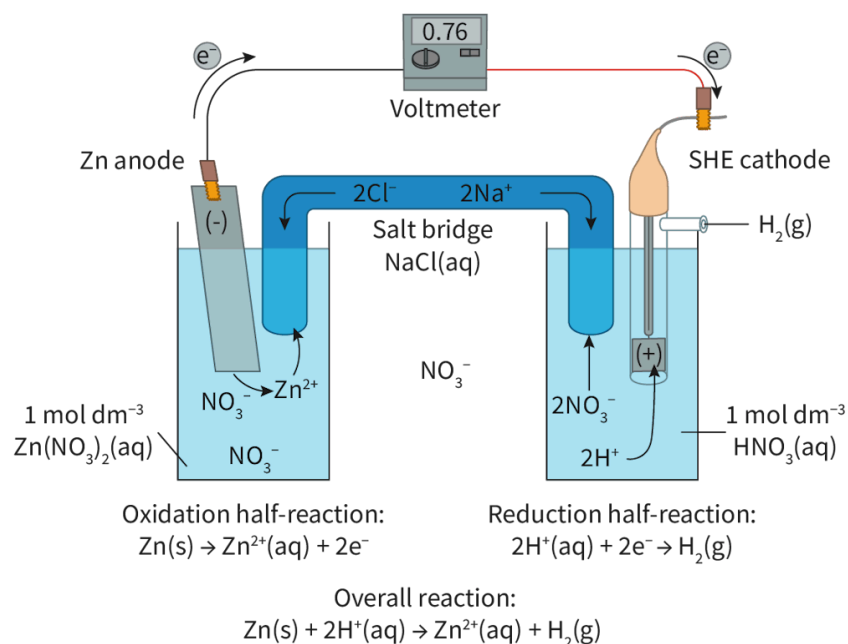


Figure 2. Determining the standard electrode potential for zinc using the SHE.

 More information for figure 2

The image is a diagram of an electrochemical cell setup to measure the standard electrode potential for zinc using the Standard Hydrogen Electrode (SHE) as the reference. On the left, a container holds the zinc anode, represented as Zn(s) , immersed in a zinc nitrate solution ($1 \text{ mol dm}^{-3} \text{ Zn(NO}_3)_2(\text{aq})$). At the anode, zinc undergoes oxidation according to the half-reaction: $\text{Zn(s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$. On the right is another container housing the SHE cathode where the reduction half-reaction occurs: $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$.

A voltmeter connects the two half-cells indicating the electric potential, with a salt bridge containing NaCl solution (2Cl^- and 2Na^+) linking them to allow for the movement of ions and complete the circuit. Arrows indicate electron flow from the zinc anode to the SHE cathode, and the setup is designed to measure the potential difference when zinc is oxidized, making the zinc half-cell act as the anode while the SHE half-cell acts as the cathode.

The overall electrochemical reaction can be summarized as: $\text{Zn(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$.

[Generated by AI]

This allows us to determine the standard electrode potentials for various species and rank these species against one another in terms of how easily they are oxidised or reduced. As you will notice in **Table 1**, these reactions are listed as reduction half-reactions of metal ions forming a reduced metal product.

Table 1. Examples of various standard electrode potentials.

Half-reaction	$E^\ominus$
$\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}$	-0.23 V
$\text{Cu}^+ + \text{e}^- \rightarrow \text{Cu}$	$+0.52\text{ V}$
$\frac{1}{2}\text{Cl}_2 + \text{e}^- \rightarrow \text{Cl}^-$	$+1.36\text{ V}$

These values, when ranked in order from most negative to most positive, produce a reactivity series, which can be found in the DP chemistry data booklet. The values are always given as the reagent species being reduced within the reduction half-reaction. The number of moles of electrons and reactants and products need not be equal to zero.

Species that have a more negative standard electrode potential will be more easily oxidised. This also means that they will be stronger reducing agents, and that they will be placed higher in the reactivity series.

Species with a more positive standard electrode potential will be more easily reduced. They will be stronger oxidising agents and will also be ranked lower in the reactivity series.

Lithium has the most negative standard electrode potential of all metals, at -3.03 volts, compared to gold which has one of the most positive values at $+1.50$ volts.

Use **Interactive 1** to rank the metals in order of how easily they are oxidised.

Easily oxidised



Easily reduced

☒ Check**Interactive 1. Ranking metals according to ease of oxidation.****Making connections**

You previously studied the concept of ionisation energy (see [section R3.1.2—3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/conjugate-acids-bases-id-46053/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/conjugate-acids-bases-id-46053/>) and variable oxidation states of transition metals (see [section R3.1.9—10](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ion-dissociation-constants-of-weak-acids-and-bases-hl-id-46058/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ion-dissociation-constants-of-weak-acids-and-bases-hl-id-46058/>)). Recall that transition metals have variable oxidation states due to the fact that their ionisation energies are close in magnitude. It is interesting to note that there is some correlation between the sign of the standard electrode potential of a certain species and its ionisation energy or energies.

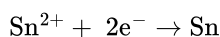
Note that lithium has the lowest first ionisation energy of all metal species and also has the most negative standard electrode potential!

Standard electrode potentials, like changes in enthalpy and equilibrium constants are examples of state functions. This means that if you reverse a particular half-cell reaction then the standard electrode potential for that reaction will be reversed in sign, but its magnitude will stay the same. Though individual half-reactions are reversible, the direction of the overall net reaction for an electrochemical cell depends on what specific combination of electrodes and solutions are being used to construct the cell. The setup of the cell determines the direction in which the reaction will proceed, and from this setup one can deduce which species are undergoing reduction and oxidation.

Study skills

Standard electrode potentials can sometimes seem confusing. It is important to remember that these potentials are reduction potentials for reduction half- reactions that produce a product with a more negative oxidation state.

When tin (Sn) is said to have a standard electrode potential value of -0.14 V this is for the half-reaction:

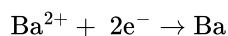


not for tin itself being reduced.

Worked example 1

Barium has an $E^{\ominus} = -2.90\text{ V}$.

Give the half-cell reaction that has this value of standard electrode potential.



where:

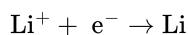
$$E^{\ominus} = -2.90\text{ V}$$

The most commonly occurring cation of barium is Ba^{2+} . Therefore, the half-reaction that forms barium from those barium ions and two electrons has a standard electrode potential of -2.90 V .

Worked example 2

Lithium metal easily undergoes oxidation when compared to all other metals.

Give the standard electrode half-cell reaction for lithium and predict the sign for its standard electrode potential.



where:

$$E^{\ominus} = \text{a large negative value}$$

Standard electrode potentials are always listed as reduction half-reactions, and so the half-reaction for lithium involves a lithium cation gaining one electron to form a lithium atom.

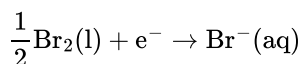
Since lithium is more easily oxidised than all other metals, you can deduce that it must have an value that is large and negative, placing it at the very top of the reactivity series.

Worked example 3

Bromine can be reduced to form bromide ions in solution.

Give the half-reaction for the reduction of bromine to produce one mole of bromide ions.

Predict the sign for the value for the half-reaction.



where:

$E^\ominus$ = a large positive value

In this case the bromide ion will have a -1 oxidation state and be aqueous on the products side of the equation. The ion is formed from bromine gas, which is a diatomic molecule in liquid form at SATP, and has an oxidation state of 0. To balance the reaction to produce one mole of bromide ions, there must only be a half mole of liquid bromine as the reactant gains one mole of electrons.

This reaction will have a large positive standard electrode potential.

Bromine is easily reduced to form bromide ions (recall that bromine has a high electron affinity), which means it will be found towards the bottom of the reactivity series and have a positive value.

Now that you have a better understanding of standard electrode potentials, let's apply these new concepts.

Activity

- **IB learner profile attribute:** Inquirers
- **Approaches to learning:** Thinking skills — Designing procedures and models, Experimenting with new strategies for learning
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Individual/pair activity

Using the simulator, you (and a partner if working in pairs) will set up an electrochemical reaction using two half cells. Once you have constructed the appropriate voltaic cells you will record the necessary qualitative and quantitative data so as to answer a series of questions related to standard electrode potentials.

Your task

1. Access the simulator via the following link:
<https://teachchemistry.org/classroom-resources/voltaic-cells> 
(<https://teachchemistry.org/classroom-resources/voltaic-cells>)
2. Under the heading Galvanic/Voltaic Cells click on Start →
3. Create a data table in Excel or a digital notebook, consisting of a column with the heading 'Metal Electrodes' and containing four different metal electrodes (Ag, Cu, Zn and Mg), a second column with the heading 'Voltage' and a third column with the heading 'Molecular Half-reaction'.
4. Using the simulator, construct a voltaic cell consisting of a hydrogen electrode in HCl in the left beaker and a silver electrode in $\text{AgNO}_3(\text{aq})$ in the right beaker.

5. Start the simulation by clicking the button on the voltmeter from off to on.
6. Record any quantitative data (including measurement uncertainties).
7. Click on the 'See Molecular Scale' button and then record the half-reaction that is taking place at the surface of the silver electrode.
8. Repeat the experiment three more times using the Cu, Zn and Mg electrodes and keeping the hydrogen electrode constant in the left-hand beaker.

Use the data recorded during the activity to answer these questions.

1. Explain how it is possible to determine the standard electrode potentials ($E^\ominus$) for each metal in this experiment, using the data you (and your partner) collected.
2. Describe the percent uncertainty of your results and describe how the percent uncertainty might change if you were to conduct this experiment in the laboratory and not using a simulator.
3. List the half-cell reactions occurring at the metal electrode in order from most easily oxidised to most easily reduced.
4. Hypothesise as to what effect reversing the beakers for the hydrogen and metal electrodes would have on the standard electrode potential's magnitude and/or sign.
5. Compare and contrast your results with other members of your class.

Gibbs free energy and standard electrode potentials (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R3.2.13) Predict whether a reaction is spontaneous in the forward or reverse direction from $E^\ominus$ data.
- (R3.2.14) Determine the value for $\Delta G^\ominus$ from $E^\ominus$ data.

Calculations involving standard electrode potentials

Now that you have a better understanding of how we determine the standard electrode potential for a certain half-cell reaction and the placement of a species within the reactivity series, you can take this information even further and extend it to more complex voltaic and electrolytic cells.

Take for example primary and secondary batteries (see [section R3.2.6–7](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/voltaic-cells-and-rechargeable-batteries-id-46490/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/voltaic-cells-and-rechargeable-batteries-id-46490/>)). These types of batteries, often referred to simply as disposable and rechargeable, must provide a consistent output of electricity to operate a device. The amount of electrical charge generated from a fresh disposable battery differs from the energy produced by a depleted rechargeable battery. But why are some batteries rechargeable, and others not? What determines the voltage produced by a particular battery, and how does the chemical composition of the battery affect this value?

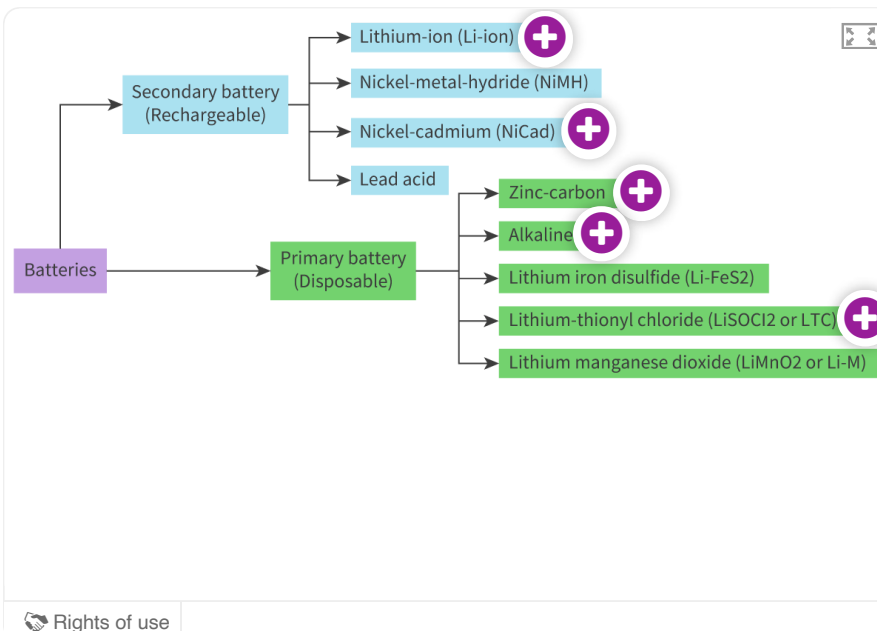
International Mindedness

Lithium ion batteries are extraordinarily useful in electrically powered vehicles (EVs) as lithium is the most easily oxidised species of metal.

EVs are essential to modernising the automotive industry as the global community moves to phase out classic combustion engine vehicles in order to reduce greenhouse gas emissions.

But where does this lithium come from and what impact does the mining of this metal have on the global ecosystem? Can the global demand for lithium be met as we move towards having only EVs on our roads?

Some examples are given in the flow chart in **Interactive 1**.



Interactive 1. Examples of Primary and Secondary Batteries.

More information for interactive 1

An interactive flowchart illustrates two major types of batteries: primary (disposable) and secondary (rechargeable), along with several examples for each category. Some examples feature a plus sign, representing an interactive hotspot. Selecting the hotspot reveals photographs of the respective battery types.

Secondary Batteries (Rechargeable): These batteries are rechargeable and can be used multiple times. A reversible chemical reaction allows electrons to be restored to the battery through charging.

Examples of secondary batteries shown in the flowchart are as follows:

1. Lithium-ion (Li-ion) with a hotspot: Photo of a rectangular lithium-ion battery labeled lithium-ion battery 3.7 V, 3000 mAh. The battery also shows a recycling symbol with a crossed-out trash can, which means it should not be thrown in the regular trash can.
2. Nickel-metal hydride (NiMH)
3. Nickel-cadmium (NiCad) with a hotspot: Photo of a cylindrical rechargeable battery labeled Nickel cadmium made by Berec. The full text on the battery reads: Rechargeable sealed cell; 1.2v, 2.0Ah; charge at 250 mA for 12 hours or as advised; NCC 200 C Size; KAH 27/50.
4. Lead acid

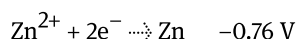
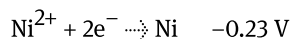
Primary Batteries (Disposable): These batteries are designed for one-time use. Once the chemicals are spent, they cannot be recharged.

Examples of primary batteries shown in the flowchart are as follows:

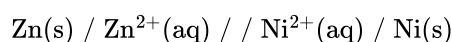
1. Zinc-carbon with a hotspot: Photo of an AA-size cylindrical battery labeled KODAK Super Heavy Duty ZINC AA R6P 1.5V.
2. Alkaline with a hotspot: Photo of a cylindrical Kodak Xtralife AAA alkaline battery labeled, Alkaline Battery AAA LR03 1.5V.
3. Lithium iron disulphide (Li-FeS₂)
4. Lithium thionyl chloride (LiSOCl₂ or LTC) with a hotspot: Photo of a rectangular Sony Lithium Thionyl Chloride Battery with wires attached.
5. Lithium manganese dioxide (LiMnO₂ or Li-M)

This flowchart supports the understanding of electrochemical cells by showing real-world applications of electrode potentials and redox reactions. By analyzing the chemical compositions and structures of these batteries, viewers can relate the theoretical concepts of voltage and energy capacity to actual battery performance.

If you take the standard electrode potentials for nickel and zinc it is clear that if the two half-reactions were to be combined to form a voltaic cell, then zinc would be preferentially oxidised and nickel would be reduced:



Recall the concept of electrochemical cell notation (cell diagrams) (see [section R3.2.3–4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ease-of-oxidation-and-reduction-id-46488/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ease-of-oxidation-and-reduction-id-46488/>)). In this example, we would write the cell diagram as:



since the zinc is being oxidised (at the anode) and nickel is being reduced (at the cathode).

It is possible to then determine the overall standard cell potential for this reaction, by subtracting the standard electrode potential of the anode (oxidised species) from the standard electrode potential for the cathode (reduced species).

$$E_{\text{cell}}^{\ominus} = E^{\ominus}(\text{cathode or reduced species}) - E^{\ominus}(\text{anode or oxidised species})$$

For this particular example involving zinc and nickel, the calculation would be.

$$\begin{aligned} E_{\text{cell}}^{\ominus} &= E^{\ominus}(\text{nickel}) - E^{\ominus}(\text{zinc}) \\ &= -0.23 - (-0.76) \\ &= +0.53 \text{ V} \end{aligned}$$

The standard cell potentials do not need to be multiplied by any coefficient even if coefficients are needed to balance the chemical reaction. As well, by convention the standard cell potentials are always given as reduction potentials and so the values are never inverted within the calculation even if a species is undergoing oxidation.

Positive standard cell potentials are associated with spontaneous reactions, whereas negative cell potentials are associated with non-spontaneous reactions. Non-spontaneous electrochemical reactions include such processes as the electrolysis of water and electroplating, which you will study in more detail in [sections R3.2.15](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrolysis-of-water-and-aqueous-solutions-hl-id-46495/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrolysis-of-water-and-aqueous-solutions-hl-id-46495/>) and [R3.2.16](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electroplating-hl-id-46496/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electroplating-hl-id-46496/>).

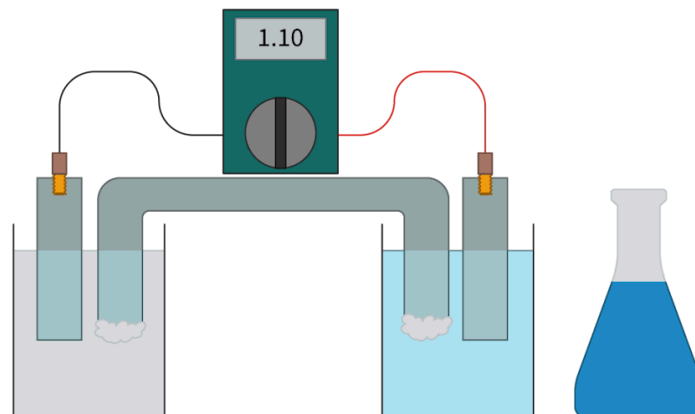


Figure 1. The experimental setup for a voltaic cell involving copper and zinc electrodes in solutions of copper sulfate and zinc sulfate.

 More information for figure 1

The image shows an experimental setup for a voltaic cell. On the left, a beaker contains a copper electrode immersed in a solution, likely copper sulfate, indicated by a light blue color. On the right, another beaker holds a zinc electrode in a light blue solution, likely zinc sulfate. These two beakers are connected by a U-shaped tube, suggesting a salt bridge, allowing ions to flow between the two solutions to maintain the charge balance. A voltmeter, placed at the center top, displays a reading of 1.10 volts, measuring the potential difference between the two electrodes. Wires connect the electrodes to the voltmeter, with the left side having a black wire and the right side a red wire, indicating polarity. A partially filled blue laboratory flask is placed next to the zinc beaker, indicating the presence of a chemical reagent possibly involved in the reaction.

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Worked example 1

The redox reaction between zinc and copper in solutions of zinc sulfate and copper(II) sulfate produces a spontaneous electrochemical reaction.

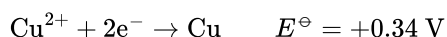
Using the standard electrode potentials given in [section 19](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/standard-reduction-potentials-at-29815-k-id-45121/>) of the DP chemistry data booklet, justify why this reaction is classified as spontaneous based on its overall standard cell potential.

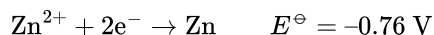
$$\begin{aligned} E_{\text{cell}}^{\ominus} &= E^{\ominus} \text{ Cu (reduced species)} - E^{\ominus} \text{ Zn (oxidised species)} \\ &= +0.34 - (-0.76) \\ &= +1.10 \text{ V} \end{aligned}$$

As the $E_{\text{cell}}^{\ominus}$ is positive, the overall voltaic reaction is defined as being spontaneous.

In order to complete this question you must understand that since copper is placed in a solution of copper(II) sulfate the half-cell reaction for copper is:



The half-reaction for zinc is:



Therefore, zinc will preferentially be oxidised over copper.

The anode will be zinc and the cathode will be copper.

Gibbs free energy

It is interesting to realise that many electrochemical reactions are reversible processes. During the day when you are using your cell phone, its battery is acting as a spontaneous voltaic cell, where the flow of electrons from the anode to the cathode is powering your device. At night, when you charge your phone, the electrical current from the wall outlet supplies the flow of electrons that allows for the chemical reaction inside your battery to proceed in the reverse (and non-spontaneous) direction. When you wake up in the morning the battery is fully 'charged' and the voltaic cell is ready to be used once again. That is of course if you remember to charge it.

How does this relate to Gibbs free energy? How is it possible to determine the from $\Delta G^{\ominus}$ from the standard cell potential of an electrochemical reaction?



Figure 2. Charging a battery favours the non-spontaneous reaction.

Credit: Priscila Zambotto, Getty Images

This is not the first time that you have studied the idea of spontaneous versus non-spontaneous chemical reactions. You may recall that changes in Gibbs free energy allow you to determine if a reaction is spontaneous or not in a particular direction (see [section R1.4.2-3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-spontaneous-changes-hl-id-44766/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-spontaneous-changes-hl-id-44766/>)). If the change in Gibbs free energy ($\Delta G^{\ominus}$) for a reaction is negative, then the reaction is said to be spontaneous in the forward direction under standard conditions.

Nature of Science

Aspect: Theories

It is interesting to note that the concept of Gibbs free energy is present in many areas of chemistry, despite being originally developed by Josiah Willard Gibbs as a theory related to thermodynamics. From biochemistry, to thermodynamics, to equilibria, to electrochemical cells, Gibbs free energy allows us to tie together concepts like enthalpy, entropy, equilibrium position, rates of reaction, and electromotive forces under one unifying umbrella. The greatest discoveries made in science relate to concepts and theories that transcend the field in which they were developed and help to create unifying fundamental principles within all of the sciences.

It is possible to determine the change in Gibbs free energy from the standard cell potential of a particular electrochemical reaction by using the equation:

Table 1. Equation for change in Gibbs free energy.

Equation	Symbols	Units
$\Delta G^\ominus = -nFE^\ominus$	$\Delta G^\ominus$ is the change in standard Gibbs free energy	joules per (J mol ⁻¹)
	n is the number of moles of electrons transferred	
	F is the Faraday constant ($9.65 \times 10^4 \text{ C mol}^{-1}$)	Coulombs per (C mol ⁻¹)
	$E^\ominus$ is the standard cell potential	Volts (V)

From this equation it is possible to observe that the change in Gibbs free energy and standard cell potential will always have values that are opposite to one another for a given reaction. If a reaction is spontaneous in the forward direction, then the standard cell potential will be positive and the associated change in Gibbs free energy will be negative, and vice versa for the reverse reaction which by default would be non-spontaneous. Recall that as a reaction approaches equilibrium the change in Gibbs free energy starts to approach zero (see [section R1.4.4a \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-equilibrium-hl-id-44767/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-equilibrium-hl-id-44767/)). This is true for both thermodynamic and electrochemical systems.

Study skills

It can be difficult to recall all the appropriate units and constants used for various types of calculations.

Note that the Faraday constant can be found in the DP chemistry data booklet.

When performing this particular calculation using

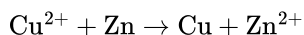
$$\Delta G^\ominus = -nFE^\ominus$$

it is important to understand that a potential difference of one volt is equivalent to one joule per coulomb.

$$1 \text{ volt} = 1 \text{ J C}^{-1}$$

Worked example 2

In the worked example from earlier in this section it was discovered that the voltaic reaction between copper and zinc produces a standard cell potential ($E_{\text{cell}}^{\ominus}$) of +1.10 V for the reaction:



Calculate the value of $\Delta G^{\ominus}$ for this reaction.

Note that two electrons are being transferred from the zinc to the copper ions, therefore $n = 2$.

$$1 \text{ V} = 1 \text{ J C}^{-1}$$

$$\begin{aligned}\Delta G^{\ominus} &= -nFE^{\ominus} \\ &= -2 \times 9.64 \times 10^4 \times 1.10 \\ &= -2.12 \times 10^5 \text{ J mol}^{-1} \text{ or } -212 \text{ kJ mol}^{-1}\end{aligned}$$

Now you will be able to take these concepts and apply them to other situations.

Activity

- **IB learner profile attribute:**
 - Inquirer
 - Reflective
- **Approaches to learning:**
 - Thinking skills — Reflecting on the credibility of results
 - Research skills — Comparing, contrasting and validating information
- **Time required to complete activity:** 20–30 minutes
- **Activity type:** Individual/pair activity

Using the simulator, build four different voltaic cells with a variety of combinations of electrodes and electrolytes (all at a concentration of 1.0 M), and compare your experimental values to the theoretical standard cell potentials which you will calculate yourself using the standard cell potentials provided in the DP chemistry data booklet.

1. Access the simulator via the link: <https://teachchemistry.org/classroom-resources/galvanic-voltaic-cells-2> (<https://teachchemistry.org/classroom-resources/galvanic-voltaic-cells-2>)
2. Under the heading Galvanic/Voltaic Cells 2 click on Start →
3. Create a data table in Excel or using a digital notebook.
 - (a) Create a table with three columns and five rows. The first column should have the heading 'Cell Combinations' for the reactions between the species, zinc and silver, zinc and magnesium, copper and silver, and lastly silver and magnesium.
 - (b) The second column should have the heading 'Standard Cell

Potential' and a third column with 'Cell Notation'

4. Using the simulator, construct a voltaic cell consisting of a zinc electrode in a 1.00 M zinc nitrate solution in the left beaker and a silver electrode in 1.00 M $\text{AgNO}_3(\text{aq})$ in the right beaker.
5. Start the simulation by clicking the button on the voltmeter from off to on.
6. Record any quantitative data (including measurement uncertainties) for the standard cell potential.
7. Repeat the experiment three more times using the other combinations of electrodes and solutions that you recorded in your data table.

Use the data recorded during the activity to answer these questions.

1. Using the standard electrode potentials in the DP chemistry data booklet, calculate the theoretical standard cell potentials for the four simulated reactions that you ran.
2. Complete the data table by inserting the correct cell notation for each of the four reactions that were performed in this investigation. Be careful to understand which species is being oxidised and which is being reduced.
3. Explain why it is necessary to keep the concentrations of the aqueous solutions at 1 mol dm^{-3} .
4. List the reactions in order from most spontaneous to least spontaneous, using calculations of the change in Gibbs free energy to justify why they are listed in this order.
5. Deduce how the results of your investigation may have been different in terms of accuracy and precision if you had conducted this experiment using a hands-on experimental setup instead of a simulation.

Electrolysis of water and aqueous solutions (HL)

Higher level (HL)

Learning outcomes

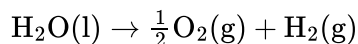
By the end of this section you should be able to:

- Deduce from standard electrode potentials, the products of the electrolysis of aqueous solutions.
- Demonstrate that electrolytic processes include the electrolysis of water and of aqueous solutions.
- Describe the effects of concentration and the nature of the electrode for the electrolysis of NaCl(aq) and CuSO₄(aq).

As you learned in [section R3.2.13–14](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/14%20gibbs-free-energy-and-standard-electrode-potentials-hl-id-44784/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/14%20gibbs-free-energy-and-standard-electrode-potentials-hl-id-44784/>) electrochemical reactions can occur spontaneously or non-spontaneously. Non-spontaneous electrochemical reactions require a constant supply of energy to drive the reaction forward, in the form of an external power supply. Recall that it is also possible to oxidise and reduce the component ions in a molten salt through the use of an external power supply, despite the reaction being non-spontaneous (see [section R3.2.8](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrolytic-cells-id-46491/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrolytic-cells-id-46491/>)). In fact, electrolysis reactions can occur not only with molten salts but in aqueous solutions as well. How does the presence of water in the solution change the products of electrolysis? To answer this question, we will begin by looking at the electrolysis of water.

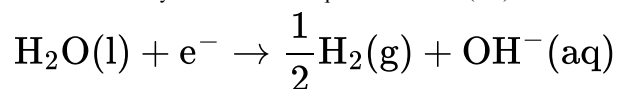
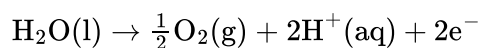
Electrolysis of water: A non-spontaneous redox process

If it is possible for aqueous solutions to undergo electrolysis reactions, then it stands to reason that it should be possible for the water in the aqueous solution to be oxidised and reduced. This is referred to as the electrolysis of water, which occurs according to the overall (net) chemical reaction:



This net reaction can be further subdivided into the possible reactions taking place at the anode and the cathode.

Reduction half-reaction (at the cathode):


Oxidation half-reaction (at the anode):


All half-reactions and their standard electrode potentials are provided in [section 19 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/standard-reduction-potentials-at-29815-k-id-45121/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/standard-reduction-potentials-at-29815-k-id-45121/) of the DP chemistry data booklet.

Figure 1 shows the apparatus used for the electrolysis of water, which is known as a Hofmann voltameter. The water to be electrolysed is mixed with dilute sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$, which is necessary because pure water is a poor conductor of electricity. The addition of the dilute sulfuric acid (dilute sodium hydroxide solution, $\text{NaOH}(\text{aq})$, can also be used) allows the solution to conduct electricity due to the presence of mobile ions. So rather than pure water being electrolysed, it is actually a dilute solution sulfuric acid or sodium hydroxide. Based on the net equation for the reaction shown above, what is the ratio of $\text{H}_2(\text{g})$ to $\text{O}_2(\text{g})$ produced?

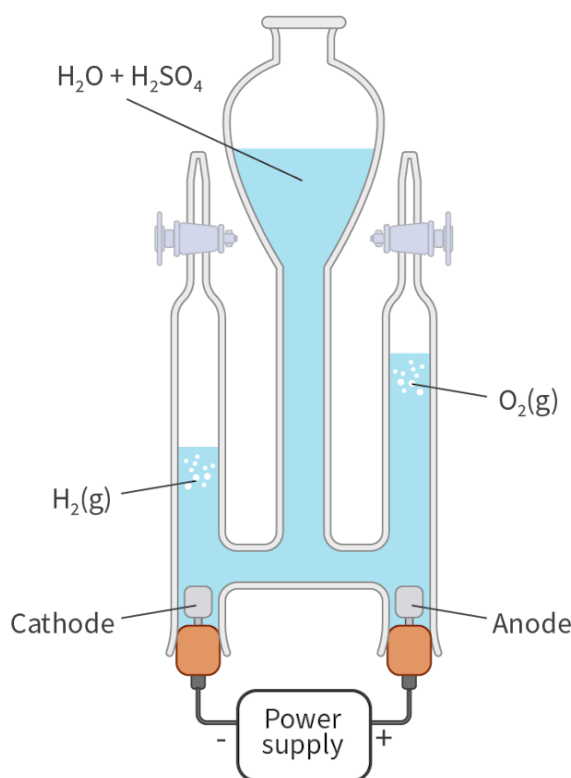


Figure 1. The electrolysis of water using a Hoffman voltameter. What is the ratio of hydrogen gas to oxygen gas produced in the reaction?

 More information for figure 1

The image is a diagram of a Hoffman voltameter used for the electrolysis of water. It illustrates the setup with a central reservoir containing water mixed with sulfuric acid, represented as $\text{H}_2\text{O} + \text{H}_2\text{SO}_4$. The central container is connected to two side tubes: one labeled with $\text{H}_2(\text{g})$, representing hydrogen gas, and the other with $\text{O}_2(\text{g})$, representing oxygen gas. Hydrogen gas is produced at the cathode, while oxygen gas is produced at the anode, both part of the power supply labeled at the bottom of the setup. The hydrogen is

collected on the left, and oxygen on the right, with labeled outlets for each gas. The diagram shows the collection of gases in the respective tubes, illustrating the 2:1 volume ratio of hydrogen to oxygen produced in the reaction.

[Generated by AI]

The addition of a universal indicator to the electrolyte shows interesting qualitative data, as the indicator turns from green to orange at the anode (due to the production of hydrogen ions) and from green to blue at the cathode (due to the evolution of hydroxide ions). Other qualitative and quantitative observations can be made by collecting the gas phase products at each electrode. From the net reaction for the electrolysis of water it is possible to see that hydrogen gas is produced at twice the rate of oxygen gas, as two moles of hydrogen gas are produced for every mole of oxygen gas. This is also reflected in the volumes of the gases produced in **Figure 1** - the ratio of hydrogen to oxygen is 2:1.

Worked example 1

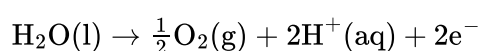
The electrolysis of water is conducted in a lab using a dilute sodium hydroxide solution, two graphite electrodes with attaching wires to a 9 V battery. When the reaction takes place, small bubbles start to form at both electrodes. Suggest two possible experimental techniques that could be used to determine the nature of the gases being evolved at each of the electrodes.

- Use a universal indicator, pH paper strips, or a pH probe to determine the change in pH at each electrode. This would allow you to determine the rate of formation of the hydroxide ions and hydronium ions at each electrode.
- Collection of each gas at either electrode using an eudiometer would be possible. It would be possible to measure the rate of formation of each gas evolved.
- Conducting a flaming splint test and a glowing splint test on each gas sample would allow you to identify the hydrogen and oxygen gas.

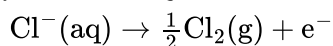
The effect of concentration on the electrolysis of aqueous solutions

There is a special consideration required when an aqueous solution of sodium chloride, NaCl(aq), is electrolysed. Unlike the electrolysis of molten sodium chloride – which has only one possible product at the anode – the presence of water in the aqueous solution somewhat complicates the situation. This is because, as discussed previously, water itself can undergo oxidation to produce oxygen gas, O₂(g).

The half equation for the oxidation of water is as follows:



And the half equation for the oxidation of chloride ions is:

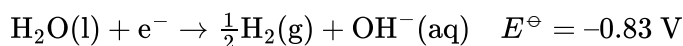
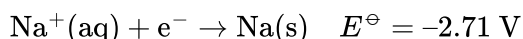


So which species will undergo oxidation at the anode and which gas will be produced? The answer to this question depends on the concentration of the solution.

- In dilute solutions of sodium chloride, the water will be oxidised leading to the production of oxygen gas at the anode.
- In concentrated solutions, the chloride ions will be preferentially oxidised leading to the formation of chlorine gas.

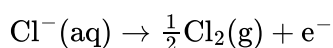
Changes in the concentration of the salt solution have a large impact on which species is oxidised during electrolysis. In dilute solutions, water molecules are oxidised to produce oxygen gas. Whereas, in concentrated solutions of sodium chloride, chloride ions are oxidised to produce chlorine gas. So to summarise, if the concentrated solution contains a halide ion (Cl^{-} , Br^{-} or I^{-}) then the halogen (Cl_2 , Br_2 or I_2) is given off at the anode. However, if the halide ions are present in a very low concentration, oxygen gas will be given off at the anode instead.

The two possible reactions at the cathode are the reduction of sodium ions or the reduction of water:

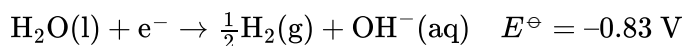


H_2O is a stronger oxidising agent than Na^{+} ions (H_2O has a more positive $E^{\ominus}$ value), meaning that it is more easily reduced. Therefore, at the cathode, water undergoes reduction to form hydrogen gas. The electrolysis of concentrated sodium chloride solution (also known as brine) is the basis of a major chemical industry noted for its efficient use of all of the products. The electrolysis of brine produces hydrogen and chlorine gases, but the product of greatest economic importance is the alkali, sodium hydroxide. Note that sodium hydroxide is not produced at either the cathode or the anode of the cell, instead, it exists as the solution formed in the reaction.

In general, metal ions with a less positive $E^{\ominus}$ value than water will not be reduced at the cathode when part of an aqueous solution. And metal ions with a more positive $E^{\ominus}$ value than water will be reduced instead of water. Take, for example, a concentrated aqueous solution of copper(II) chloride, $\text{CuCl}_2(\text{aq})$. Because of the high concentration of chloride ions, the product at the anode is chlorine gas.



At the cathode, the two possible half equations are the reduction of water:



or the reduction of copper(II) ions:



The $E^{\ominus}$ value for the reduction of copper(II) ions is more positive than that for the reduction of water. For this reason, copper(II) ions are reduced to form copper metal.

Use **Interactive 1** to investigate the gases given off at the anode and the cathode.

Chlorine gas

Hydrogen gas

Oxygen gas

Copper solid

☒ Check

Interactive 1. Electrolysis of aqueous solutions at different concentrations.

Figure 2 shows a concept diagram of the electrolysis of aqueous solutions at different concentrations.

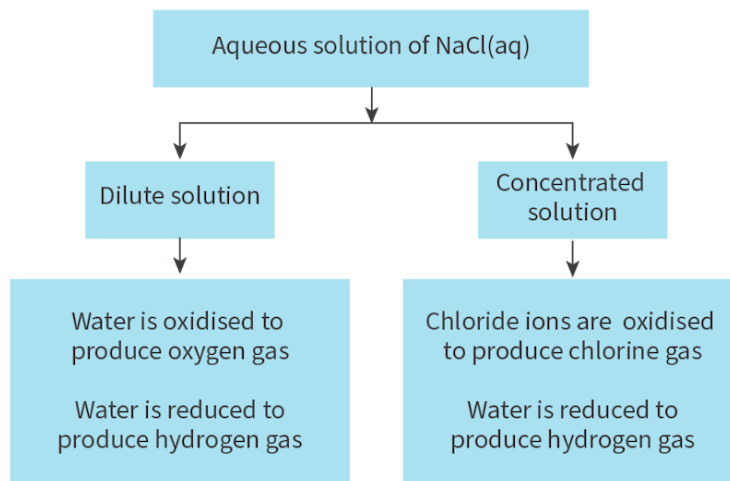


Figure 2. A concept diagram of the electrolysis of aqueous solutions.

 More information for figure 2

The diagram illustrates the electrolysis of an aqueous NaCl (sodium chloride) solution, divided into two paths based on concentration. At the top, a blue box labeled "Aqueous solution of NaCl(aq)" is shown. Two arrows branch downward from this box.

On the left, the path is labeled "Dilute solution." Below, a box contains the text: "Water is oxidised to produce oxygen gas. Water is reduced to produce hydrogen gas."

On the right, the path is labeled "Concentrated solution." Below, a box contains the text: "Chloride ions are oxidised to produce chlorine gas. Water is reduced to produce hydrogen gas."

Each process is connected by arrows, indicating the flow of chemical reactions based on the concentration of the solution.

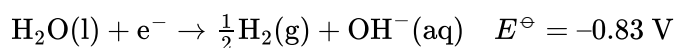
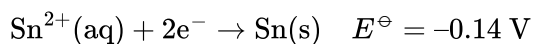
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Worked example 2

Tin(II) bromide is highly soluble in water. Predict the products formed via the electrolysis of a concentrated solution of $\text{SnBr}_2(\text{aq})$ at SATP.

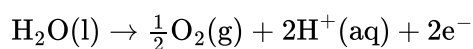
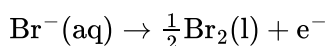
The possible half-reactions occurring at the cathode and anode must be assessed based on their standard electrode potentials to see which species is preferentially oxidised and which is preferentially reduced. In addition, because the solution contains aqueous halide ions, $\text{Br}^-(\text{aq})$, the concentration must also be considered.

Cathode (reduction):



Tin ions have a more positive (or less negative) $E^\ominus$ value so will be preferentially reduced to form solid tin at the cathode.

Anode (oxidation):



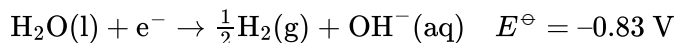
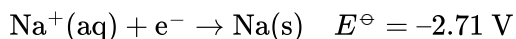
In a concentrated solution, bromide ions will be preferentially oxidised to form liquid bromine at the anode.

Worked example 3

A concentrated sodium iodide solution can undergo electrolysis at SATP to produce hydrogen gas and iodine. Determine the half-equations and products at each electrode.

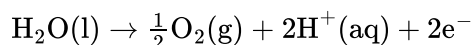
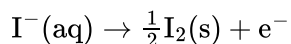
The possible half-reactions occurring at the cathode and anode must be assessed based on their standard electrode potentials to see which species is preferentially oxidised and which is preferentially reduced.

Cathode (reduction):



Hydrogen gas will be evolved at the cathode as the hydrogen is preferentially reduced over the sodium ions at the cathode.

Anode (oxidation):



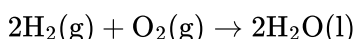
Because of their high concentration, iodide ions will be preferentially oxidised at the anode to form iodine.

Making connections

You previously learned that the use of hydrogen fuel cells, which are voltaic cells, allows for the continuous production of an electrical current (see section R1.3.5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fuel-cells-id-44783/>)). This type of fuel cell produces an electrical current simply by subdividing the spontaneous redox reaction between oxygen and hydrogen to produce water and harnessing the flow of electrons between the two half-cells in the wire.

It is interesting to note that the hydrolysis of water and the reaction taking place in a hydrogen fuel cell are the reverse of the same reaction.

Where one process



is spontaneous and releases energy, the hydrolysis of water is a non-spontaneous process and requires the input of energy.

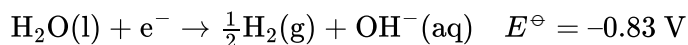
Types of electrodes and electrolytes on the electrolysis of salt solutions

Up until this point you have looked extensively at the electrolysis of aqueous salt solutions consisting of a metal cation and a halogen anion. As you have seen salts form not only from metal cations and halogen anions, but a variety of polyatomic species (see section R3.1.12 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-of-salt-solutions-hl-id-46060/>)). It is possible for these types of salt solutions to undergo electrolysis, but how they undergo electrolysis changes somewhat depending on the type of electrode used to electrolyse the solution.

Take for example the electrolysis of copper(II) sulfate using an inert electrode like platinum. Platinum is able to carry a charge into the electrolyte but not participate in the chemical reaction. Graphite can also be used as an inert electrode as it is relatively unreactive, has a high melting point, and is a good electrical conductor

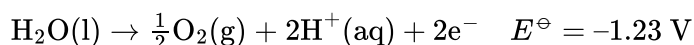
due to the delocalised electrons in its molecular structure that provides for good electron mobility. It is essential to use electrodes which are highly conductive, have high melting points, and yet are not preferentially oxidised or reduced over the species in solution.

At the cathode the possible reactions are:



Copper(II) ions will be preferentially reduced and copper solid will be produced at the cathode.

At the anode the only possible reaction is:



The sulfur atom in SO_4^{2-} has an oxidation state of +6, which is its highest oxidation state, and therefore cannot be further oxidised. This is true for many other polyatomic oxyanions like chlorate, nitrate, and phosphate to name a few.

Worked example 4

The electrolysis of a copper(II) sulfate solution takes place using two graphite electrodes. What would be possible signs that a chemical reaction is taking place once the electrolysis starts to occur?

Use a universal indicator, pH paper strips, or a pH probe to determine the change in pH would show a decrease in the pH of the electrolyte as hydronium ions are produced.

- Production of a goldish orange colour on the cathode, as copper solid is deposited.
- Production of oxygen gas bubbles at the anode. If collected the gas could be identified using a glowing splint test.
- The intense blue colour of the copper(II) sulfate solution would decrease over time due to the removal of the copper(II) ions via reduction.

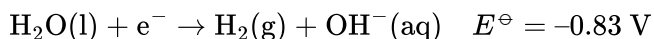
Worked example 5

An aqueous solution of concentrated copper(II) sulfate undergoes electrolysis using inert carbon electrodes.

State the half-reactions that occur at the cathode and the anode, and the overall balanced chemical reaction for the electrolysis of concentrated copper(II) sulfate.

At the cathode, the two following reduction reactions could be taking place:



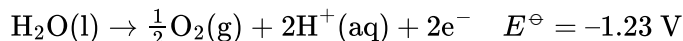


As the copper(II) ions have a more positive standard electrode, they will be preferentially reduced compared to the water.

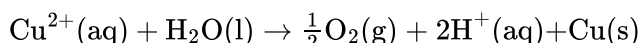
At the anode, two possible oxidation half-reactions should be able to occur, that of water or the oxidation of the sulfate ion.

Note: The sulfur atom in SO_4^{2-} has an oxidation state of +6, which is its highest oxidation state, and therefore cannot be further oxidised.

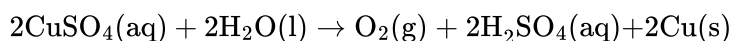
This means that at the anode, water is being preferentially oxidised:



The overall net-ionic equation is therefore the sum of these two half reactions:



Adding in the spectator sulfate ions and doubling the coefficients produces the reaction:



If a reactive electrode like copper is used instead of an inert electrode then the solid copper electrode will be preferentially oxidised at the anode and the electrode will produce copper(II) ions in solution instead of generating oxygen gas.

This means that the pH of the electrolyte would remain constant, as would the concentration of copper(II) ions in solution. Copper would still be plated onto the cathode and the copper anode would be slowly eroded.

Aspect: Experiments

An extremely important aspect of scientific research often has to do with improving the protocols of other scientists. Developing new techniques allows us to improve the yield of chemical reactions and is an essential part to most areas of chemical research.

For example, the direct-current electrolysis of concentrated potassium hydroxide solutions often needs to occur at high temperatures (up to 400 °C) and under extremely high pressures. From your studies of chemical kinetics, it is easy to see the advantages of running an electrolysis reaction at high temperatures and pressures, including increasing the conductivity of the electrolyte and increasing the rates of electrochemical reactions at the electrode surfaces.

Researchers often need to design experiments in order to meet the constraints of the chemical reactions they are studying. In this example, Jason C. Ganley of Howard University was able to better understand the type of anode materials that would work best under these conditions,

including nickel, lithiated nickel, cobalt-plated nickel and other metal alloys in this article [↗](#)

(<https://www.sciencedirect.com/science/article/abs/pii/S0360319909003577?via%3Dihub>).

Activity

- **IB learner profile attribute:**
 - Communicator
 - Reflective
- **Approaches to learning:** Communication skills — Clearly communicating complex ideas in response to open-ended questions; Using terminology, symbols and communication conventions consistently and correctly
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Individual/group activity

In this activity you will use your understanding of electrochemical reactions and standard cell potentials to reflect on the conditions necessary to produce an electrolysis reaction of an aqueous solution.

Your task

You will analyse the electrolysis reaction and describe the properties and components of your chosen system. The ability to communicate complex concepts correctly and clearly to others is a vital skill. As well, in preparing your ideas and concepts for explanation you often develop a better understanding of the processes.

Table 1 contains the half-reactions and associated standard electrode potentials for a variety of different species that can combine to produce an ionic salt. You should select a solution that when electrolysed would produce solid metal at the cathode and oxygen gas at the anode.

Step 1: Select from **Table 1** two species that would form an appropriate ionic salt.

Step 2: Sketch a diagram of the electrolytic cell for your chosen salt in aqueous solution.

- Include the type of electrode required (inert or reactive).
- The half-reactions occurring at each electrode.
- The approximate molecular concentration of the aqueous solution (very dilute, dilute or concentrated).
- The products being produced at each electrode.
- The overall balanced reaction for your chosen electrolysis process.

Table 1. Half-reactions and associated standard elec

Oxidised species	Reduced species
$X_2(g) + 2e^-$	$2X^-(aq)$
$Y^{2+}(aq) + 2e^-$	$Y(s)$
$W^{3+}(aq) + 3e^-$	$W(s)$
$Z_2(q) + 2e^-$	$2Z^-(aq)$

Note: The standard electrode potentials and half-reactions for the oxidation and reduction of water are found in the DP chemistry data booklet.

Extension

Share your sketch with the other members of your group and compare and contrast the half-reactions each of your group members chose. Discuss any similarities and differences between your chosen systems and look for any inconsistencies or areas where other students might need greater clarity.

Electroplating (HL)

Higher level (HL)

Learning outcomes

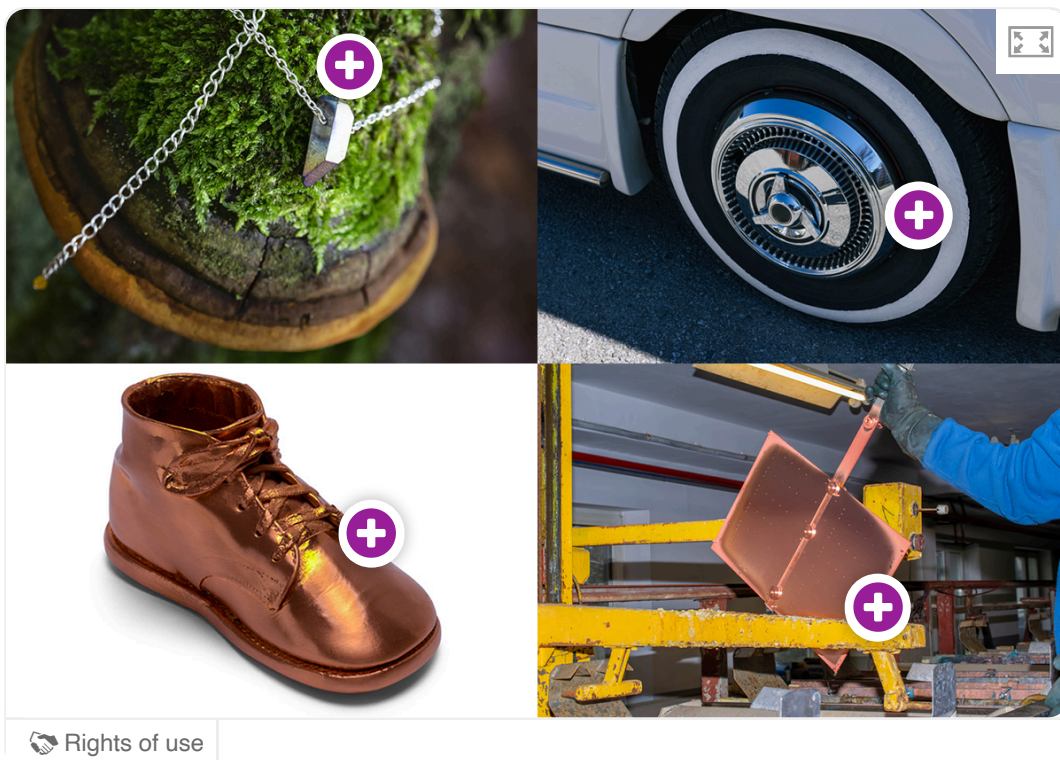
By the end of this section you should be able to deduce equations for the electrode reactions during electroplating.

Electroplating: Depositing metallic ions on surfaces

Many of the objects we have around us are made of materials that have been chemically coated or plated in a thin coat of solid metal. Sometimes this is done to improve the appearance of the object, or to improve its physical properties like resilience, lustre, or hardness.

Take for example gold tone jewellery. Have you ever purchased a somewhat inexpensive pair of earrings just to notice that after a few months the gold seems to be wearing off and another metal is being exposed underneath or has left a greenish stain on your skin? This is an example of electroplating one metal onto another. But how does this occur? Is this thin coating of metal simply painted on, or is it somehow bonded to the material underneath?

Use **Interactive 1** to investigate some of the materials that are used in electroplating.



Interactive 1. Electroplating Different Objects.

More information for interactive 1

A collage of four images, each featuring an object with a plus icon indicating a hotspot which when clicked, pops up additional information.

The top left image is a silver chain necklace with a pendant resting on a moss-covered tree trunk. The hotspot reads, Crystal quartz electroplated on a silver chain.

The top right image is a close-up of a car tire with a white sidewall and a shiny chrome hubcap. The hotspot reads, A chrome hubcap.

The bottom left image is a baby shoe coated in a reflective copper finish. The hotspot reads, A bronzed baby shoe.

The bottom right image is a worker handling a copper-plated industrial component in a workshop setting, with machinery and metal structures in the background. The hotspot reads: A copper electroplating machine for printed circuit boards.

The answer lies in how electroplating occurs. Electroplating involves the use of an electrolysis reaction, driving a non-spontaneous electrochemical reaction through the use of a power supply. Once connected to the power supply, the process allows for one metal to be transferred within a solution onto the surface of an electrode.

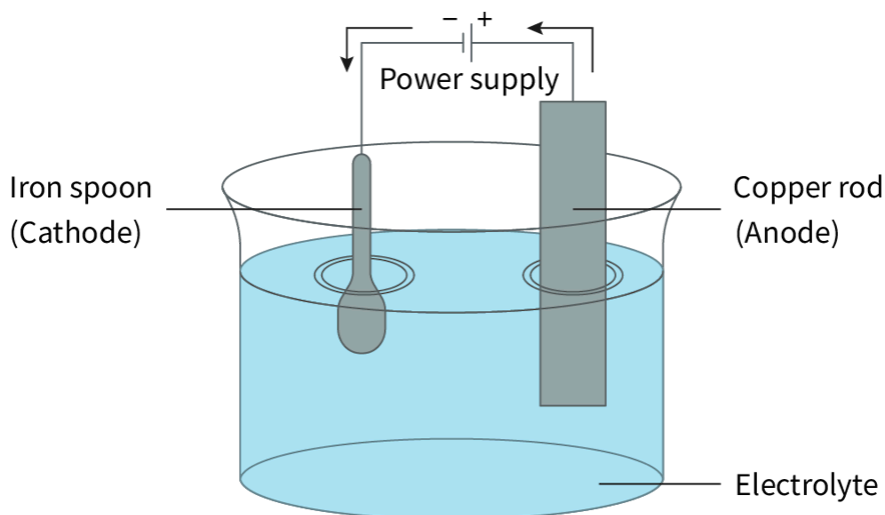


Figure 1. Plating an iron spoon with copper.

 More information for figure 1

The image is a diagram of an electroplating setup. It shows a cylindrical container filled with a blue solution labeled as 'Electrolyte.' Inside the container, there is an iron spoon on one side labeled as 'Iron spoon (Cathode).' On the opposite side, there is a copper rod labeled as 'Copper rod (Anode).' Both the spoon and the rod are partially submerged in the solution but extend above it.

Above the container, there's a representation of a power supply. The spoon is connected to the negative terminal and the copper rod to the positive terminal, suggesting the flow of current. Arrows indicate the direction of current from the power supply to the electrodes. This setup is used to plate the iron spoon with a layer of copper transferred from the copper rod through the electrolysis process.

[Generated by AI]

An electroplating system must consist of:

- a power supply (as the reaction is non-spontaneous)
- a conductive cathode (the object you wish to plate)
- a metal anode
- an electrolytic solution containing a common metal ion to the anode.

The cathode can be any solid object so long as it is electrically conductive. This electrode can be made of an inert material like graphite or platinum or a reactive metal.

When the reaction is assembled and the current is connected, electrons flow from the power supply into the cathode, just as in any other electrolysis reaction. The metal cations in solution then migrate to the cathode where they collect the electrons on the surface of the electrode and bond to the cathode as a thin layer of metal atoms. This process of layering a thin sheet of atoms of one metal, through reduction, over a surface is known as plating. If the cathode is made of metal then the plated material is alloyed onto the surface of the metal electrode.

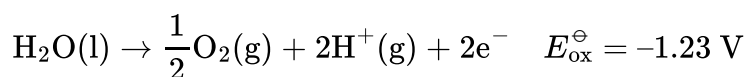
How Does Electroplating Work | Reactions | Chemistry | ...



Video 1. Electroplating reactions in action.

As the plating process occurs and metallic ions are reduced and pulled out of solution to be deposited on the cathode, the anode slowly starts to undergo an oxidation reaction. If the anode is made of a reactive metal then the anode is slowly eroded, as it is oxidised to form metallic ions in solution.

You may recall that when inert electrodes like platinum or graphite are used in electrolysis reactions, the oxygen in the water molecules from the aqueous solution will be preferentially reduced and you will see oxygen gas being evolved according to the reaction:



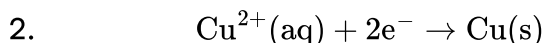
In electroplating reactions, the anode is a reactive metal and is preferentially oxidised over the water molecules.

Worked example 1

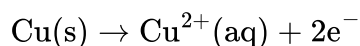
Stainless-steel (an alloy of carbon and iron) is often plated with copper for decorative purposes. The setup involves a solution of copper(II) sulfate, a copper electrode, a stainless steel object you wish to plate, and a power supply.

1. State which electrode is the anode and which is the cathode in this reaction.
2. Give the half-reaction that is allowing for the plating of the stainless-steel electrode.
3. Describe the process that is occurring at the anode, using a half-reaction.
4. Describe two experimental techniques that would allow you to observe that a chemical reaction is occurring.

1. The stainless steel electrode acts as the cathode and a copper electrode would be needed to act as the anode.



3. The anode is the location of the oxidation half-reaction, and the copper electrode becomes pitted or eroded.



4. Change in mass of the cathode and anode.
Change in colour of the electrolyte solution.

Activity

- **IB learner profile attribute:**
 - Inquirer
 - Reflective
- **Approaches to learning:**
 - Thinking skills — Combining different ideas in order to create new understandings
 - Research skills — Comparing, contrasting and validating information, Using search engines and libraries effectively
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Pair/group activity

Electroplating is a highly ubiquitous chemical process. In your daily life you come across hundreds of examples of electroplated items without even realising it. These items are often overlooked and how they are created is sometimes poorly understood by students and adults alike.

Your task

In this activity you will research to identify an object that you use in your daily life that is fabricated using electroplating. You will want to determine, and then explain, how this particular object is manufactured using electroplating.

It may be interesting to choose an object that is manufactured in your region or community. You may wish to extend your research to include the environmental impact of electroplating and a description of how solutions that contain zinc and other heavy metal ions are toxic and have to be disposed of in an appropriate manner.

Sketch a simple model showing the set-up for this type of electroplating reaction and give the equations for the half reactions taking place at each electrode.

Extension

If you were interested in performing a similar electroplating experiment using the lab equipment available in your school, what methods would you take to quantitatively measure the rate or yield of the reaction?

Determine the theoretical cell potential and change in Gibbs free energy for this reaction.